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(54) **CAR CONSTRUCTS AND METHODS OF TREATMENT**

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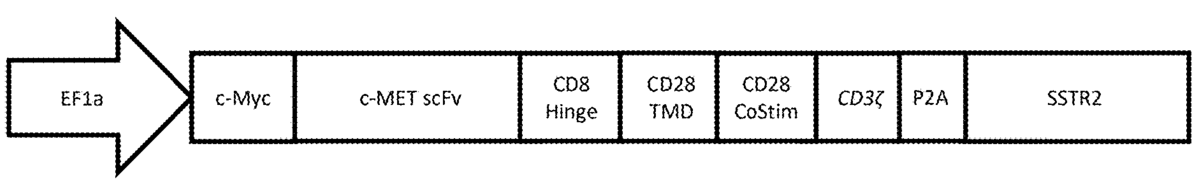
(57) **ABSTRACT**

The present disclosure relates to low affinity chimeric antigen receptors (CARs) and CAR-T cells, which provide cytotoxicity against tumors overexpressing the proto-oncogene c-MET and alleviate on-target, off-tumor toxicities. The CAR-T cells of the present disclosure comprise low affinity anti-c-MET scFvs, which facilitate enhanced anti-tumor activity and a reduced rate of tumor relapse.

Specification includes a Sequence Listing.



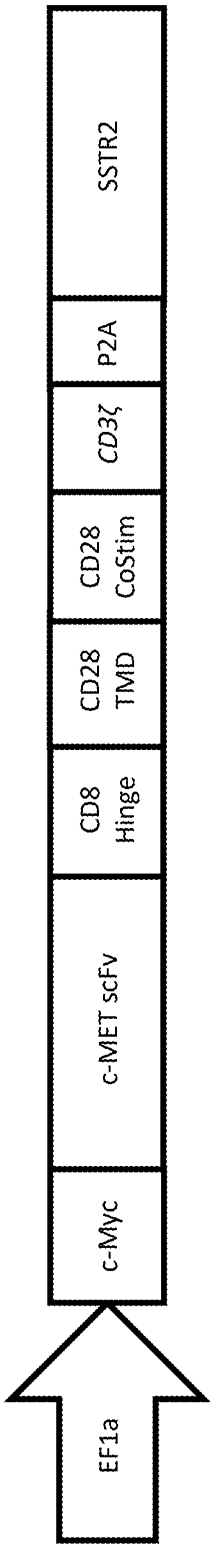
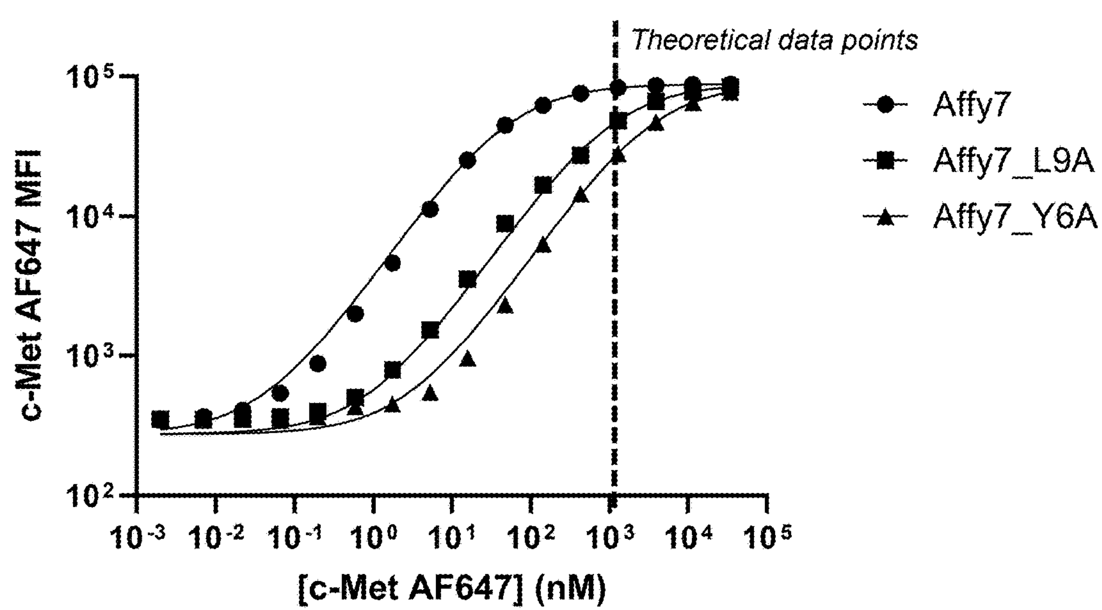

EF1a
c-Myc
c-MET scFv
CD8 Hinge
CD28 TMD
CD28 CoStim
CD3ζ
P2A
SSTR2

FIG. 1

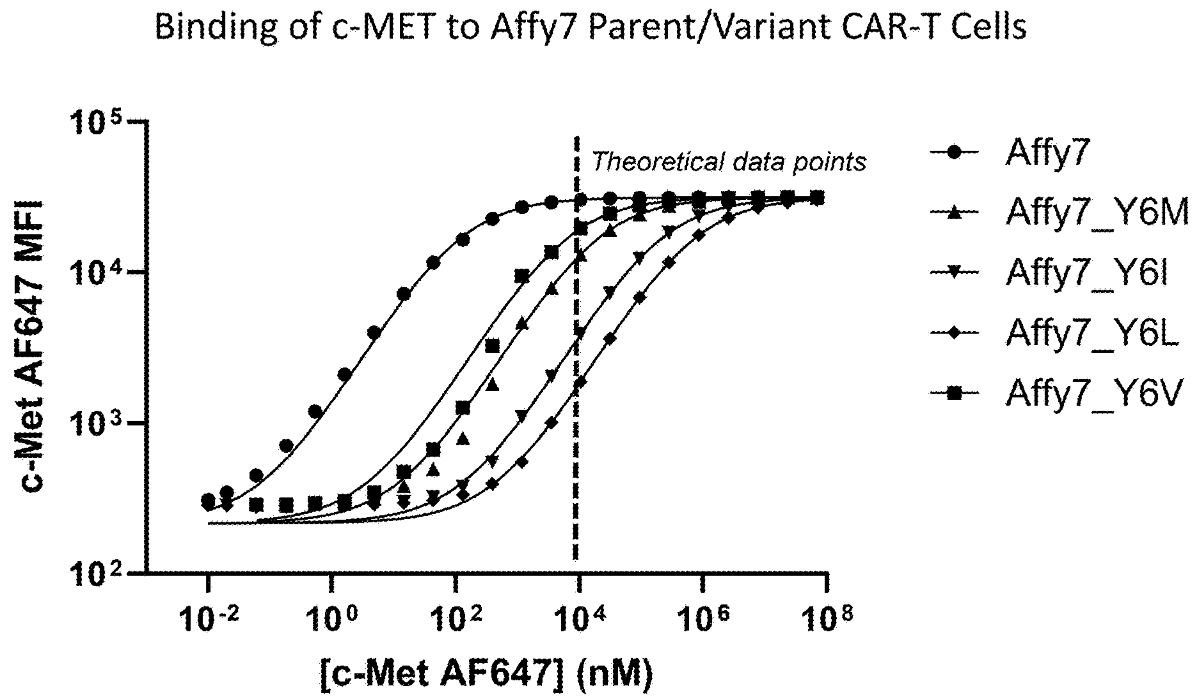
Binding of c-MET to Affy7 Parent/Variant CAR-T Cells



Parent/Variant	EC50 (uM)	Fold Change*
Affy7	0.05	N/A
Affy7_L9A	1.0	20
Affy7_Y6A	3.3	66

*Fold change relative to "parent" Affy7

FIG. 2

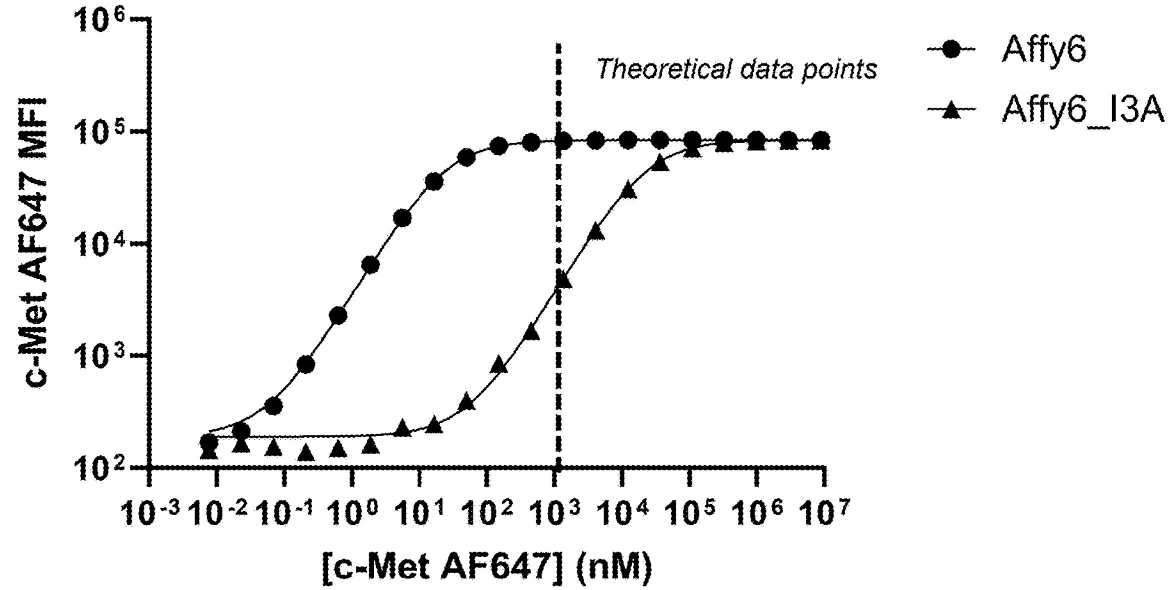


Parent/Variant	EC50 (uM)	Fold Change*
Affy7	0.1	N/A
Affy7_Y6V	5.3	53
Affy7_Y6M	17.2	172
Affy7_Y6I	183.1	1831
Affy7_Y6L	619.2	6192

*Fold change relative to "parent" Affy7

FIG. 3

Binding of c-MET to Affy6 Parent/Variant CAR-T Cells

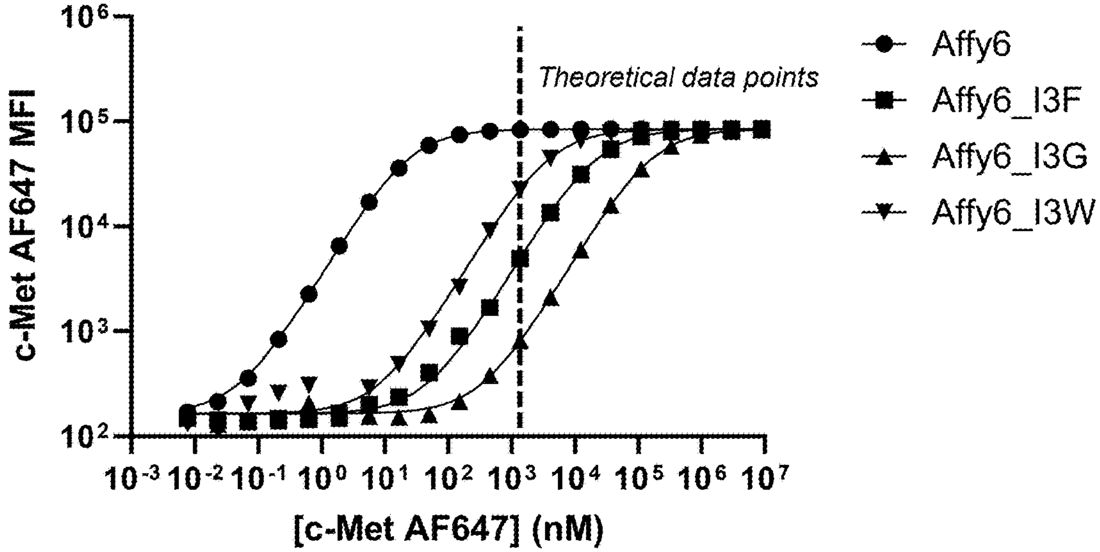


Parent/Variant	EC50 (uM)	Fold Change*
Affy6	0.02	N/A
Affy6_I3A	20.9	956

*Fold change relative to "parent" Affy6

FIG. 4

Binding of c-MET to Affy6 Parent/Variant Cells



Parent/Variant	EC50 (uM)	Fold Change*
Affy6	0.02	N/A
Affy6_I3W	3.6	167
Affy6_I3F	20.5	937
Affy6_I3G	150.8	6899

*Fold change relative to “parent” Affy6

FIG. 5

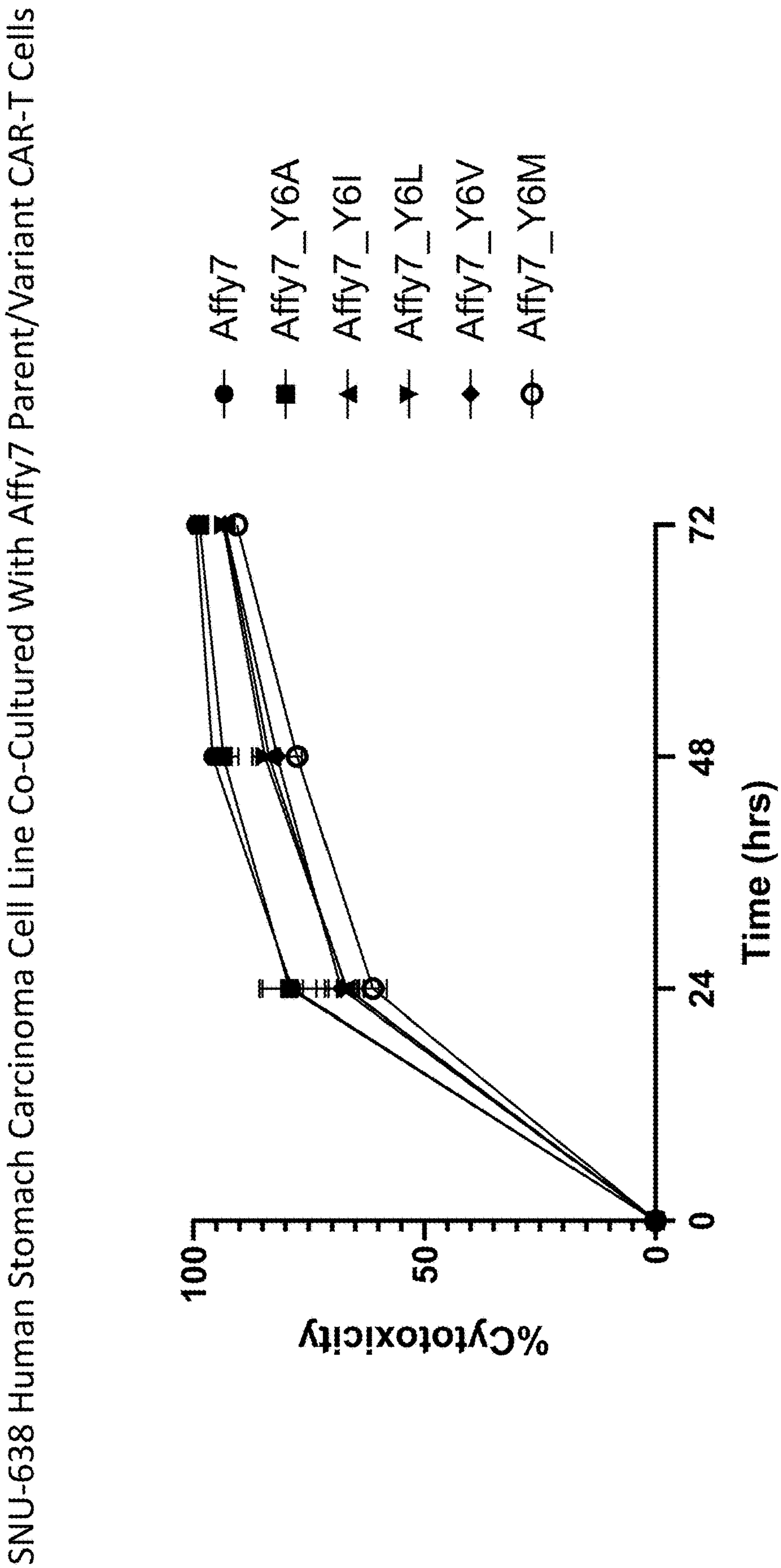

SNU-638 Human Stomach Carcinoma Cell Line Co-Cultured With Affy7 Parent/Variant CAR-T Cells
%Cytotoxicity
100
50
0
0
24
48
72
Time (hrs)
Affy7
Affy7_Y6A
Affy7_Y6I
Affy7_Y6L
Affy7_Y6V
Affy7_Y6M

FIG. 6

Hs746T Human Stomach Carcinoma Cell Line Co-Cultured With Affy7 Parent/Variant CAR-T Cells

FIG. 7A

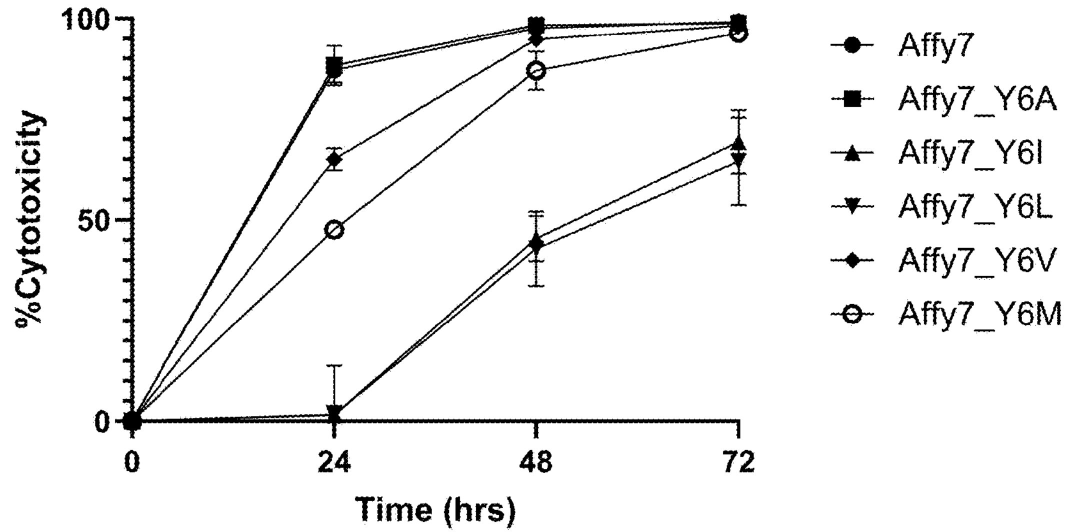
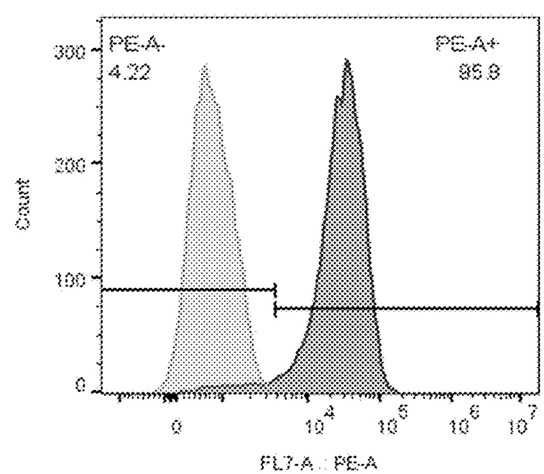


FIG. 7B



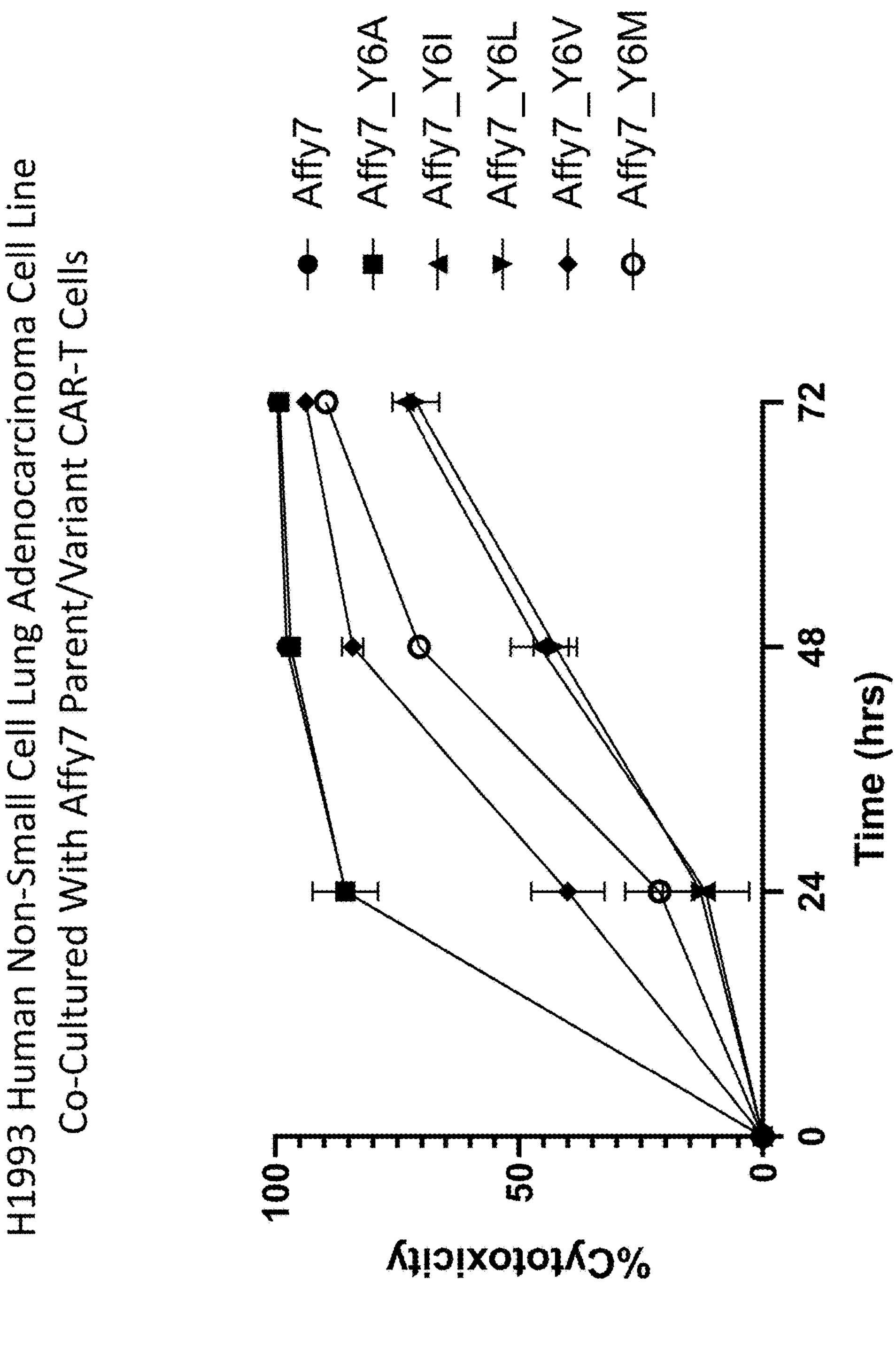

H1993 Human Non-Small Cell Lung Adenocarcinoma Cell Line
Co-Cultured With Affy7 Parent/Variant CAR-T Cells
%Cytotoxicity
100
50
0
0
24
48
72
Time (hrs)
Affy7
Affy7_Y6A
Affy7_Y6I
Affy7_Y6L
Affy7_Y6V
Affy7_Y6M

FIG. 8

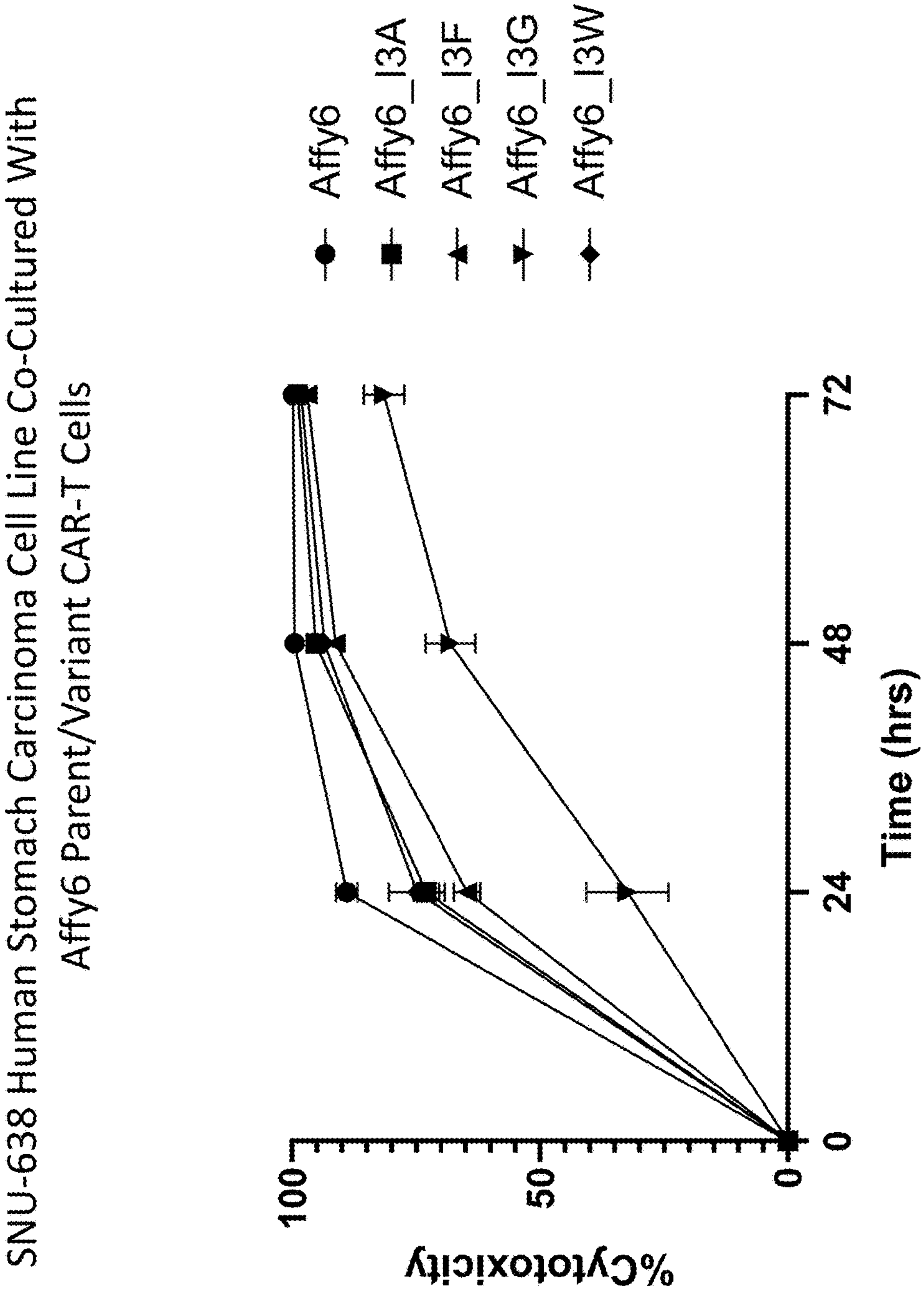

SNU-638 Human Stomach Carcinoma Cell Line Co-Cultured With
Affy6 Parent/Variant CAR-T Cells
%Cytotoxicity
100
50
0
0
24
48
72
Time (hrs)
Affy6
Affy6_I3A
Affy6_I3F
Affy6_I3G
Affy6_I3W

FIG. 9

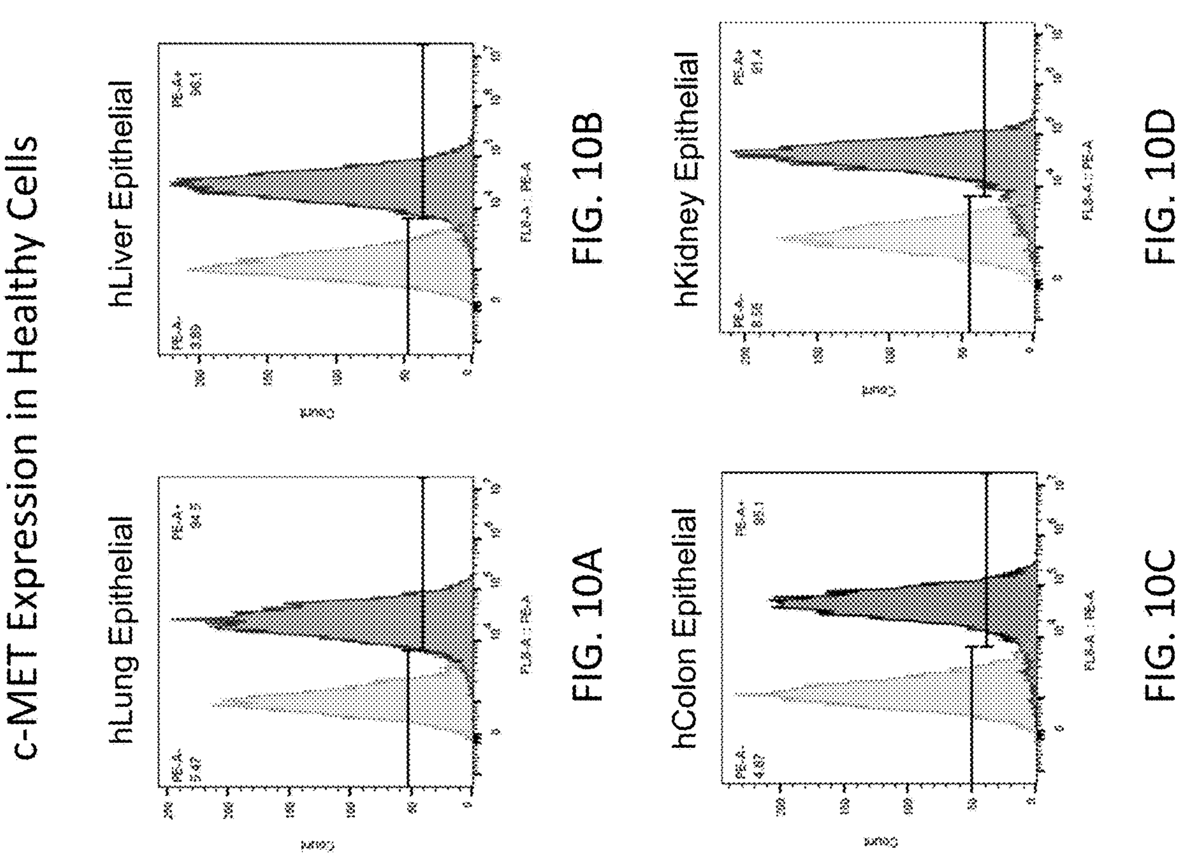


FIG. 10A

FIG. 10B

FIG. 10C

FIG. 10D

Primary Human Lung Cells Co-Cultured With Affy7 Parent/Variant CAR-T Cells

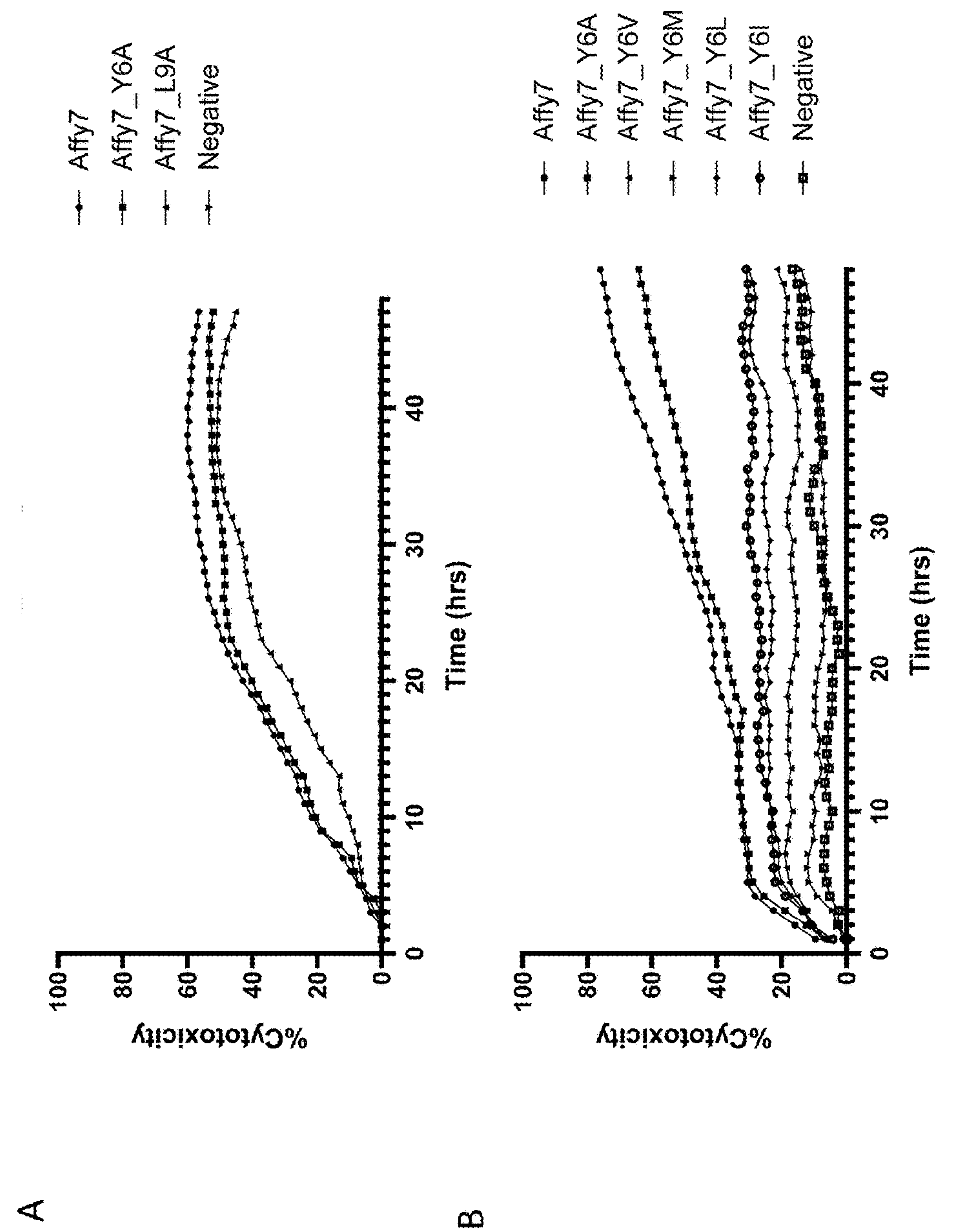


FIG. 11

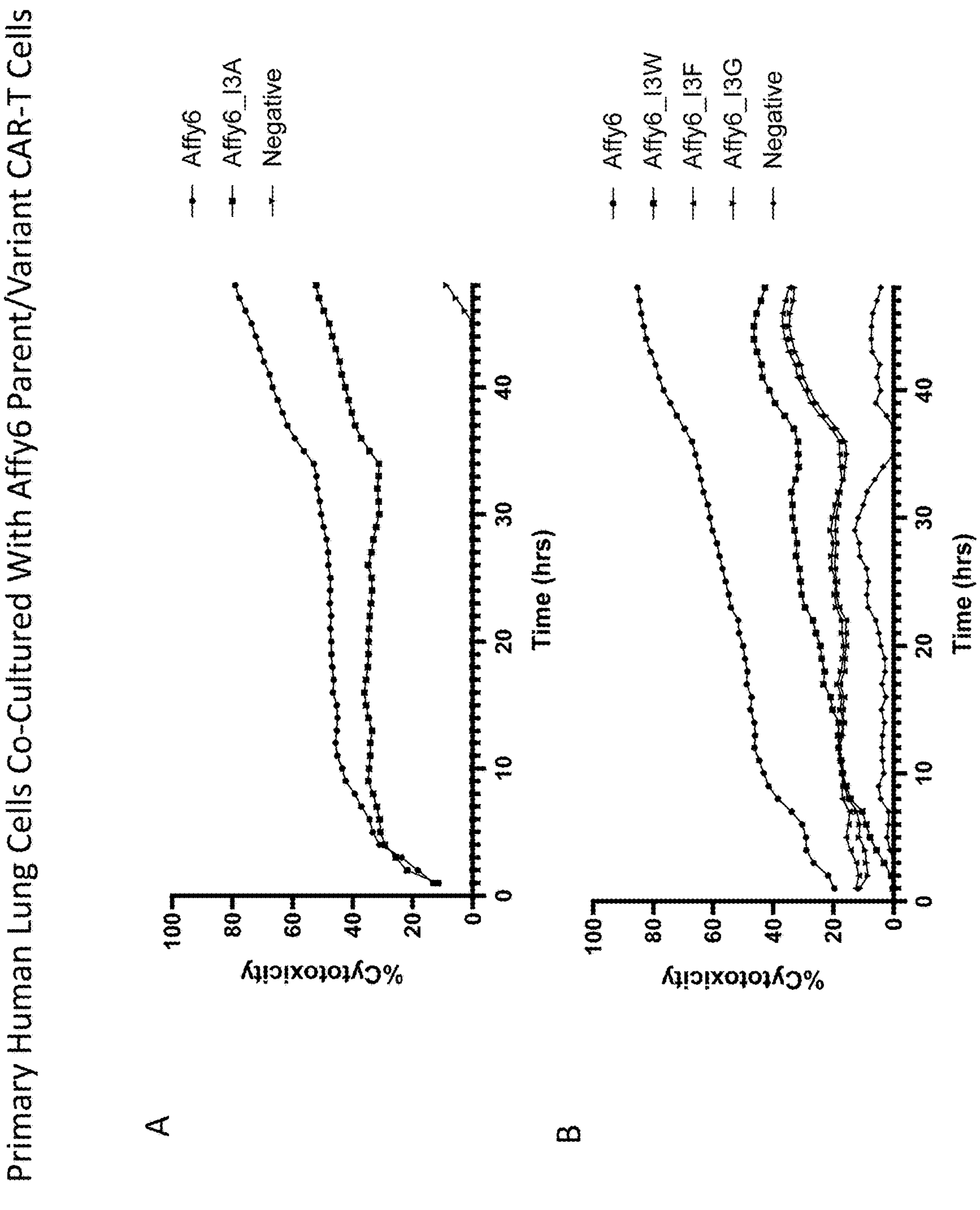

Primary Human Lung Cells Co-Cultured With Affy6 Parent/Variant CAR-T Cells
A
Affy6
Affy6_I3A
Negative
%Cytotoxicity
Time (hrs)
B
Affy6
Affy6_I3W
Affy6_I3F
Affy6_I3G
Negative
%Cytotoxicity
Time (hrs)

FIG. 12

Primary Human Liver Cells Co-Cultured With Affy7 Parent/Variant CAR-T Cells

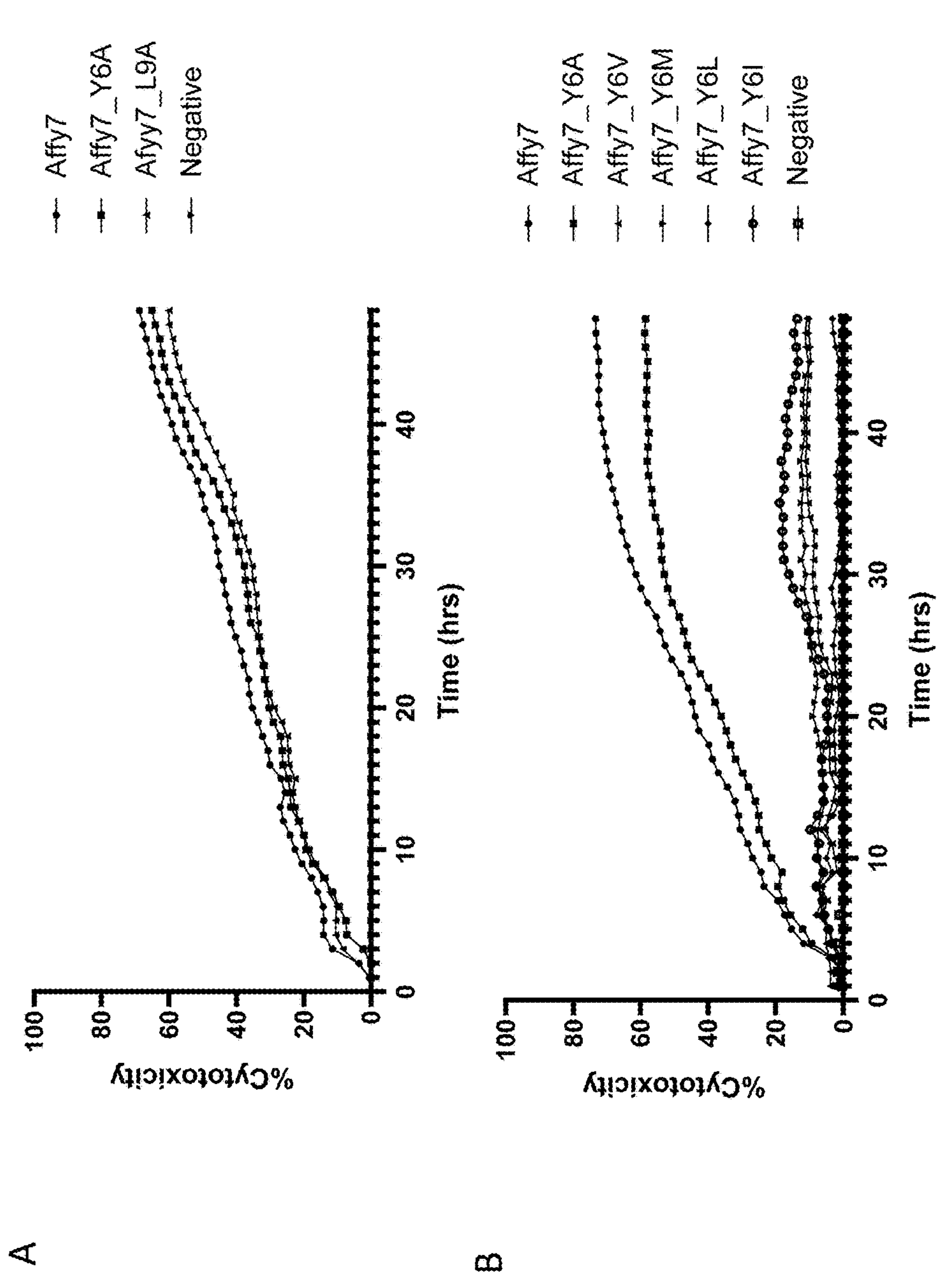


FIG. 13

Primary Human Liver Cells Co-Cultured With Affy6 Parent/Variant CAR-T Cells

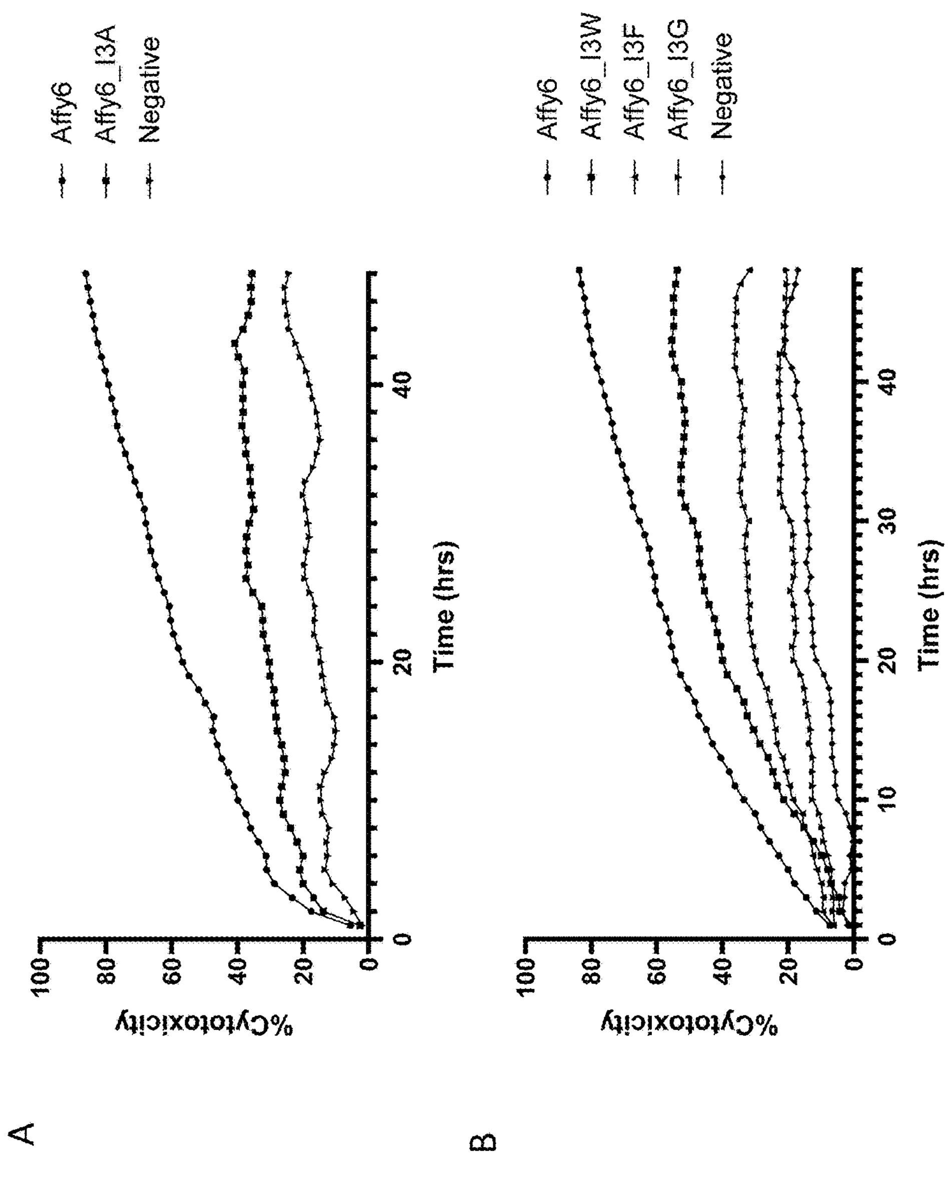


FIG. 14

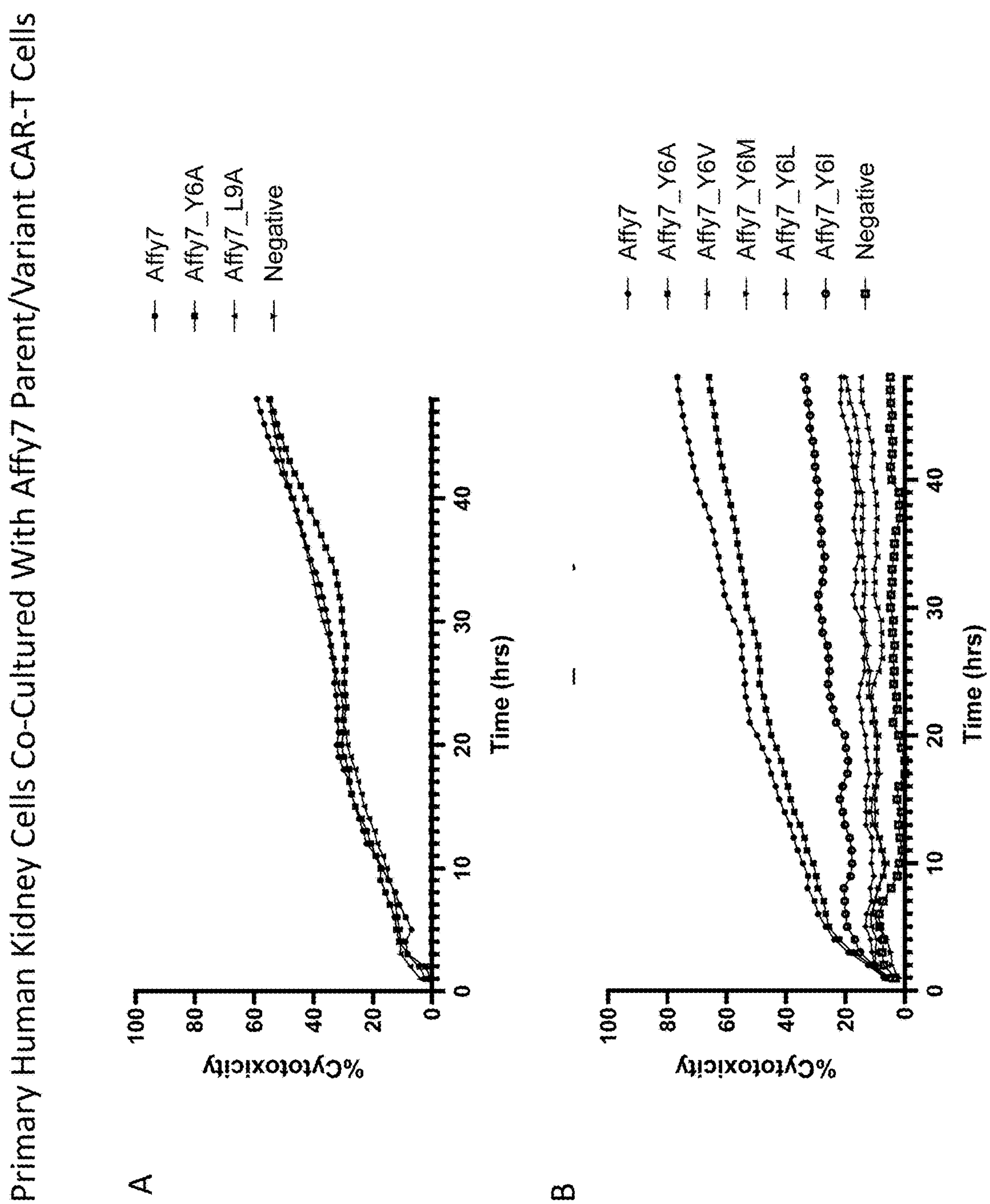

Primary Human Kidney Cells Co-Cultured With Affy7 Parent/Variant CAR-T Cells
A
%Cytotoxicity
Time (hrs)
Affy7
Affy7_Y6A
Affy7_L9A
Negative
B
%Cytotoxicity
Time (hrs)
Affy7
Affy7_Y6A
Affy7_Y6V
Affy7_Y6M
Affy7_Y6L
Affy7_Y6I
Negative

FIG. 15

Primary Human Kidney Cells Co-Cultured With Affy6 Parent/Variant CAR-T Cells

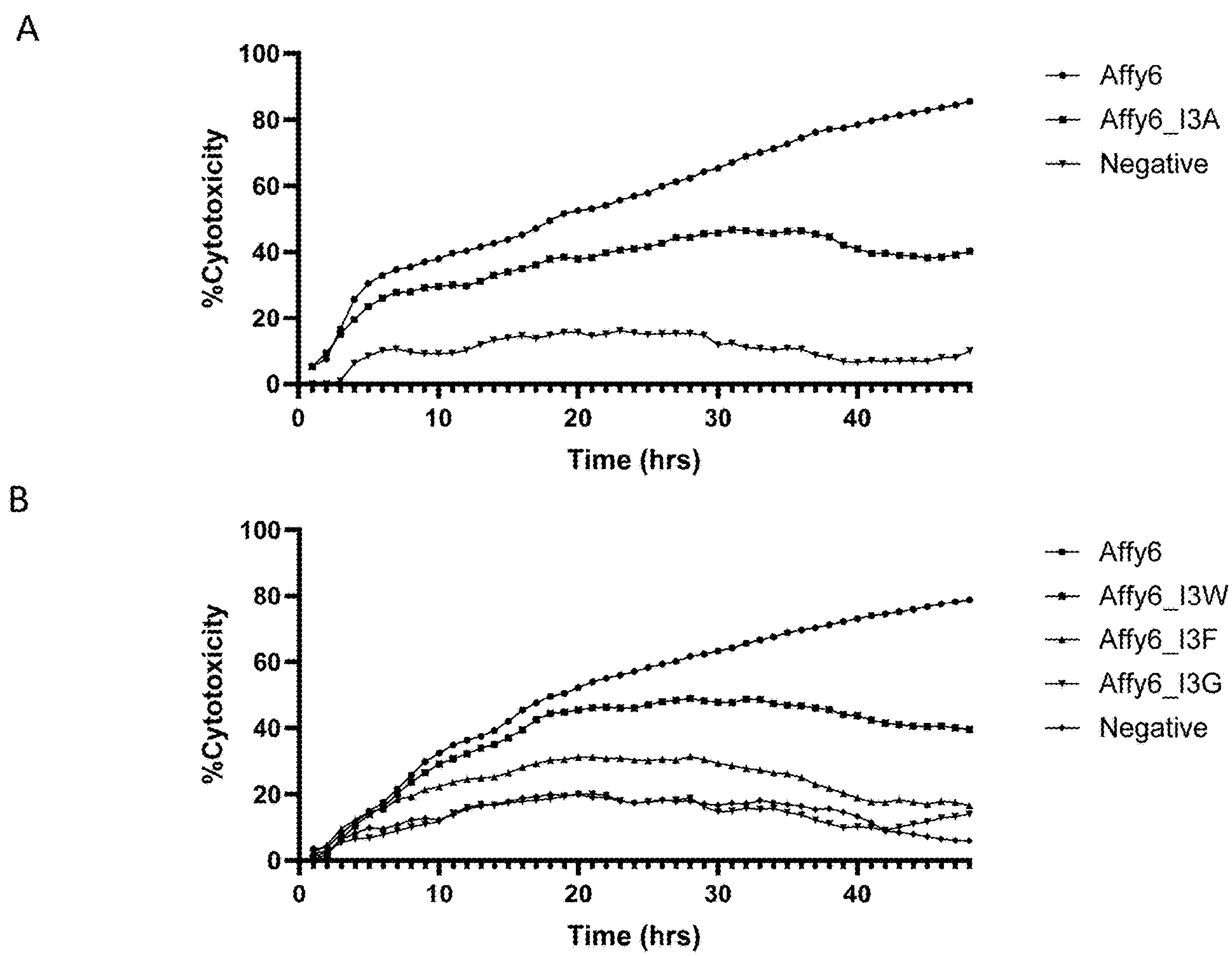


FIG. 16

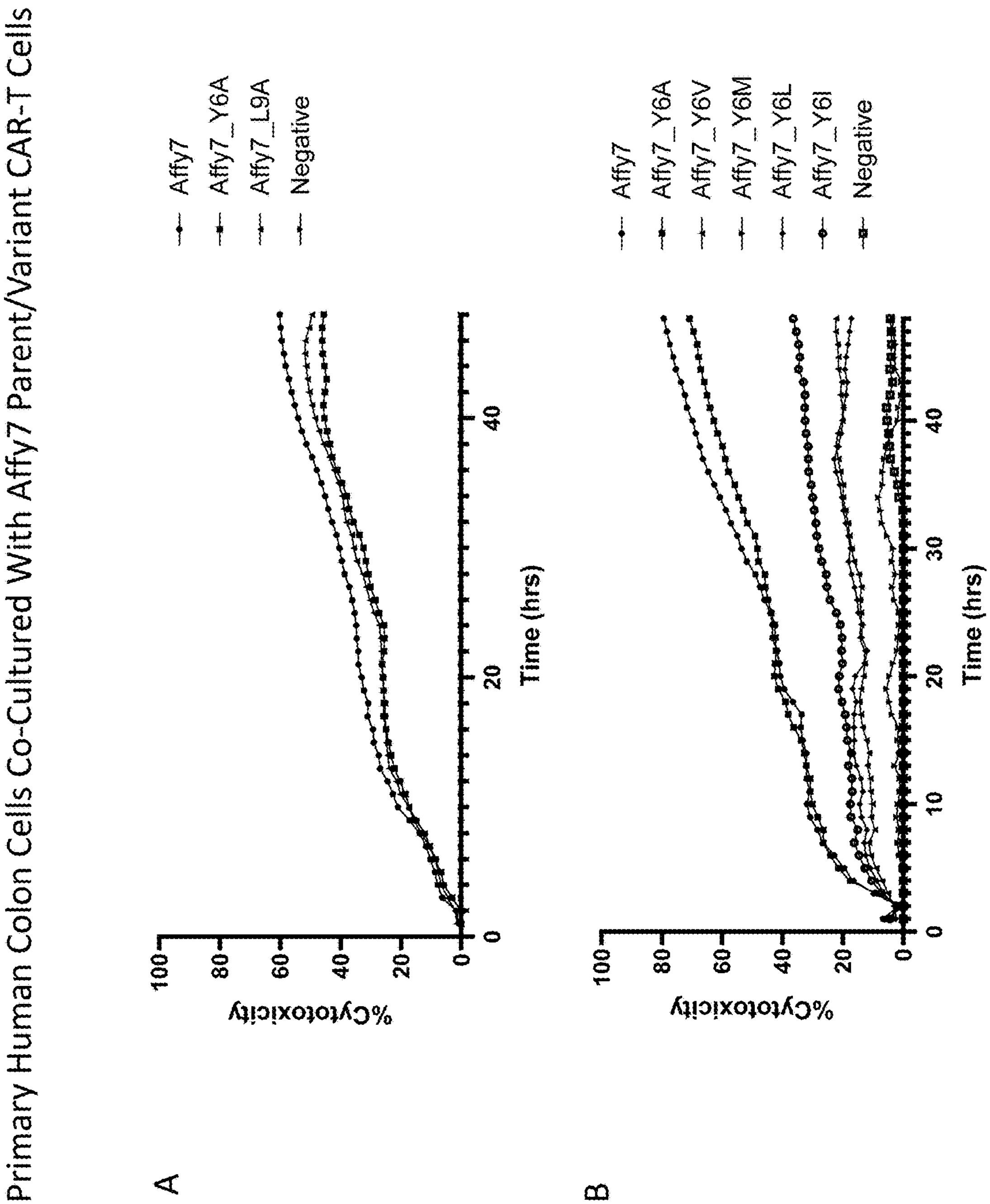

Primary Human Colon Cells Co-Cultured With Affy7 Parent/Variant CAR-T Cells
A
%Cytotoxicity
Time (hrs)
Affy7
Affy7_Y6A
Affy7_L9A
Negative
B
%Cytotoxicity
Time (hrs)
Affy7
Affy7_Y6A
Affy7_Y6V
Affy7_Y6M
Affy7_Y6L
Affy7_Y6I
Negative

FIG. 17

Primary Human Colon Cells Co-Cultured With Affy6 Parent/Variant CAR-T Cells

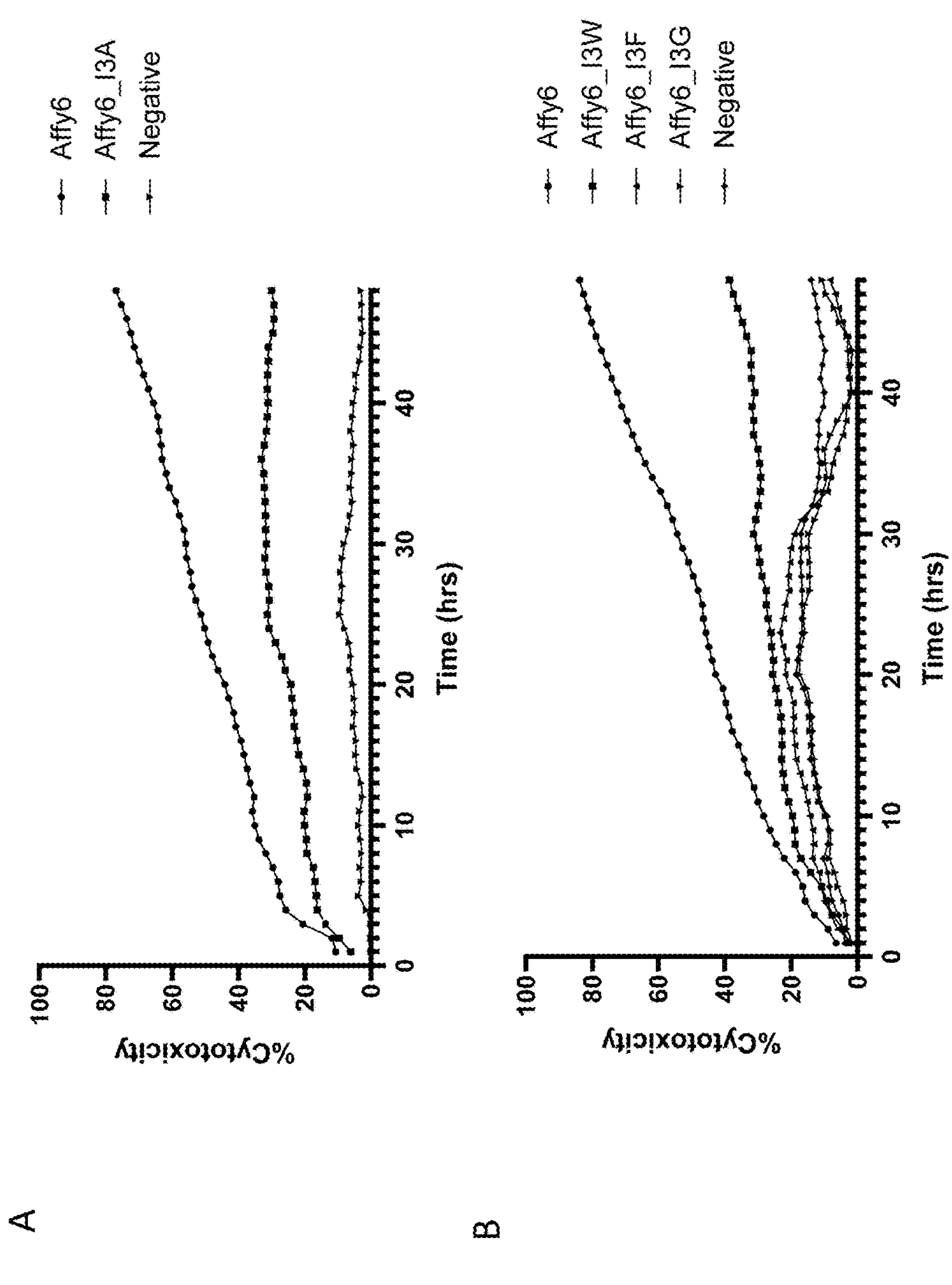


FIG. 18

CAR CONSTRUCTS AND METHODS OF TREATMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Patent Application No. 63/616,278, filed on Dec. 29, 2023, the entire content of which is incorporated herein by reference.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in XML file format and is hereby incorporated by reference in its entirety. Said XML copy, created on Jan. 20, 2025, is named 115568-0030_SL.xml and is 246,882 bytes in size.

FIELD

[0003] The present disclosure relates to novel c-MET CAR constructs, c-MET CAR-expressing cells and methods of their use in the treatment of cancer.

BACKGROUND

[0004] Adoptive cell transfer therapy is a type of immunotherapy involving ex vivo expansion of autologous or allogeneic immune cells and subsequent infusion into a patient. The immune cells may be modified ex vivo to specifically target malignant cells. Modifications include engineering of T cells to express chimeric antigen receptors (CARs). The advantage of CAR-T technology compared with chemotherapy or therapeutic antibodies is that reprogrammed engineered T cells can proliferate and persist in the patient and work like a living drug.

[0005] The ideal CAR target antigen would be a native, surface-exposed tumor neoantigen that is highly expressed and is undetectable in healthy tissues. However, many commonly targeted solid tumor antigens are also expressed by non-tumor tissues, albeit at lower levels. CAR molecules with high affinity for such antigens can lead to collateral targeting of healthy tissues resulting in on-target, off-tumor toxicity, a major limiting factor to the progress of CAR T cell therapy to date.

[0006] C-MET is a member of the receptor tyrosine kinase family of proteins, and the product of the proto-oncogene MET. See, e.g., https://www.ncbi.nlm.nih.gov/gene/4233. The encoded preproprotein is proteolytically processed to generate alpha and beta subunits that are linked via disulfide bonds to form the mature receptor. Binding of its ligand, hepatocyte growth factor (HGF), induces dimerization and activation of the receptor, which plays a role in cellular survival, embryogenesis, and cellular migration and invasion. Id. Amplification and overexpression of this gene are associated with multiple human cancers. As a proto-oncogene, abnormal activation of c-MET can promote the development and progression of multiple carcinomas such as liver, lung, colon, rectal, breast, pancreatic, ovarian, prostate, renal hepatocellular, gastric, and various head and neck carcinomas, in addition to cancers of the nervous system such as glioblastoma.

[0007] High affinity c-MET CAR-T cells recognize c-MET-expressing cells, including both normal epithelial tissues with low levels of c-MET and carcinomas with considerably higher c-MET expression levels. The recognition of antigen both on normal, non-target cells as well as on cancer cells can lead to severe toxicities and T cell exhaustion. CAR-T cell exhaustion has been recognized as a major cause of nonresponse and relapse associated with CAR-T cell therapy. T cells expressing chimeric antigen receptors (CARs) at high levels undergo tonic, antigen independent signaling due to receptor clustering, and function poorly as a result of T cell exhaustion, as evidenced by high levels of PD-1, TIM-3, LAG-3 expression, diminished antigen induced cytokine production, poor CAR-T cell persistence, and excessive programmed cell death. These problems are particularly acute in CAR-T treatments in patients with solid tumors, particularly carcinomas.

[0008] It is therefore of great interest to develop CARs and CAR-T expressing cells facilitating more selective killing of tumor cells, reduced T cell exhaustion and T cell activation, improved in vivo persistence, and reduced systemic toxicities.

SUMMARY OF THE INVENTION

[0009] Previously, it was observed that Met is expressed in normal human tissue (liver, gastric, thyroid and kidney) and neoplastic human tissues (Di Renzio et al. (1991) 6(11): 1997-2003). The inventors observed that CAR-T cells expressing anti-c-MET scFvs have the undesirable effect of killing healthy cells and tissue. Accordingly, it has now been unexpectedly found that affinity tuned anti-c-MET scFvs described herein avoid this problem and exhibit reduced killing of healthy cells (lung, liver, colon, and kidney epithelial) while maintaining efficacy of high c-met expressing cancer cell lines. The present disclosure is based, at least in part, on the development of affinity tuned chimeric antigen receptors (CARs) with variant heavy chain variable (V_H) region CDR3 domains providing reduced binding affinity to c-MET as compared to a corresponding binding domain from the parent antibody.

[0010] Accordingly, one aspect of the present disclosure features a chimeric antigen receptor (CAR) comprising: (a) an affinity-tuned extracellular antigen binding domain; (b) a co-stimulatory domain; and an (c) activation domain. The affinity-tuned extracellular antigen binding domain comprises an anti-c-MET antibody. In some embodiments, the anti-c-MET antibody is a single-chain antibody fragment (scFv).

[0011] In some embodiments, the extracellular binding domain comprises a heavy chain variable (V_H) region comprising complementary determining 1 (HC CDR1), complementary determining 2 (HC CDR2), and complementary determining 3 (HC CDR3) regions; and a light chain variable (V_L) region comprising complementary determining 1 (LC CDR1), complementary determining 2 (LC CDR2), and complementary determining 3 (LC CDR3) regions, wherein: the V_H region comprises HC CDR1 and HC CDR2 regions comprising amino acid sequences set forth SEQ ID NOs: 1 and 2, respectively, and an HC CDR3 region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 3-8; and the V_L region comprises LC CDR1, LC CDR2, and LC CDR3 regions comprising amino acid sequences set forth in SEQ ID NOs: 9-11, respectively. In certain embodiments, the HC CDR3 comprises an amino acid sequence set forth in SEQ ID NO: 5 or SEQ ID NO: 8.

[0012] In some embodiments, the V_H and V_L region amino acid sequences are selected from: (a) SEQ ID NOs: 13 and 19, respectively; (b) SEQ ID NOs: 14 and 19, respectively;

(c) SEQ ID NOs: 15 and 19, respectively; (d) SEQ ID NOs: 16 and 19, respectively; (e) SEQ ID NOs: 17 and 19, respectively; or (f) SEQ ID NOs: 18 and 19, respectively. In some embodiments, the extracellular binding domain comprises an anti-c-MET scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 29-34.

[0013] In some embodiments, the V_H region in the extracellular binding domain (e.g., anti-c-MET single-chain antibody) comprises HC CDR1 and HC CDR2 regions comprising the amino acid sequences set forth in SEQ ID NOs: 70 and 71, respectively, and an HC CDR3 region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 72-75; and the V_L region comprises LC CDR1, LC CDR2, and LC CDR3 regions comprising amino acid sequences set forth in SEQ ID NOs: 76-78, respectively.

[0014] In some embodiments, the extracellular binding domain comprises V_H and V_L region amino acid sequences selected from: (a) SEQ ID NOs: 80 and 84, respectively; (b) SEQ ID NOs: 81 and 84, respectively; (c) SEQ ID NOs: 82 and 84, respectively; or (d) SEQ ID NOs: 83 and 84, respectively. In some embodiments, extracellular binding domain comprises an anti-c-MET scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 92-95.

[0015] In some embodiments, the anti-c-MET scFv comprises a linker between the V_H and V_L regions. In some embodiments, the linker comprises the amino acid sequence (GxS)n in which x is an integer between 1-6, inclusive, and n is an integer between 1-10, inclusive. In some embodiments, the linker comprises the amino acid sequence (GGGS)n in which n is an integer between 1-10, inclusive (SEQ ID NO: 143). In one embodiment, the linker comprises the amino acid sequence multimer GGGGSGGGGSGGGGS (SEQ ID NO: 121).

[0016] In some embodiments, the costimulatory domain is from 4-1BB (CD 137), CD28, OX40, ICOS, GITR, CD27, CD30, CD40, DAP 10, DAP12, BAFFR, HVEM, ICAM-1, lymphocyte function-associated antigen-1 (LFA-1), CD2, CDS, CD7, CD287, LIGHT, NKG2C, NKG2D, SLAMF7, NKp80, NKp30, NKp44, NKp46, CD 160, B7-H3, a ligand that specifically binds with CD83, or a combination thereof. In one embodiment, the extracellular antigen binding domain (or scFv) comprises a costimulatory domain from 4-1BB (SEQ ID NO: 122), CD28 (SEQ ID NO: 123) or OX-40 (SEQ ID NO: 124). In another embodiment, the extracellular binding domain (or scFv) comprises costimulatory domains from 4-1BB (SEQ ID NO: 122) and CD28 (SEQ ID NO: 123). In some embodiments, the costimulatory domain comprises a mutant CD28 costimulatory domain as described in U.S. Patent Application No. 2020/0129554.

[0017] In some embodiments, the activation domain comprises one or more immunoreceptor tyrosine-based activation motifs (ITAMs). Examples of ITAM containing primary activation domain include, but are not limited to, those derived from CD3 zeta, common FcR gamma (FCER1G), Fc gamma RIIa, FcR beta (Fc epsilon lb), CD3 gamma, CD3 delta, CD3 epsilon, CD5, CD22, CD79a, CD79b, CD278 (ICOS), FcεRI, and CD66d. In one embodiment, the activation domain is from CD3 zeta (SEQ ID NO: 125).

[0018] In some embodiments, the CAR further comprises a hinge domain, a transmembrane domain, or a combination thereof, which is located between the extracellular binding domain and the costimulatory domain. In some embodiments, the CAR disclosed herein may further comprise a hinge domain, which may be linked to the C-terminus of the extracellular antigen binding domain and to the N-terminus of the transmembrane domain. In some embodiments, the CAR comprises a hinge domain from CD8 alpha (SEQ ID NO: 126), CD28 (SEQ ID NO: 127), or IgG4 (SEQ ID NO: 128).

[0019] In some embodiments, the CAR comprises a transmembrane domain from a cell surface receptor, which is C-terminal to the extracellular antigen binding domain and N-terminal to the costimulatory domain. Exemplary transmembrane domains, include but are not limited to those derived from an alpha, beta or zeta chain of a T cell receptor, CD28, CD3 epsilon, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137, CD154, CD271, CD278 (ICOS), TNFRSF18 (GITR), TNFRSF19, or killer cell immunoglobulin-like receptor (KIR). In certain embodiments, the transmembrane domain is from CD8 alpha (SEQ ID NO: 129), CD28 (SEQ ID NO: 130), ICOS (SEQ ID NO: 131), or GITR (SEQ ID NO: 132).

[0020] In exemplary embodiments, the c-MET CAR comprises an amino acid sequence set forth in any one of SEQ ID NOs: 43-48 or 102-105.

[0021] In some embodiments, the c-MET CAR is a bispecific CAR or a tandem CAR (tanCAR), which further comprises a second extracellular antigen binding domain. In some embodiments, the second extracellular antigen binding domain is a second scFv. In some embodiments, the second extracellular binding domain is a binding domain of a protein ligand. In some embodiments, the second extracellular antigen binding domain comprises an αL subunit I domain of human lymphocyte function-associated antigen-1 (LFA-1) targeting ICAM-1. In one embodiment, the binding domain comprises an amino acid sequence set forth in SEQ ID NO: 134 or a mutant thereof. In some embodiments, the second extracellular binding domain targets a tumor-associated antigen (TAA), particularly one that is overexpressed in solid tumors and metastatic tumors.

[0022] Any of the CARs disclosed herein may further comprise a signal peptide located at the N-terminus of the CAR precursor protein. This may include, for example, a CD8 alpha chain signal peptide (SEQ ID NO: 162).

[0023] In another aspect, the present disclosure provides nucleic acids encoding the c-MET scFvs and CARs described herein. In some embodiments, the nucleic acid encoding an anti-c-MET scFv comprises a nucleotide sequence set forth in any one of SEQ ID NOs: 36-41 or 92-95. In some embodiments, the nucleic acid encoding a c-MET CAR comprises a nucleotide sequence set forth in any one of SEQ ID NOs: 57-62, 64-69, 112-115, or 117-120. In some embodiments, the nucleic acid is a plasmid. In some embodiments, the nucleic acid is a vector, such as a lentivirus vector. In some embodiments, the nucleic acid is an RNA.

[0024] In some embodiments, the nucleic acid further comprises a human somatostatin receptor 2 (SSTR2) coding sequence. In some embodiments, the nucleic acid encodes a full length SSTR2 polypeptide (aa 1-381; SEQ ID NO: 137). In other embodiments, the nucleic acid encodes a truncated SSTR2 polypeptide (aa 1-314; SEQ ID NO: 136). In certain embodiments, the nucleic acid encodes an SSTR2-CAR fusion protein comprising an amino acid cleavage sequence between the SSTR2 and CAR coding sequences. In some embodiments, the SSTR2 coding sequence is 5' of the CAR coding sequence. In other embodiments, the CAR coding sequence is 5' of the SSTR2 coding sequence. In some embodiments, the amino acid cleavage sequence is encoded by a self-cleaving 2A peptide sequence from porcine teschovirus-1 (P2A), equine rhinitis A virus (E2A), *Thosea asigna* virus (T2A), or foot-and-mouth disease virus (F2A), or a combination thereof. In certain embodiments, the amino acid cleavage sequence comprises an amino acid sequence set forth in any one of SEQ ID NOs: 139-142.

[0025] In another aspect, the present disclosure provides a population of immune cells expressing one or more of the c-MET CARs disclosed herein. In some embodiments, the population of cells comprises one or more of the nucleic acids encoding an anti-c-MET CAR and/or an additional nucleic acid described herein. In some embodiments, the population of c-MET expressing cells further expresses a human somatostatin receptor 2 (SSTR2) marker protein. In some embodiments, the population of cells are immune cells. In some embodiments, the immune cells are T cells, natural killer (NK) cells, tumor infiltrating lymphocytes, dendritic cells, macrophages, B cells, neutrophils, eosinophils, basophils, mast cells, myeloid-derived suppressor cells, stem cells, precursors thereof, subtypes thereof, or a combination thereof; optionally wherein the immune cell is a human immune cell. In some embodiments, the immune cells are T cells. In some embodiments, the T cells additionally express an SSTR2 marker. In some embodiments, the population of cells further comprises a second population of immune cells. In some embodiments, the second population of immune cells expresses a second CAR or other polypeptide of interest.

[0026] In another aspect, the present disclosure provides a cell therapy-based method of treating cancer comprising administering to a subject in need thereof a population of immune cells described herein. In some embodiments, the subject is a human patient. In some embodiments, the cancer is a solid tumor. In some embodiments, the solid tumor is a carcinoma. In some embodiments, the carcinoma is a small cell lung carcinoma, non-small cell lung carcinoma, squamous cell lung carcinoma, large cell lung carcinoma, pancreatic carcinoma, pancreatic ductal carcinoma, prostate carcinoma, esophageal carcinoma, breast carcinoma, ovarian carcinoma, prostate carcinoma, colorectal carcinoma, bladder carcinoma, cervical carcinoma, hepatocellular carcinoma, renal hepatocellular carcinoma, gastric carcinoma, papillary carcinoma, adrenocortical carcinoma, pituitary carcinoma, a head and neck carcinoma, an adenocarcinoma thereof, or a squamous cell carcinoma thereof. In some embodiments, the cancer is a glioblastoma or mesothelioma. In some embodiments, the cancer is metastatic.

[0027] In some embodiments, the method further comprises administering the cell therapy to a human patient prior to, concurrently, or after a therapy against the cancer to reduce tumor burden. In some embodiments, the therapy is a chemotherapy, an immunotherapy, a radiotherapy, or a surgery. In some embodiments, prior to the cell therapy, the subject has been administered a lymphodepleting treatment to condition the subject for the cell therapy. In some embodiments, the lymphodepleting treatment comprises administering to the subject fludarabine and/or cyclophosphamide.

[0028] In some embodiments, the therapy comprises administration of an immune checkpoint inhibitor, such as pembrolizumab (Keytruda™), ipilimumab (Yervoy™) nivolumab (Opdivo™), or atezolizumab (Tecentriq™). In some embodiments, the cell therapy comprises administration of a therapeutic antibody selected from the group consisting of abagovomab, adecatumumab, afutuzumab, alemtuzumab, altumomab, amatuximab, anatumomab, arcitumomab, bavituximab, bectumomab, bevacizumab, bivatuzumab, blinatumomab, brentuximab, cantuzumab, catumaxomab, cetuximab, citatuzumab, cixutumumab, clivatuzumab, conatumumab, daratumumab, drozitumab, duligotumab, dusigitumab, detumomab, dacetuzumab, dalotuzumab, ecromeximab, elotuzumab, ensituximab, ertumaxomab, etaracizumab, farietuzumab, ficlatuzumab, figitumumab, flanvotumab, futuximab, ganitumab, gemtuzumab, girentuximab, glembatumumab, ibritumomab, igovomab, imgatuzumab, indatuximab, inotuzumab, intetumumab, ipilimumab, iratumumab, labetuzumab, lexatumumab, lintuzumab, lorvotuzumab, lucatumumab, mapatumumab, matuzumab, milatuzumab, minretumomab, mitumomab, moxetumomab, namatumab, naptumomab, necitumumab, nimotuzumab, nofetumomab, ocaratuzumab, ofatumumab, obinutuzumab, olaratumab, onartuzumab, oportuzumab, oregovomab, panitumumab, parsatuzumab, patritumab, pemtumomab, pertuzumab, pintumomab, pritumumab, racotumomab, radretumab, rilotumumab, rituximab, robatumumab, satumomab, sibrotuzumab, siltuximab, simtuzumab, solitomab, tacatuzumab, taplitumomab, tenatumomab, teprotumumab, tigatuzumab, tositumomab, trastuzumab, tucotuzumab, ublituximab, veltuzumab, vorsetuzumab, votumumab, zalutumumab, CC49 and 3F8.

[0029] In some embodiments, the method of cell therapy further comprises administration of a tyrosine kinase inhibitor capable of inhibiting TCR signaling and/or CAR signaling, such as dasatinib, ponatinib, saracatinib, bosutinib, nilotinib, or a combination thereof. In a particular embodiment, the inhibitor is dasatinib. In some embodiments, the tyrosine kinase inhibitor is administered for a period of time sufficient to restore at least partial T cell function and then discontinued. In some embodiments, the tyrosine kinase inhibitor is administered continuously. In other embodiments, the tyrosine kinase inhibitor is administered intermittently. In some embodiments, the tyrosine kinase inhibitor is administered intermittently to facilitate periods of T cell inactivation (e.g., during pharmaceutical composition administration) and periods of T cell activation (e.g., during absence of pharmaceutical composition administration). In some embodiments, the tyrosine kinase inhibitor is administered intermittently so that the concentration is maintained below a threshold level required to block CAR-T cell function.

[0030] In another aspect, the present disclosure provides a method for treating a cancer and monitoring CAR-T cell distribution in a patient. The method comprises the steps of: (a) incubating a population of CAR-T cells expressing SSTR2 with a radioactive label that binds to SSTR2; (b) intravenously infusing the labeled CAR-T cells into a patient in an amount of 10^6-10^8 cells/kg patient; and detecting the labeled CAR-T cell distribution by PET/CT imaging, wherein the labeled CAR-T cells are infiltrated into cancer cells to kill the cancer cells. In some embodiments, the label is radioactively labeled DOTATOC or radioactively labeled DOTATATE. In certain embodiments, DOTATOC or DOTATATE is radiolabeled with ^{68}Ga. In certain embodiments, the cancer is thyroid cancer, gastric cancer, pancreatic cancer, or breast cancer.

[0031] In another embodiment, a method for treating cancer and monitoring CAR-T cell distribution in a patient, comprises the steps of: (a) intravenously a population of

CAR-T cells expressing SSTR2 into a patient, where the CAR-T cells have been transduced to express at least 100,000 molecules of SSTR2 per T cell; (b) injecting into the patient a radioactive label that binds to SSTR2 at least one hour prior to PET/CT imaging, and (c) detecting the labeled CAR-T cell distribution by PET/CT imaging, wherein the labeled CAR-T cells are infiltrated into cancer cells to kill the cancer cells. In certain embodiments, the cancer is thyroid cancer, gastric cancer, pancreatic cancer, or breast cancer.

[0032] In another aspect, the present disclosure provides method of producing a population of genetically engineered immune cells. The method comprises introducing into a population of immune cells a nucleic acid coding for a CAR disclosed herein, and optionally one or more exogenous nucleic acids. In some embodiments, the immune cells include T cells, natural killer (NK) cells, tumor infiltrating lymphocytes, dendritic cells, macrographs, B cells, neutrophils, eosinophils, basophils, mast cells, myeloid-derived suppressor cells, stem cells, precursors thereof, or a combination thereof; optionally wherein the immune cell is a human immune cell. In some embodiments, the population of genetically engineered immune cells are CAR-T cells. In some embodiments, the method further comprises the step of expanding a population of CAR-T cells in the presence of a tyrosine kinase inhibitor capable of inhibiting TCR signaling and/or CAR signaling. In some embodiments, the tyrosine kinase is dasatinib. In other embodiments, the tyrosine kinase is ponatinib, saracatinib, bosutinib, nilotinib, or a combination thereof.

[0033] The details of one or more embodiments of the invention are set forth in the description below. Other features or advantages of the present invention will be apparent from the following drawings and detailed description of several embodiments, and also from the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] FIG. **1** is a schematic depiction of a CAR expression construct used in the Examples described herein.

[0035] FIG. **2** depicts binding curves reflecting the binding of c-MET to parent Affy7 CAR-T-cells and to two affinity-reduced Affy7 variant CAR-T cells, including the results of this analysis expressed in EC50 and fold change in binding relative to the parent Affy7 CAR-T cells.

[0036] FIG. **3** depicts binding curves reflecting the binding of c-MET to parent Affy7 CAR-T-cells and to four affinity-reduced Affy7 variant CAR-T cells, including the results of this analysis expressed in EC50 and fold change in binding relative to the parent Affy7 CAR-T cells.

[0037] FIG. **4** depicts binding curves reflecting the binding of c-MET to parent Affy6 CAR-T-cells and to two affinity-reduced Affy6 variant CAR-T cells, including the results of this analysis expressed in EC50 and fold change in binding relative to the parent Affy6 CAR-T cells.

[0038] FIG. **5** depicts binding curves reflecting the binding of c-MET to parent Affy6 CAR-T-cells and to four affinity-reduced Affy6 variant CAR-T cells, including the results of this analysis expressed in EC50 and fold change in binding relative to the parent Affy6 CAR-T cells.

[0039] FIG. **6** depicts in vitro cytotoxicity of Affy7 parent CAR-T cells and five affinity-reduced variant Affy7 CAR-T cell populations against an SNU-638 human stomach carcinoma cell line.

[0040] FIG. **7**A depicts in vitro cytotoxicity of Affy7 parent CAR-T cells and five affinity-reduced Affy7 variant CAR-T cell populations against an Hs746T human stomach carcinoma cell line. FIG. **7**B shows expression of c-met on the cell surface of HS746T cells.

[0041] FIG. **8** depicts in vitro cytotoxicity of Affy7 parent CAR-T cells and five affinity-reduced Affy7 variant CAR-T cell populations against an H1993 human non-small cell lung adenocarcinoma cell line.

[0042] FIG. **9** depicts in vitro cytotoxicity of Affy6 parent CAR-T cells and four affinity-reduced Affy6 variant CAR-T cell populations against an SNU-638 human stomach carcinoma cell line.

[0043] FIGS. **10**A-**10**D show expression levels of c-MET as determined by flow cytometry in normal, healthy human epithelial cells from lung (**10**A), liver (**10**B), colon (**10**C), and kidney (**10**D).

[0044] FIG. **11**, panels A and B depict in vitro cytotoxicity of Affy7 parent CAR-T cells and seven affinity-reduced Affy7 variant CAR-T cell populations against primary human hung cells.

[0045] FIG. **12**, panels A and B depict in vitro cytotoxicity of Affy6 parent CAR-T cells and five affinity-reduced Affy6 variant CAR-T cell populations against primary human hung cells.

[0046] FIG. **13**, panels A and B depict in vitro cytotoxicity of Affy7 parent CAR-T cells and seven affinity-reduced Affy7 variant CAR-T cell populations against primary human liver cells.

[0047] FIG. **14**, panels A and B depict in vitro cytotoxicity of Affy6 parent CAR-T cells and five affinity-reduced Affy6 variant CAR-T cell populations against primary human liver cells.

[0048] FIG. **15**, panels A and B depict in vitro cytotoxicity of Affy7 parent CAR-T cells and seven affinity-reduced Affy7 variant CAR-T cell populations against primary human kidney cells.

[0049] FIG. **16**, panels A and B depict in vitro cytotoxicity of Affy6 parent CAR-T cells and five affinity-reduced Affy6 variant CAR-T cell populations against primary human kidney cells.

[0050] FIG. **17**, panels A and B depict in vitro cytotoxicity of Affy7 parent CAR-T cells and seven affinity-reduced Affy7 variant CAR-T cell populations against primary human colon cells.

[0051] FIG. **18**, panels A and B depict in vitro cytotoxicity of Affy6 parent CAR-T cells and five affinity-reduced Affy6 variant CAR-T cell populations against primary human colon cells.

DETAILED DESCRIPTION OF THE INVENTION

[0052] The present disclosure is based on the unexpected discovery that chimeric antigen receptor (CAR)-T cells expressing an affinity tuned anti-c-MET scFv with reduced affinity for c-MET can effectively kill c-MET overexpressing cancer cells, while avoiding the undesirable effect of killing healthy cells and tissue.

I. Chimeric Antigen Receptor with Reduced Affinity Anti-c-MET Binding Domain

[0053] One aspect of the present disclosure provides a chimeric antigen receptor (CAR) comprising an affinity-tuned anti-c-MET scFv with reduced target affinities to avoid targeting of healthy tissue with basal c-MET expression while maintaining sufficient avidity for targeting tumor tissues with high c-MET expression. The c-MET CARs disclosed herein comprise an artificial (non-naturally occurring) receptor having a binding specificity for the c-MET proto-oncogene, which is capable of triggering immune responses in immune cells upon binding to c-MET, particularly cells overexpressing c-MET.

[0054] The CARs disclosed herein comprise the c-MET binding domain, one or more costimulatory domains and an activation domain comprising a plurality of immunoreceptor tyrosine-based activation motifs (ITAMs), such as a CD3ζ signaling domain (also referred to as CD3 zeta). The CAR may also have a hinge domain, a transmembrane domain, or a combination thereof. In some embodiments, the transmembrane domain is located between the extracellular antigen binding domain and the costimulatory domain and the hinge domain may be located between the transmembrane domain and the costimulatory domain.

[0055] In one embodiment, provided herein is a CAR comprising an anti-c-MET scFv, one or more costimulatory domains from 4-1BB, CD28, and/or OX-40, and an ITAM-containing activation domain, such as a CD3ζ signaling domain.

Extracellular Antigen Binding Domain

[0056] The CAR constructs disclosed herein comprise an affinity tuned extracellular antigen binding domain with reduced affinity for the c-MET proto-oncogene. The affinity tuned antigen binding domain is derived from an antibody and comprises a heavy chain variable (V_H) region and a light chain variable (V_L) region. The V_H and V_L regions herein are further subdivided into regions of hypervariability, also known as "complementarity determining regions" ("CDR"), interspersed with regions that are more conserved, which are known as "framework regions" ("FR"). Each V_H and V_L is typically composed of three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. The extent of the framework region and CDRs can be precisely identified using methodology known in the art, for example, by the Kabat definition, the Chothia definition, the AbM definition, and/or the contact definition, all of which are well known in the art. See, e.g., Kabat, E. A., et al. (1991) *Sequences of Proteins of Immunological Interest, Fifth Edition*, U.S. Department of Health and Human Services, NIH Publication No. 91-3242, Chothia et al., (1989) Nature 342:877; Chothia, C. et al. (1987) J. Mol. Biol. 196:901-917, Al-lazikani et al (1997) J. Molec. Biol. 273:927-948; and Almagro, J. Mol. Recognit. 17:132-143 (2004). See also the Human Genome Mapping Project Resources at the Medical Research Council in the United Kingdom and the antibody rules described at the Bioinformatics and Computational Biology group website at University College London. Amino acid and nucleotide sequences corresponding to the VH, VL and CDR sequences are shown in Table 18. The CDR regions in Table 18 were identified through NCBI Blast.

[0057] In some embodiments, the V_H and V_L regions are "humanized." Humanized variable regions (or antibodies) are derived from chimeric immunoglobulins, human immunoglobulins, immunoglobulin chains, or antigen-binding fragments thereof in which minimal sequences are derived from a non-human immunoglobulin. In some embodiments, the humanized variable regions (or antibodies) comprise residues from CDRs of a non-human species (donor antibody) such as mouse, rat, or rabbit having the desired specificity, affinity, and capacity, where the framework region (FR) residues from the non-human species are replaced by corresponding human residues. Furthermore, the humanized binding region may comprise residues that are found neither in the recipient antibody nor in the imported CDR or framework sequences, but are included to further refine and optimize antibody performance. In general, the humanized antibody will comprise substantially all of at least one, and typically two, variable domains, in which all or substantially all of the CDR regions correspond to those of a non-human immunoglobulin and all or substantially all of the FR regions are those of human immunoglobulin consensus sequences.

[0058] In some embodiments, the EC50 of the c-MET binding domain in the c-MET CAR to c-MET is greater than 5 μM, greater than 15 μM, greater than 30 μM, greater than 60 μM, greater than 150 μM, greater than 300 μM, greater than 600 μM but less than 800 μM, or any range thereof. In other embodiments, the fold change in EC50 relative to a corresponding wild-type c-MET binding domain in the c-MET CAR (i.e. from which the reduced affinity binding domain is derived) is greater than 50-fold, greater than 150-fold, greater than 500-fold, greater than 1000-fold, greater than 2000-fold, greater than 6000-fold but less 8000-fold, or any range thereof.

[0059] In some embodiments, the extracellular binding domain (e.g., anti-c-MET scFv) comprises a V_H region having HC CDR1 and HC CDR2 regions comprising amino acid sequences set forth SEQ ID NOs: 1 and 2, respectively, and an HC CDR3 region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 3-8; and a V_L region having LC CDR1, LC CDR2, and LC CDR3 regions comprising amino acid sequences set forth in SEQ ID NOs: 9-11, respectively. In certain embodiments, the HC CDR3 comprises an amino acid sequence set forth in SEQ ID NO: 5 or SEQ ID NO: 8. In some embodiments, the V_H and V_L region amino acid sequences are selected from: (a) SEQ ID NOs: 13 and 19, respectively; (b) SEQ ID NOs: 14 and 19, respectively; (c) SEQ ID NOs: 15 and 19, respectively; (d) SEQ ID NOs: 16 and 19, respectively; (e) SEQ ID NOs: 17 and 19, respectively; or (f) SEQ ID NOs: 18 and 19, respectively. In some embodiments, the extracellular binding domain comprises an anti-c-MET scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 29-34. In some embodiments, the extracellular binding domain comprises an anti-c-MET scFv encoded by a nucleotide sequence set forth in any one of SEQ ID NOs: 36-41.

[0060] In some embodiments, the extracellular binding domain (e.g., anti-c-MET single-chain antibody) comprises a V_H region having HC CDR1 and HC CDR2 regions comprising amino acid sequences set forth in SEQ ID NOs: 70 and 71, respectively, and an HC CDR3 region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 72-75; and a V_L region having LC CDR1, LC CDR2, and LC CDR3 regions comprising amino acid sequences set forth in SEQ ID NOs: 76-78, respectively. In some embodiments, the V_H and V_L region amino acid sequences are selected from: (a) SEQ ID NOs: 80 and 84, respectively; (b) SEQ ID NOs: 81 and 84, respectively; (c) SEQ ID NOs: 82 and 84, respectively; or (d) SEQ ID NOs: 83 and 84, respectively. In some embodiments, extracellular binding domain comprises an anti-c-MET scFv comprising an amino acid sequence set forth in any one of SEQ ID NOs: 92-95.

[0061] In some instances, the extracellular antigen binding domain is a single-chain antibody fragment (scFv) comprising V_H and V_L regions connected by a peptide linker. In some embodiments, the scFv has a $V_H \rightarrow V_L$ orientation. In other embodiments, the scFv has a $V_L \rightarrow V_H$ orientation. In some embodiments, the anti-c-MET scFv comprises a linker between the V_H and V_L regions. Peptide linkers used in scFv constructs are well known in the art and include, for example, the amino acid sequence (GxS)n in which x is an integer between 1-6, inclusive, and n is an integer between 1-10, inclusive. In some embodiments, the linker comprises the amino acid sequence (GGGS)n in which n is an integer between 1-10, inclusive (SEQ ID NO: 143). In one embodiment, the linker comprises the amino acid sequence multimer GGGGSGGGGSGGGGS (SEQ ID NO: 121).

[0062] In some embodiments, the affinity tuned extracellular binding domain against c-MET comprises a single domain antibody (sdAb), such as a VHH fragment, a single chain Fab fragment, a single chain Fab' fragment, or a c-MET binding peptide.

Other CAR Components

[0063] In addition to the affinity tuned extracellular antigen binding domain disclosed above, the CAR constructs disclosed herein further comprise one or more costimulatory domains and an activation domain comprising one or more ITAMs (as described below), such as the CD3ζ signaling domain (which contains 3), or a combination thereof.

[0064] The c-MET CARs disclosed herein comprise one or more costimulatory domains. Costimulatory domains typically enhance and/or alter the nature of the response to activation of the activation domain. Co-stimulatory domains suitable for use in the CARs of the present disclosure are typically receptor-derived polypeptides. In some embodiments, the co-stimulatory domains homodimerize. In some embodiments, the co-stimulatory domain may be the intracellular portion of a transmembrane protein (i.e., the co-stimulatory domain may be derived from the transmembrane protein). In exemplary embodiments, the costimulatory domains are from 4-1BB (CD 137), CD28, OX40, ICOS, GITR, DAP 10, DAP12, CD27, CD30, CD40, BAFFR, HVEM, ICAM-1, LFA-1 (CD11a/CD18), CD2, CDS, CD7, CD287, LIGHT, NKG2C, NKG2D, SLAMF7, NKp80, NKp30, NKp44, NKp46, CD 160, B7-H3, and a ligand that specifically binds with CD83.

[0065] The c-MET CARs disclosed herein further comprises an activation domain. As used herein, the term "activation domain" refers to an intracellular signaling domain that can trigger the production of one or more cytokines upon activation; increases antibody-dependent cellular cytotoxicity (ADCC) and cell death; and/or increased activation and/or proliferation of $CD8^+$ T cells, $CD4^+$ T cells, natural killer T cells, γδ T cells, and/or neutrophil proliferation. In some embodiments, the activation domain comprises at least one (e.g., 1, 2, 3, 4, 5, 6, etc.) immunoreceptor tyrosine-based activation motif (ITAM or ITAMa) with the sequence $Yxx[L/I]x_{(6-8)}Yxx[L/I]$) present in the cytoplasmic tail of an immune signaling receptor or associated subunit to induce cell signaling. The ITAM domains for use in the CARs disclosed herein can include signaling domains from several types of immune signaling receptors, including CD3, B7 family costimulatory intracellular signaling proteins such as molecules and tumor necrosis factor receptor (TNFR) superfamily receptors; signaling domains used by NK and NKT cells, such as NKp30 (B7-H6), DAP12, NKG2D, NKp44, NKp46, DAP10, and CD3 zeta; and signaling domains from ITAM-containing human immunoglobulin receptors, such as FcαRI, FcγRI, FcγRIIA, FcγRIIIA, FcγRIIC, and FcRL5. Thus, in certain embodiments, the activation domain is from CD3 zeta (SEQ ID NO: 125), Fc epsilon receptor gamma (FCER1G), Fc gamma RIIa, FcR beta (Fc epsilon lb), CD3 gamma, CD3 delta, CD3epsilon, CD5, CD22, CD79a, CD79b, CD278 (also known as "ICOS"), FcεRI, or CD66d.

[0066] The c-MET CARs disclosed herein further comprises a transmembrane domain, a hinge (or spacer) domain, or both. The transmembrane domain facilitates insertion of the CAR into the cell membranes and can be inserted between the binding domain and a costimulatory domain. In some embodiments, the transmembrane domain can be inserted between the hinge region and the co-stimulatory domain. Any transmembrane domains and/or hinge domains commonly used in CAR constructs can be used here.

[0067] In some embodiments, the c-MET CAR further comprises a hinge between the extracellular binding domain (e.g., scFv) and the transmembrane domain. In this orientation, the hinge domain provides additional distance between the antigen binding domain and the cell membrane surface on which the CAR is expressed. In some examples, the hinge domain may be from CD28, CD8, or an IgG, such as IgG1 or IgG4. In some embodiments, the hinge domain comprises an amino sequence set forth in any one of SEQ ID NOs: 126-128. In some embodiments, the transmembrane domain is obtained from a suitable cell-surface receptor, such as the transmembrane domain of a cell surface receptor of the alpha, beta or zeta chain of the T cell receptor, CD28, CD3 epsilon, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137, CD154, CD271, TNFRSF19, and killer cell immunoglobulin-like receptor (KIR). In certain embodiments, the transmembrane membrane domain is from CD8 alpha, CD28, ICOS, or GITR, and optionally comprises an amino acid sequence set forth in SEQ ID NOs: 129-132, respectively.

[0068] In exemplary embodiments, the c-MET CAR comprises an amino acid sequence set forth in any one of SEQ ID NOs: 43-48, 50-54, 102-105, or 107-110.

[0069] In another aspect, the present disclosure provides a bispecific c-MET-CAR (or tandem c-MET-CAR (tanCAR)) comprising an additional extracellular antigen binding domain to provide improved specificity. In some embodiments, the present disclosure further contemplates a multi-specific c-MET-CAR targeting three of more antigens.

[0070] In one embodiment, the second extracellular binding domain in the bispecific CAR or tanCAR comprises an antibody fragment, such as an scFv. In some embodiments, within each antibody or antibody fragment (e.g., scFv) of a bispecific antibody molecule, the VH can be upstream or downstream of the VL. In some embodiments, the upstream antibody or antibody fragment (e.g., scFv) is arranged with its VH (VH_1) upstream of its VL (VL_1) and the downstream antibody or antibody fragment (e.g., scFv) is arranged with its VL (VL_2) upstream of its VH (VH_2), such that the overall bispecific antibody molecule has the arrangement VH_1-VL_1-VL_2-VH_2. In other embodiments, the upstream antibody or antibody fragment (e.g., scFv) is arranged with its VL (VL_1) upstream of its VH (VH_1) and the downstream antibody or antibody fragment (e.g., scFv) is arranged with its VH (VH_2) upstream of its VL (VL_2), such that the overall bispecific antibody molecule has the arrangement VL_1 VH_1-VH_2-VL_2. In some embodiments, a linker is disposed between the two antibodies or antibody fragments (e.g., scFvs), for example, between VL_1 and VL_2 if the construct is arranged as VH_1-VL_1-VL_2-VH_2, or between VH_1 and VH_2 if the construct is arranged as VL_1-VH_1-VH_2-VL_2. The linker may be a linker as described herein, e.g., a (Gly_3-Ser)n or (Gly_4-Ser)n linker, wherein n is 1, 2, 3, 4, 5, or 6. In general, the linker between the two scFvs should be long enough to avoid mispairing between the domains of the two scFvs. In some embodiments, a linker is disposed between the VL and VH of the first scFv. In some embodiments, a linker is disposed between the VL and VH of the second scFv. In constructs that have multiple linkers, any two or more of the linkers may be the same or different. Accordingly, in some embodiments, a bispecific CAR or tanCAR comprises VLs, VHs, and may further comprise one or more linkers in an arrangement as described herein.

[0071] In some embodiments, immune effector cells can be genetically modified one or more CARs recognizing target cells with combinatorial Boolean logic: one can engineer T cells with multi-receptor circuits that function as AND gates (requiring two antigens to be present), OR gates (requiring the presence of one of two possible antigens), and NOT gates (high expression of one antigen, low expression of another) to increase tumor selectivity by limiting cross-reactivity with healthy tissues that also express the CAR/TCR target antigen. Exemplary target antigen combinations for AND and AND-NOT logic gates are disclosed in Table 1 of WO 2022/036133 where an "AND" precedes or follows a target antigen present on the surface of a target cancer cell and a "NOT" precedes an antigen that that is not present on the surface of a target cancer cell, but may be present on the surface of a non-cancerous cell.

[0072] Where a target antigen pair (or triple) provides for an AND logic gate, two (or three) antigens must be present on the surface of a target cancer cell in order for a genetically modified cytotoxic immune cell of the present disclosure to kill the target cancer cell, where in this case the genetically modified cytotoxic immune cell is genetically modified to express two or three antigen-triggered polypeptides, each recognizing one of the target antigens of the target antigen pair/triplet. For example, where a target antigen pair provides an AND gate logic, each of the target antigens of the target antigen pair must be present on the surface of a target cancer cell in order for a genetically modified cytotoxic immune cell of the present disclosure to kill the target cancer cell.

[0073] Where a target antigen pair/triple provides an AND-NOT logic gate (or, correspondingly, a NOT-AND logic gate), a genetically modified cytotoxic immune cell of the present disclosure: a) is activated to kill a target cancer cell that expresses the AND target cell surface antigen (e.g., the first target cell surface antigen), but not the NOT target cell surface antigen (e.g., the second and/or third target cell surface antigen), on its cell surface; and b) is inhibited from killing a non-cancerous cell if the non-cancerous cell expresses both the AND target cell surface antigen and the NOT target cell surface antigen(s) on its cell surface. In these cases, the genetically modified cytotoxic immune cell must express at least a first antigen-triggered polypeptide that specifically binds the AND target antigen of the target antigen pair and a second triggered polypeptide that specifically binds the NOT antigen of the target antigen pair. For example, in some cases, binding of an antigen-triggered polypeptide to the NOT target cell surface antigen (expressed on a non-cancerous cell) inhibits T cell activation. In this manner, unintended/undesired killing of a non-cancerous cell is reduced, because the target cancer cell expressing the AND target antigen and not the NOT target antigen will be preferentially killed over the non-cancerous cell expressing both the AND target antigen and the NOT target antigen. Since the cancer cell does not express the NOT target cell surface antigen (expressed on a non-cancerous cell), binding of the first antigen-triggered polypeptide to the AND target antigen (present on the cancer cell surface) results in activation of the genetically modified cytotoxic T cell and killing of the cancer cell.

[0074] In some embodiments, the second extracellular binding is an scFv. In other embodiments, the second extracellular binding domain is a binding domain of a protein ligand. In some embodiments, the second extracellular binding domain targets a tumor-associated antigen (TAA), particularly one that is overexpressed in solid tumors and metastatic tumors of renal, lung, thyroid, and gastrointestinal tissues. Exemplary solid tumor antigen targets for the second binding domain may include but are not limited to target antigens selected from ICAM-1, EpCAM, CEACAM5, EGFRvIII, mesothelin, CS-1, GD2, Tn Ag, PSMA, TAG72, CD44v6, CEA, KIT, IL-13Ra2, GD3, CD171, IL-lRa, PSCA, VEGFR2, Lewis Y, CD24, PDGFR-beta, SSEA-4, folate receptor alpha, ERBB2, Her2/neu, MUC1, EGFR, NCAM, Ephrin B2, CAIX, LMP2, sLe, HMWMAA, o-acetyl-GD2, folate receptor beta, TEM1/CD248, TEM7R, FAP, Legumam, HPV E6 or E7, CLDN6, TSHR, GPRC5D, ALK, Polysialic acid, PLACl, globoH, NY-BR-1, UPK2, HAVCR1, ADRB3, PANX3, GPR20, Ly6k, OR51E2, TARP, and GFRa4.

[0075] In certain embodiments, the second extracellular antigen binding domain comprises an ICAM-1 binding domain. In one embodiment, the ICAM-1 binding domain comprises the I domain of the αL subunit (e.g., SEQ ID NO: 134) of human lymphocyte function-associated antigen-1 (LFA-1), which is known to target ICAM-1. A wild-type (WT) I domain encompasses amino acid residues 130-310 (SEQ ID NO: 134) of the 1145 amino acid long mature αL integrin subunit protein (SEQ ID NO: 133, which corresponds to amino acid residues 26-1170 of GenBank Accession No. NP_002200).

[0076] In some embodiments, the ICAM-1 binding domain comprises an I domain mutant disclosed in U.S. Pat. No. 10,428,136, which is incorporated herein by reference in its entirety. Such mutants differ in their affinity for ICAM-1. For example, I domain mutants having one mutation at F292A (Kd 20 μM), F292S (Kd 1.24 μM), L289G (Kd 196 nM), F265S (Kd 145 nM), and F292G (Kd 119 nM), or having two mutations at K287C/K294C (Kd 100 nM) in the wild-type I domain are suitable for the present invention. The above numbering of the amino acid residues is in reference to the amino acid sequence of the mature αL integrin of SEQ ID 133. Thus, in some embodiments, the I domain in the bispecific or multi-specific c-MET-CAR set forth in SEQ ID NO: 134 includes one mutation of F292A, F292S, L289G, F265S, or F292G, or with two mutations at K287C/K294C.

[0077] In a specific example, the CAR construct comprises, from the N-terminus to C-terminus, a c-Myc tag, an anti-c-MET scFv described herein, a CD8 hinge domain, a CD28 transmembrane domain, a CD28 co-stimulatory domain, and a CD3ζ signaling domain. In some embodiments, the CAR construct further comprises a ribosome skipping element (e.g., P2A) following the CD3ζ signaling domain and an SSTR2 coding region operably linked (i.e. fused in frame) to the ribosome skipping element. Exemplary c-MET-CARs suitable for use according to the present disclosure are set forth in SEQ ID NOs: 43-48 (encoded by SEQ ID NOs: 57-62, respectively), SEQ ID NOs: 50-55 (encoded by SEQ ID NOs: 64-69, respectively), SEQ ID NOs: 102-105 (encoded by SEQ ID NOs: 112-115, respectively), and SEQ ID NOs: 107-110 (encoded by SEQ ID NOs: 117-120, respectively). Additionally, the CAR typically comprises a signal peptide at the N-terminus of the CAR precursor protein, which is operably linked to the c-MET binding domain.

[0078] In some embodiments, the other CAR components in section (B) may be variants of their naturally occurring counterparts. In some embodiments, any of the other CAR components may contain an amino acid sequence at least 80% (e.g., at least 85%, 90%, 95%, 98%, 99% or above) identical to its natural counterpart. As used herein, the "percent identity" of two amino acid sequences is determined using the algorithm of Karlin and Altschul Proc. Natl. Acad. Sci. USA 87:2264-68, 1990, modified as in Karlin and Altschul Proc. Natl. Acad. Sci. USA 90:5873-77, 1993. Such an algorithm is incorporated into the NBLAST and XBLAST programs (version 2.0) of Altschul, et al. J. Mol. Biol. 215:403-10, 1990. BLAST protein searches can be performed with the XBLAST program, score=50, wordlength=3 to obtain amino acid sequences homologous to the protein molecules of interest. Where gaps exist between two sequences, Gapped BLAST can be utilized as described in Altschul et al., Nucleic Acids Res. 25(17):3389-3402, 1997. When utilizing BLAST and Gapped BLAST programs, the default parameters of the respective programs (e.g., XBLAST and NBLAST) can be used.

[0079] In some examples, the costimulatory or activation domain contain up to 15 (e.g., up to 12, 10, 8, 6, 5, 4, 3, 2, or 1) amino acid residue substitutions relative to the wild-type counterpart. In some examples, the amino acid residue substitutions are conservative amino acid residue substitutions. As used herein, a "conservative amino acid substitution" refers to an amino acid substitution that does not alter the relative charge or size characteristics of the protein in which the amino acid substitution is made. Variants can be prepared according to routine methods for altering polypeptide sequences known to those skilled in the art Conservative substitutions of amino acids may include substitutions within the following groups: ((a) A→G, S; (b) R→K, H; (c) N→Q, H; (d) D→E, N; I C→S, A; (f) Q→N; (g) E4→D, Q; (h) G4→A; (i) H→N, Q; (j) I→L, V; (k) L→I, V; (l) K→R, H; (m) M→L, I, Y; (n) F→Y, M, L; (o) P→A; (p) S→T; (q) T→S; I W→Y, F; (s) Y→W, F; and (t) V→I, L.

II. Genetically Modified Immune Cells

[0080] In another aspect, the present disclosure provides a population of immune cells comprising genetically modified immune cells (e.g., T cells) expressing one or more of the chimeric antigen receptor (CAR) constructs disclosed herein. Such modified immune cells express a CAR which specifically binds c-MET, thereby eliminating the target disease cells via, e.g., the effector activity of the immune cells. In some embodiments, the population of immune cells comprises genetically modified cytotoxic effector cells (CAR-T cells) independently expressing a single CAR, a bispecific or multispecific CAR, or multiple CARs (e.g., dual CAR-T).

[0081] In some embodiments, the immune cells for gene transduction herein may be T cells, natural killer (NK) cells, tumor infiltrating lymphocytes, dendritic cells, monocytes, macrophages, B cells, neutrophils, eosinophils, basophils, mast cells, myeloid-derived suppressor cells, mesenchymal stem cells, precursors thereof, subtypes thereof, or combinations thereof.

[0082] T cells may be selected from the group consisting of cytotoxic T-lymphocytes (CD8+), including Tc1, Tc2, Tc9, Tc17, and Tc22 T cells; helper T-lymphocytes (CD4+), including Th1, Th2, Th17, Th9, and Tfh T cells; antigen-inexperienced naïve T cells (T_N), stem cell memory T cells (Tscm or T_{SCM}), central memory T cells (Tcm or T_{CM}), effector memory T cells (Tem or T_{EM}), effector T cells (Teff, T_{EFF} or T_E), precursors to an exhausted T cell (Tpex or T_{PEX}), or exhausted T cells (Tex or T_{EX}), central memory T cells, effector memory T cells, tissue resident memory T cells, virtual memory T cells, natural killer T cells (NKT cells), FOXP3+ T cells, FOXP3− T cells). T cells may be purified from peripheral blood lymphocytes by methods known to those skilled in the art. In some embodiments, T cells may be cultured, expanded, differentiated or de-differentiated to obtain particular T cell subsets prior to or following transduction, such as antigen-inexperienced naïve T cells (T_N), stem cell memory T cells (Tscm or T_{SCM}), and/or central memory T cells (Tcm or T_{CM}).

[0083] In some embodiments, the immune cells are stem cells or are derived from stem cells. The stem cells can be adult stem cells, non-human embryonic stem cells, more particularly non-human stem cells, mesenchymal stem cells, cord blood stem cells, progenitor cells, bone marrow stem cells, induced pluripotent stem cells, totipotent stem cells or hematopoietic stem cells. Representative human cells are CD34+ cells. In other embodiments, the immune cell is derived from the differentiation of a population of induced pluripotent cells (iPSCs).

[0084] In some embodiments, the immune cells are harvested directly from a subject, e.g., a human subject. The cells are genetically modified as described herein and the genetically engineered immune cells are infused back into the same subject, for example, in a CAR-T cell therapy. In this case, the genetically engineered immune cells are autologous to the subject receiving the CAR-T cell therapy. In another embodiment, the immune cells are harvested directly from a donor subject, modified, and the genetically engineered immune cells are infused into a recipient subject in need of therapy, e.g., a CAR-T cell therapy. In this case, the donor immune cells are HLA-matched to the recipient subject, i.e., the cells are allogeneic to the recipient subject. In some embodiments, the immune cells are harvested and isolated from the peripheral blood of the subject (e.g., peripheral blood lymphocytes) and expanded in vitro prior to the genetic modifications disclosed herein.

[0085] In some embodiments, the CAR-expressing immune cells, including CAR-T cells, are transduced to additionally express one or more gene products, including, but not limited to ICAM-1. In some embodiments, the immune cells (e.g., T cells) are transduced to additionally express human somatostatin receptor 2 (SSTR2).

[0086] In some embodiments, the population of immune cells comprising the CAR-expressing cells further includes a second population of immune cells. In some embodiments, the second population of immune cells includes non-transduced immune cells, immune cells expressing another CAR, and/or immune cells expressing another gene product.

[0087] In some embodiments, the population of immune cells comprising the CAR disclosed herein may comprise at least about 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of the total immune cell population, or a range between two of the foregoing amounts. In one embodiment, about 50-70% of the immune cells may express the CAR.

III. Methods of Preparing Modified Immune Cells

[0088] The c-MET CAR and optionally other exogenous nucleic acids can be introduced into suitable immune cells by routine methods and/or approaches. To generate modified immune cells expressing the c-MET-directed CARs described herein, coding sequences from one or more CARs, and optionally other gene products, may be cloned into a suitable expression vector (e.g., including but not limited to lentiviral vectors, retroviral vectors, adenoviral vectors, adeno-associated vectors, PiggyBac transposon vector and Sleeping Beauty transposon vector) and introduced into host immune cells using conventional recombinant technology known to those skilled in the art. As a result, modified immune cells of the present disclosure may comprise one or more exogenous nucleic acids encoding at least one CAR and optionally one or more other gene products described herein. In some instances, the coding sequences of such molecules are integrated into the genome of the cell for expression using viral expression vectors (e.g., lentivirus vectors) or by gene editing into suitable target sites. In other instances, the coding sequences of such molecules are not integrated into the genome of the cell.

[0089] An exogenous nucleic acid comprising a coding sequence of interest may further comprise a suitable promoter, which can be in operable linkage to the coding sequence. A promoter, as used herein, refers to a nucleotide sequence in a nucleic acid to which RNA polymerase can bind to initiate the transcription of a DNA coding region into mRNA, which will then be translated into the corresponding protein. A promoter is considered to be "operably linked" to a coding sequence when it is in a correct functional location and orientation relative to the coding sequence to control ("drive") transcriptional initiation to produce the mRNA for translating the protein. In some instances, the promoter described herein can be constitutive, which initiates transcription independent other regulatory factors. In some instances, the promoter described herein can be inducible, which is dependent on regulatory factors for transcription.

[0090] Additionally, the exogenous nucleic acids described herein may further contain, for example, some or all of the following: a selectable marker gene, such as the neomycin gene for selection of stable or transient transfectants in mammalian cells; enhancer/promoter sequences from the immediate early gene of human CMV for high levels of transcription; transcription termination and RNA processing signals from SV40 for mRNA stability; SV40 polyoma origins of replication and ColE1 for proper episomal replication; versatile multiple cloning sites; and T7 and SP6 RNA promoters for in vitro transcription of sense and antisense RNA.

[0091] In some instances, one or more nucleic acids encoding the CAR(s) and/or other gene products disclosed herein can be inserted into a suitable expression cassette in a multi-cistronic manner such that the various molecules are expressed as separate polypeptides. In some examples, an internal ribosome entry site (IRES) can be inserted between two coding sequences to achieve this goal. Alternatively, a nucleotide sequence coding for a self-cleaving peptide (e.g., T2A or P2A) can be inserted between two coding sequences as described above.

[0092] For example, in some embodiments, T cells may be transduced with a nucleic acid encoding a second gene product, such as SSTR2, and a CAR disclosed herein. In certain embodiments, the nucleic acid encodes an SSTR2 polypeptide (aa 1-381; SEQ ID NO: 137) or a truncated SSTR2 polypeptide (aa 1-314; SEQ ID NO: 138). In a specific embodiment, the nucleic acid encodes a SSTR2-CAR fusion protein comprising an amino acid cleavage sequence between the CAR coding sequence and the SSTR2 coding sequence. The cleavage sequence may encode a self-cleaving 2A peptide from porcine teschovirus-1 (P2A; SEQ ID NO: 139), equine rhinitis A virus (E2A; SEQ ID NO: 140), *Thosea asigna* virus (T2A; SEQ ID NO: 141), or foot-and-mouth disease virus (F2A; SEQ ID NO: 142), or a combination thereof.

[0093] SSTR2 expressing cell compositions and methods for using SSTR2 as a reporter for CAR-T cell monitoring and use in cancer are disclosed in U.S. Pat. No. 10,577,408, which is expressly incorporated by reference herein.

IV. Therapeutic Applications

(A) Adoptive CAR-T Cell Therapy

[0094] In one aspect, immune cell populations comprising the modified immune cells as described herein may be used in an adoptive immune cell therapy (e.g., CAR-T) for treating a target disease, such as a solid tumor or tumor characterized by overexpression of c-MET. In an embodiment, a method for treating cancer comprises administering to a subject in need thereof, a population of immune cells comprising genetically engineered CAR-expressing cells (e.g., T cells) described herein in an amount suitable for treating the cancer.

[0095] To practice the therapeutic methods described herein, an effective amount of the immune cell population comprising the genetically modified immune cells described herein may be administered to a subject in need of cancer treatment via a suitable route of administration (e.g., intravenous infusion). One or more of the immune cell populations may be mixed with a pharmaceutically acceptable carrier to form a pharmaceutical composition prior to administration, which is also within the scope of the present disclosure. The immune cells may be autologous to the subject, e.g., obtained from the subject in need of the treatment, modified to express the c-MET CAR construct and optionally one or more additional exogenous gene products. The resultant modified immune cells can then be administered to the same subject. Administration of autologous cells to a subject may result in reduced rejection of the immune cells as compared to administration of non-autologous cells. Alternatively, the immune cells can be allogeneic cells, e.g., the cells are obtained from a first subject, modified as described herein and administered to a second subject that is different from the first subject but of the same species. For example, allogeneic immune cells may be derived from a human donor and administered to a human recipient who is different from the donor.

[0096] The subject to be treated may be a mammal (e.g., human, mouse, pig, cow, rat, dog, guinea pig, rabbit, hamster, cat, goat, sheep, or monkey) suffering from cancer, particularly a human patient with a cancer characterized by overexpression of c-MET.

[0097] The term "an effective amount" as used herein refers to the amount of each active agent required to confer therapeutic effect on the subject, either alone or in combination with one or more active agents. Effective amounts may vary, as recognized by those skilled in the art, depending on the particular condition being treated, the severity of the condition, toxicity consideration, previous trial results, individual patient parameters including age, physical condition, size, gender and weight, the duration of treatment, route of administration, excipient usage, co-usage (if any) with other active agents and like factors within the knowledge and expertise of the health practitioner. The quantity to be administered depends on the subject to be treated, including, for example, the capacity of the individual's immune system to produce a cell-mediated immune response. Effective amounts of the genetically engineered CAR-T cells required to be administered depend on the judgment of the practitioner. However, suitable dosage ranges are readily determinable by one skilled in the art.

[0098] The term "treating" as used herein refers to the application or administration of a cell composition to a subject with cancer with the purpose to cure, heal, alleviate, relieve, alter, remedy, ameliorate, improve, affect progression of the cancer and its symptoms.

[0099] An effective amount of the immune cells may be administered to a human patient in need of the treatment via a suitable route, e.g., intravenous infusion. In some instances, about 1×10^6 to about 1×10^8 CAR+ T cells may be given to a human patient (e.g., a leukemia patient, a lymphoma patient, or a multiple myeloma patient). In some examples, a human patient may receive multiple doses of the immune cells. For example, the patient may receive two doses of the immune cells on two consecutive days. In some instances, the first dose is the same as the second dose. In other instances, the first dose is lower than the second dose, or vice versa.

[0100] In any of the treatment methods involving the use of the modified immune cells disclosed herein, the subject may be administered IL-2 concurrently with the cell therapy. More specifically, an effective amount of IL-2 may be given to the subject via a suitable route before, during, or after the cell therapy. In some embodiments, IL-2 is given to the subject after administration of the immune cells. In some embodiments, prior to the cell therapy, the subject receives a lymphodepleting treatment to condition the subject for the cell therapy.

[0101] In some embodiments, the cancer for treatment is a solid tumor. In some embodiments, the solid tumor is a carcinoma. In certain embodiments, the carcinoma is an adenocarcinoma or squamous cell carcinoma thereof. Examples of carcinomas include small cell lung carcinoma, non-small cell lung carcinoma, squamous cell lung carcinoma, large cell lung carcinoma, pancreatic carcinoma, pancreatic ductal carcinoma, prostate carcinoma, esophageal carcinoma, breast carcinoma, ovarian carcinoma, prostate carcinoma, colorectal carcinoma, bladder carcinoma, cervical carcinoma, hepatocellular carcinoma, renal hepatocellular carcinoma, gastric carcinoma, papillary carcinoma, adrenocortical carcinoma, pituitary carcinoma, a head and neck carcinoma, an adenocarcinomas thereof, a squamous cell carcinomas thereof, and a metastatic cancer thereof. In some embodiments, the solid cancer is a glioblastoma or mesothelioma.

[0102] In some embodiments, the solid tumor expresses one or more antigen selected from the group consisting of: EpCAM, CEACAM5, EGFRvIII, mesothelin, CS-1, GD2, Tn Ag, PSMA, TAG72, CD44v6, CEA, KIT, IL-13Ra2, GD3, CD171, IL-1Ra, PSCA, VEGFR2, Lewis Y, CD24, PDGFR-beta, SSEA-4, folate receptor alpha, ERBB2, Her2/neu, MUC1, EGFR, NCAM, Ephrin B2, CAIX, LMP2, sLe, HMWMAA, o-acetyl-GD2, folate receptor beta, TEM1/CD248, TEM7R, FAP, Legumam, HPV E6 or E7, CLDN6, TSHR, GPRC5D, ALK, Polysialic acid, PLAC1, globoH, NY-BR-1, UPK2, HAVCR1, ADRB3, PANX3, GPR20, Ly6k, OR51E2, TARP, and GFRa4. In some embodiments, these antigens may serve as target antigens for the bispecific c-MET CARs or as secondary CARs to be used in conjunction with the c-MET CARs disclosed herein.

(B) Co-Administration of Tyrosine Kinase Inhibitors.

[0103] It is known that transient inhibition or modulation of TCR signaling and/or CAR signaling in human T cells can prevent or reverse T cell exhaustion and restore T cell function to CAR-T cells undergoing functional exhaustion. In particular, in vivo treatment of the CAR-T cells described herein with certain tyrosine kinase inhibitors inhibiting T cell receptor signaling (e.g., a Lck tyrosine kinase inhibitor (e.g., dasatinib)) can suppress exhaustion marker expression, augment memory formation, decrease expression of PD1 on CAR-T cells, and facilitate cell survival/proliferation as described in US 2020/101108 A1 and US 2021169880 A1, the disclosures of which are expressly incorporated by reference herein. Similar findings have been obtained using small molecules having a thiazole, imidazolepyridiazine or piperazinyl-methyl-aniline structure (US 2021/0393628 A1, incorporated by reference herein).

[0104] Thus, in some embodiments, a patient receiving CAR-T treatment in accordance with the present disclosure may be additionally administered a tyrosine kinase or small molecule having a thiazole, imidazolepyridiazine or piperazinyl-methyl-aniline structure to mitigate T cell exhaustion, augment memory T cell formation, and/or maintain and facilitate cell survival and proliferation. In other embodiments, these active agents may be used to improve ex vivo cell expansion and collection of CAR-T populations that are resistant and/or less prone to T cell exhaustion. Thus, in another aspect, the present disclosure further provides cell compositions comprising a population of CAR-T cells that are expanded in the presence of particular compounds described herein.

[0105] Exemplary tyrosine kinases for administration or cell treatment include dasatinib, ponatinib, saracatinib, bosutinib, nilotinib, and combinations thereof. Exemplary small molecules having a thiazole, imidazolepyridiazine or piperazinyl-methyl-aniline structure are disclosed in US 2021/0393628 A1, the disclosures of which are expressly incorporated by reference herein.

[0106] Such methods are not limited to particular manner of administration. In some embodiments, multiple cycles of treatment are administered to the subject. In some embodiments, the pharmaceutical composition is administered intermittently. In some embodiments, the pharmaceutical composition is administered for a period of time sufficient to restore at least partial T cell function then discontinued. In some embodiments, the pharmaceutical composition is administered orally.

[0107] In some embodiments, the pharmaceutical compositions are administered iteratively for purposes of facilitating periods of CAR-T cell inactivation (e.g., during pharmaceutical composition administration) and periods of CAR-T cell activation (e.g., during absence of pharmaceutical composition administration; following clearance of the pharmaceutical composition).

[0108] In some embodiments, patients undergoing CAR-T treatment are subjected to intermittent exposure to dasatinib (or other active agents above) to reduce exhaustion, and augment the engraftment, proliferation, and persistence of CAR-T cells in vivo., as well as antitumor function of the CAR-T cells.

[0109] The terms “intermittent administration” or “administered intermittently” in connection with the tyrosine kinase inhibitors described herein refer to the use of these tyrosine kinase inhibitors in an administration regime that causes intermittent changes between a state wherein the patient has tyrosine kinase inhibitor serum levels within the therapeutic window and a state wherein the patient has tyrosine kinase inhibitor serum levels below the therapeutic window. A therapeutic window of a given tyrosine kinase inhibitor can be determined by any methods known in the art.

[0110] Alternatively, the terms “intermittent administration” and “administered intermittently” in connection with a tyrosine kinase inhibitor as used herein refer to the use of a tyrosine kinase inhibitor in an administration regime causing: (1) intermittent changes between a state where the patient has tyrosine kinase inhibitor serum levels causing complete inhibition of the tyrosine kinase and a state where the patient has tyrosine kinase inhibitor serum levels causing partial inhibition of the tyrosine kinase; (2) intermittent changes between a state where the patient has tyrosine kinase inhibitor serum levels causing complete inhibition of the tyrosine kinase and a state where the patient has tyrosine kinase inhibitor serum levels causing no inhibition of the tyrosine kinase; or (3) intermittent changes between a state where the patient has tyrosine kinase inhibitor serum levels causing partial inhibition of the tyrosine kinase and a state where the patient has tyrosine kinase inhibitor serum levels causing no inhibition of the tyrosine kinase.

[0111] Such inhibition can be measured by any methods known in the art, e.g., by measuring the activity of the tyrosine kinase itself using appropriate enzyme assays, or by measuring cellular functions downstream of the kinase. In some embodiments, a partial inhibition refers to an inhibition of at least 25% to 75% compared to a situation in the absence of the inhibitor. As used herein, “no inhibition” refers to an inhibition of less than 25%, or less than 10% compared to a situation in the absence of the inhibitor.

[0112] In the case of CAR-T cells, inhibition of less than 25% or 10% can be an inhibition of the cytotoxic lysis, cytokine secretion, and/or proliferation of the T cells. Further, the inhibition of at least 25%, but no more than 75%, can preferably be an inhibition of the cytotoxic lysis, cytokine secretion, and proliferation of the CAR-T cells.

[0113] In some embodiments, intermittent administration of dasatinib may cause intermittent changes between a state wherein the serum levels of dasatinib are above 50 nM and a state wherein the serum levels of dasatinib are at or below 50 nM. Intermittent administration may be achieved by using an administration interval longer than the terminal phase half-life of the tyrosine kinase inhibitor, longer than 2 times the terminal phase half-life of the tyrosine kinase inhibitor, or longer than 3 times, 4 times, or 5 times the terminal phase half-life of the tyrosine kinase inhibitor. For example, intermittent administration of dasatinib may be achieved using an administration interval of at least 6 hours for dasatinib or at least 12 hours for dasatinib. It will be understood by a person skilled in the art that for each administration regime, appropriate dosages of the respective tyrosine kinase inhibitors can be selected based on pharmacokinetic and pharmacodynamic experiments.

[0114] The terms “continuous administration” or “administered continuously” in connection with a tyrosine kinase inhibitor as used herein refer to the use of said tyrosine kinase inhibitor in an administration regime that causes a complete inhibition of the tyrosine kinase in a continuous manner. According to the invention, a complete inhibition refers to an inhibition of at least 75%, compared to a situation in the absence of the inhibitor. Such inhibition can be measured by any methods known in the art, e.g., by measuring the activity of the tyrosine kinase itself using appropriate enzyme assays, or by measuring cellular functions downstream of said kinase.

[0115] In the case of CAR-T cells, inhibition of at least 75% can be an inhibition of the cytotoxic lysis, cytokine secretion, and proliferation of T cells. Alternatively, the terms “continuous administration” and “administered continuously” in connection with a tyrosine kinase inhibitor described herein refer to use of the tyrosine kinase inhibitor in an administration regime that results in serum levels of the tyrosine kinase which are continuously within the therapeutic window. In some embodiments, continuous administration of dasatinib encompasses any administration wherein the serum levels of dasatinib are constantly maintained at or above 50 nM. In one embodiment, dasatinib is administered continuously, where the administration comprises oral administration of 50-200 mg dasatinib every 6-8 hours or 140 mg every 6 hours.

[0116] In some embodiments, the threshold serum level is within the range of 0.1 nM-1 μM, 1 nM-500 nM, 5 nM-100 nM, 10 nM-75 nM, or 25 nM-50 nM.

(C) Monitoring CAR-T Cell Distribution in a Patient

[0117] In another aspect, the present disclosure provides a method for treating cancer and monitoring CAR-T cell distribution in a patient undergoing CAR-T cell therapy. The method involves the use of CAR-T cells co-expressing human somatostatin receptor 2 (SSTR2) as a cell surface marker for monitoring CAR-T cell distribution in a patient.

[0118] SSTR2 can be used in conjunction with FDA-approved positron emission tomography (PET) radiotracers currently used in clinics to probe for overexpressed SSTR2 in neuroendocrine tumors, specifically ^{68}Gallium conjugates of DOTATOC and DOTATATE. Single-photon emission computed tomography (SPECT)-based imaging is also available using ^{111}In-DTPAOC (Octreoscan) or ^{177}Lutetium. SSTR2 displays restricted basal expression in tissues and all major organs except in the kidneys and cerebrum making it ideal for detection of adoptively transferred CAR-T cells targeting a multitude of solid tumors.

[0119] It has previously been shown that SSTR2 facilitates rapid radiotracer uptake and this combined with swift renal clearance of unbound DOTATOC means that high quality, clinical-grade images can be obtained at one hour post DOTATOC injection. DOTATOC also has a short half-life of 68 min which, combined with its rapid clearance, delivers a low radiation dose to the patient. The fact that SSTR2 is of human origin also limits its immunogenicity which has plagued experiments using non-human genetic reporters.

[0120] SSTR2 compositions and methods for using SSTR2 as a reporter for CAR-T cell monitoring and use in cancer are disclosed in U.S. Pat. No. 10,577,408, which is expressly incorporated by reference herein.

[0121] In one embodiment, the method comprises: (a) incubating a population of CAR-T cells described herein with a radioactive label that binds to SSTR2; (b) intravenously infusing the labeled CAR-T cells into a patient in an amount of 10^4-10^8 cells/kg patient, and (c) detecting the labeled CAR-T cell distribution by positron emission tomography/computed tomography (PET/CT) imaging, wherein the labeled CAR-T cells are infiltrated into cancer cells to kill the cancer cells. In this method, SSTR2 is pre-labeled in vitro. In some embodiments, the labeled CAR-T cells are administered in an amount of 10^6-10^8 or 10^6-10^7 cells/kg of the patient.

[0122] In another embodiment, the method comprises: (a) intravenously infusing a population of CAR-T cells described herein into a patient; (b) injecting into the patient a radioactive label that binds to SSTR2 at least one hour prior to PET/CT imaging, and (c) detecting the labeled CAR-T cell distribution by PET/CT imaging, wherein the labeled CAR-T cells are infiltrated into cancer cells to kill the cancer cells. In this method, SSTR2 is labeled post-infusion in vivo. In certain embodiments, the CAR-T cells have been transduced to express at least 100,000 molecules of SSTR2 per T cell.

[0123] In some embodiments, the label is radioactively labeled DOTATOC or radioactively labeled DOTATATE, such as ^{68}Gallium-DOTATOC or ^{68}Gallium-DOTATATE. The methods for monitoring distribution of radiolabeled CAR-T cells may be used in connection with treating any of the cancers described herein, including but not limited to thyroid cancer, gastric cancer, pancreatic cancer, or breast cancer.

(D) Combination Therapies

[0124] The immune cell populations comprising e.g., the CAR-T cells described herein may be utilized in conjunction with other types of therapy or active agents for cancer, such as chemotherapy, surgery, radiation, gene therapy, and so forth. Such therapies can be administered simultaneously or sequentially (in any order) with the immunotherapy described herein. When co-administered with an additional therapeutic agent, suitable therapeutically effective dosages for each agent may be lowered due to additive or synergistic effects.

[0125] In some examples, the subject is treated with an anti-cancer therapy (e.g., those disclosed herein) to reduce tumor burden prior to the CAR-T therapy disclosed herein. For example, the subject (e.g., a human cancer patient) may be treated with chemotherapy (e.g., comprising a single chemotherapeutic agent or a combination of two or more chemotherapeutic agents) at a dose that substantially reduces tumor burden. In some instances, the chemotherapy may reduce the total white blood cell count in the subject to lower than 10^8/L, e.g., lower than 10^7/L. Tumor burden of a patient after the initial anti-cancer therapy, and/or after the CAR-T cell therapy disclosed herein may be monitored via routine methods. If a patient showed a high growth rate of cancer cells after the initial anti-cancer therapy and/or after the CAR-T therapy, the patient may be subjected to a new round of chemotherapy to reduce tumor burden followed by any of the CAR-T therapies disclosed herein.

[0126] Non-limiting examples of other anti-cancer therapeutic agents for use in combination with the modified immune cells (e.g., CAR-T cells) described herein include, but are not limited to, immune checkpoint inhibitors (e.g., PDL1, PD1, and CTLA4 inhibitors), anti-angiogenic agents (e.g., TNP-470, platelet factor 4, thrombospondin-1, tissue inhibitors of metalloproteases, prolactin, angiostatin, endostatin, bFGF soluble receptor, transforming growth factor beta, interferon alpha, interferon gamma, soluble KDR and FLT-1 receptors, and placental proliferin-related protein); VEGF antagonists (e.g., anti-VEGF antibodies, VEGF variants, soluble VEGF receptor fragments); chemotherapeutic compounds. In some embodiments, the anti-cancer therapeutic agents include pembrolizumab (Keytruda™) ipilimumab (Yervoy™), nivolumab (Opdivo™), or atezolizumab (Tecentriq™)

[0127] Exemplary chemotherapeutic compounds include pyrimidine analogs (e.g., 5-fluorouracil, floxuridine, capecitabine, gemcitabine and cytarabine); purine analogs (e.g., fludarabine); folate antagonists (e.g., mercaptopurine and thioguanine); antiproliferative or antimitotic agents, for example, vinca alkaloids; microtubule disruptors such as taxane (e.g., paclitaxel, docetaxel), vincristin, vinblastin, nocodazole, epothilones and navelbine, and epidipodophyllotoxins; and DNA damaging agents (e.g., actinomycin, amsacrine, anthracyclines, bleomycin, busulfan, camptothecin, carboplatin, chlorambucil, cisplatin, cyclophosphamide, cytoxan, dactinomycin, daunorubicin, doxorubicin, epirubicin, hexamethyhnelamineoxaliplatin, iphosphamide, melphalan, merchlorehtamine, mitomycin, mitoxantrone, nitrosourea, plicamycin, procarbazine, taxol, taxotere, teniposide, triethylenethiophosphoramide and etoposide).

[0128] In some embodiments, radiation or radiation and chemotherapy is used in combination with the cell populations comprising modified immune cells described herein. Additional useful agents and therapies can be found in Physician's Desk Reference, 59th edition, (2005), Thomson P D R, Montvale N.J.; Gennaro et al., Eds. Remington's The Science and Practice of Pharmacy 20th edition, (2000), Lippincott Williams and Wilkins, Baltimore Md.; Braunwald et al., Eds. Harrison's Principles of Internal Medicine, 15th edition, (2001), McGraw Hill, NY; Berkow et al., Eds. The Merck Manual of Diagnosis and Therapy, (1992), Merck Research Laboratories, Rahway N.J.

V. General Techniques

[0129] The practice of the present disclosure will employ, unless otherwise indicated, conventional techniques of molecular biology (including recombinant techniques), microbiology, cell biology, biochemistry, and immunology, which are within the skill of the art. Such techniques are explained fully in the literature, such as *Molecular Cloning: A Laboratory Manual*, second edition (Sambrook, et al., 1989) Cold Spring Harbor Press; *Oligonucleotide Synthesis* (M. J. Gait, ed. 1984); *Methods in Molecular Biology*, Humana Press; *Cell Biology: A Laboratory Notebook* (J. E. Cellis, ed., 1989) Academic Press; Animal Cell Culture (R. I. Freshney, ed. 1987); Introduction to Cell and Tissue Culture (J. P. Mather and P. E. Roberts, 1998) Plenum Press; Cell and Tissue Culture: Laboratory Procedures (A. Doyle, J. B. Griffiths, and D. G. Newell, eds. 1993-8) J. Wiley and Sons; Methods in Enzymology (Academic Press, Inc.); Handbook of Experimental Immunology (D. M. Weir and C. C. Blackwell, eds.): Gene Transfer Vectors for Mammalian Cells (J. M. Miller and M. P. Calos, eds., 1987); Current Protocols in Molecular Biology (F. M. Ausubel, et al. eds. 1987); PCR: The Polymerase Chain Reaction, (Mullis, et al., eds. 1994); Current Protocols in Immunology (J. E. Coligan et al., eds., 1991); Short Protocols in Molecular Biology (Wiley and Sons, 1999); Immunobiology (C. A. Janeway and P. Travers, 1997); Antibodies (P. Finch, 1997); Antibodies: a practice approach (D. Catty., ed., IRL Press, 1988-1989); Monoclonal antibodies: a practical approach (P. Shepherd and C. Dean, eds., Oxford University Press, 2000); Using antibodies: a laboratory manual (E. Harlow and D. Lane, Cold Spring Harbor Laboratory Press, 1999); The Antibodies (M. Zanetti and J. D. Capra, eds. Harwood Academic Publishers, 1995); *DNA Cloning: A practical Approach*, Volumes I and II (D. N. Glover ed. 1985); *Nucleic Acid Hybridization* (B. D. Hames & S. J. Higgins eds. (1985*; Transcription and Translation* (B. D. Hames & S. J. Higgins, eds. (1984*; Animal Cell Culture* (R. I. Freshney, ed. (1986*; Immobilized Cells and Enzymes* (IRL Press, (1986); B. Perbal, *A practical Guide To Molecular Cloning* (1984); F. M. Ausubel et al. (eds.); *Chimeric Antigen Receptor* (*CAR*) *Immunotherapy* (D. W. Lee and N. N. Shah, eds., Elsevier, 2019, ISBN:9780323661812); *Basics of Chimeric Antigen Receptor* (*CAR*) *Immunotherapy* (M. Y. Balkhi, Academic Press, Elsevier Science, 2019, ISBN: 9780128197479); *Chimeric Antigen Receptor T Cells Development and Production* (V. Picanço-Castro, K. C. R. Malmegrim, K. Swiech, eds., Springer US, 2020, ISBN: 9781071601488); *Cell and Gene Therapies* (C. Bollard, S. A. Abutalib, M.-A. Perales eds., Springer International, 2018; ISBN:9783319543680) and *Developing Costimulatory Molecules for Immunotherapy of Diseases* (M. A. Mir, Elsevier Science, 2015, ISBN:9780128026755).

[0130] The present disclosure is not limited in its application to the details of construction and the arrangements of component set forth in the description herein or illustrated in the drawings. The present disclosure is capable of other embodiments and of being practice or of being carried out in various ways. Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of "including," "comprising," or "having," "containing," "involving," and variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. As also used in this specification and the appended claims, the singular forms "a," "an," and "the" include plural references unless the context clearly dictates otherwise.

[0131] Without further elaboration, it is believed that one skilled in the art can, based on the above description, utilize the present invention to its fullest extent. The following specific embodiments are, therefore, to be construed as merely illustrative, and not limitative of the remainder of the disclosure in any way whatsoever. All publications cited herein are incorporated by reference for the purposes or subject matter referenced herein.

EXAMPLES

Example 1: Generation of Reduced Affinity c-MET CAR-T Cells

[0132] Site directed mutagenesis was used to create affinity tuned anti-c-MET variant single-chain variable fragments (scFvs) comprising variable heavy (V_H) regions with lower affinities than scFvs comprising parental versions of the V_H and variable light (V_L) chain regions. Edits to the CDR3 of the heavy chain were made using an NEB Q5 site-directed mutagenesis kit (NEB #E0554S) with individually designed primers outlined in Table 1 to create desired amino acid changes.

TABLE 1

Primers used for making antibodies

Variant	Change		Primer #	Primer*	Primer SEQ ID NO	Tm	VH SEQ ID NOS
Affy-Y6A	Y → A	TAC → **gcg**	AB_001	CTGTGCCACC**gcg**GGCTCCTACGTGAGC	133	66	13, 19
			AB_002	TAGTACACGGCGCTGTOC	134		
Affy7-L9A	L → A	TTG → **gc**G	AB_015	CGTGAGCCCT**gcg**GATTATTGGGGC	135	67	14, 19
			AB_016	TAGGAGCCGTAGGTGGCA	136		
Affy7-Y6V	Y → V	TAC → **gt**C	AB_037	CTGTGCCACC**gt**CGGCTCCTACG	137	66	15, 19
			AB_038	TAGTACACGGCGCTGTCC	138		
Affy7-Y6L	Y → L	TAC → **ct**C	AB_039	CTGTGCCACC**ct**CGGCTCCTAC	139	65	16, 19
			AB_038	TAGTACACGGCGCTGTOC	138		
Affy7-Y6I	Y → I	TAC → **at**C	AB_040	CTGTGCCACC**at**CGGCTCCTAC	140	65	17, 19
			AB_038	TAGTACACGGCGCTGTCC	138		
Affy7-Y6M	Y → M	TAC → **atg**	AB_042	CTGTGCCACC**atg**GGCTCCTACGTGAGCCC	141	65	18, 19
			AB_038	TAGTACACGGCGCTGTCC	138		

TABLE 1-continued

Primers used for making antibodies						
Variant	Change	Primer #	Primer*	Primer SEQ ID NO	Tm	VH SEQ ID NOS
Affy6-I3A	I → A TAC → **gc**C	AB_086 AB_087	CCGCTCGGAG**gc**CACCACCGAG GCGCAGTAGTACACGGCC	142 143	68	80, 84
Affy6-I3G	I → G ATC → **gg**C	AB_120 AB_117	CCGCTCGGAG**ggc**ACCACCGAGTTC GCGCAGTÅGTACACGGCC	147 145	70	83, 84
Affy6-I3F	I → F ATC → **tt**C	AB_123 AB_117	CCGCTCGGAG**ttc**ACCACCGAGT GCGCAGTAGTACACGGCC	148 146	71	82, 84
Affy6-I3W	I → W ATC → **tgg**	AB_125 AB_117	CCGCTCGGAG**tgg**ACCACCGAGTTC GCGCAGTAGTACACGGCC	149 146	67	81, 84

*mutant nucleotides are bolded in small case

[0133] Lentivirus constructs were prepared to facilitate characterization of the binding properties of the affinity tuned anti-c-MET scFvs in chimeric artificial receptors (CARs). C-MET CARs were prepared by cloning genetic sequences encoding parental (wild-type) c-MET Affy7 and Affy6 scFvs and the affinity-tuned c-MET Affy7 and Affy6 variant scFvs into a lentiviral vector between a c-myc tag and CAR element CD8 hinge. As shown in FIG. **1**, the lentiviral vector construct comprises (5' to 3') the EF1a promoter, c-myc tag, c-MET scFv, CD8 hinge, CD28 transmembrane domain, CD28 co-stimulatory domain, CD3 zeta signaling domain, porcine teschovirus-1 2A (P2A) ribosome skipping element, and a human somatostatin receptor 2 (SSTR2) reporter gene for CAR-T cell imaging.

[0134] To obtain stably transformed CAR-expressing cells, lentivirus particles were produced by transiently transfecting HEK 293T cells using the TransIT®-Lenti transfection reagent, (Mirus Bio, MIR6604). Briefly, 8 μg of transfer gene and 12 μg of LV-MAX lentiviral packaging mix (ThermoFisher, A43237) were mixed with Opti-MEM and 50 μL of Trans-IT and incubated for 20 minutes at room temperature. The resulting solutions were added dropwise to 10 cm^2 cell culture dishes seeded with 6×10^6 HEK 293T cells in 10 mL DMEM 24 hrs. previously. Transfection media was replaced 16-24 hrs. later. Media containing lentivirus was harvested at 48 hrs. post transfection, filtered through 0.45 μm filers and concentrated by adding Lenti-X concentrator (Takara, 631231) at ⅓ of final volume of supernatant. The solution was then incubated a 4° C. for at least 3 hrs. (as long as overnight) and then spun at 1,500 g for 1 hr at 4° C. The supernatant was removed, and the pellet was resuspended in 400 μL of sterile PBS and stored at −80° C.

Example 2: Evaluation of Binding Properties of Affinity-Tuned c-MET CARs

[0135] Binding of labeled c-MET to CAR-T cells expressing the affinity tuned c-MET scFvs was evaluated by transducing lentivirus anti-c-MET CAR particles into human T cells to form c-MET CAR-T cells incubated with AF647-labeled c-MET. Briefly, human T cells were transduced 24 hrs. post activation with ImmunoCult™ CD3/CD28 T cell activator (StemCell, 10971) in 10 mL of media containing IL7 (12.5 ng/mL), IL15 (12.5 ng/mL) and 10 μL of Lenti-BOOST (Mayflower Bioscience, SBPLV10103) in G-Rex® 6M wells (Scale Ready, 80660M). Following a 48 hr. incubation at 37° C., the culture media was brought up to 100 mL per well and incubated at 37° C. for 10 days.

[0136] Binding curves were generated by serial diluting AF647 labeled c-met protein (AlexaFluor™ 647 antibody labeling kit, ThermoFisher, A20186) and staining each sample with c-myc antibody (Miltenyi, 130-116-485) at a 1:800 dilution. Cells were stained on ice for 30 mins and then washed with PBS. After washing, residually bound protein on CAR T cells were measured on a flow cytometer (Beckman, CytoFLEX®). The resulting binding data was fit using a non-linear regression analysis on GraphPad Prism to calculate half effective dose concentration (EC_{50}) values. The results of this analysis are shown in FIGS. **2-5** and Tables 2-5 below.

[0137] As compared to CAR-T cells expressing the parental anti-c-MET scFv binding domains (i.e., Affy7 and Affy6), CAR-T cells expressing the affinity-tuned anti-c-MET scFvs exhibiting reduced affinities for c-MET ranging between 20 to 7636-fold relative to the parental-expressing CARs (see FIGS. **2-5** and Tables 2-5).

TABLE 2

Binding Affinities of Parental- and Variant Affy7-Based MET CAR-T Cells for c-MET (Round 1)		
Parent/Variant	EC50 (μM)	Fold Change*
Affy7	0.05	N/A
Affy7-L9A	1.0	20
Affy7-Y6A	3.3	66

TABLE 3

Binding Affinities of Parental- and Variant Affy7-Based MET CAR-T Cells for c-MET (Round 2)		
Parent/Variant	EC50 (μM)	Fold Change*
Affy7	0.1	N/A
Affy7-Y6V	5.3	53
Affy7-Y6M	17.2	172
Affy7-Y6I	183.1	1831
Affy7-Y6L	619.2	6192

TABLE 4

Binding Affinities of Parental- and Variant Affy6-Based MET CAR-T Cells for c-MET (Round 1)		
Parent/Variant	EC50 (μM)	Fold Change*
Affy6	0.02	N/A
Affy6-I3A	20.9	956

TABLE 5

Binding Affinities of Parental- and Variant Affy6-Based MET CAR-T Cells for c-MET (Round 2)		
Parent/Variant	EC50 (μM)	Fold Change*
Affy6	0.02	N/A
Affy6-I3W	3.6	167
Affy6-I3F	20.5	937
Affy6-I3G	150.8	6899

Example 3: Cytotoxicity of Reduced Affinity c-MET CAR-T Cells in Cancer Cell Lines

[0138] One of the pitfalls of CAR-T cell therapies concerns toxicities and T cell exhaustion associated with recognition of antigen both on normal, non-target cells as well as on cancer cells. However, when using reduced affinity c-MET CAR-T cells to reduce cell toxicities and T cell exhaustion, it is important to maintain effective killing of the targeted cancer cells.

[0139] To evaluate cytotoxicity of the reduced affinity c-MET CAR-T cells against human carcinoma cells, 5×10^4 target cells (SNU638, Hs746T, and H1993) stably transduced to express GFP and firefly luciferase were co-cultured with c-met CAR-T cells at an effector to target ratio of 5:1. Co-cultures were carried out in T cell culture medium containing 150 μg/mL D-Luciferin (Fisher Scientific, NC1276267) with no cytokine supplementation. Luminescence was measured using a plate reader (BioTek®, Synergy Neo2) with readings every 24 hrs. for 3 days. In each case, readings were normalized to the no T cell control group and percent cytotoxicity was reported.

[0140] As shown in FIGS. **6-9** and Tables 6-9 below, all of the c-MET CAR-T cells of the present disclosure exhibited cytotoxicities similar to or moderately reduced when compared the high-affinity parental c-MET CAR-T cells in the 3 carcinoma cell lines tested. Although the cytotoxicities were significantly reduced in the 24 hour co-cultures compared to the high-affinity parental CAR-T cells, upon further incubation for 48 and 72 hours, the cytotoxicities typically approached the cytotoxicity levels obtained with the high-affinity parental CAR-T cells (see FIGS. **6-9** and Tables 6-9 below).

TABLE 6

In Vitro Cytotoxicity of Parental- and Variant Affy7-Based MET CAR-T Cells Against SNU-638 Human Stomach Carcinoma Cells								
SNU638	0 hrs	stdev	24 hrs	stdev	48 hrs	stdev	72 hrs	stdev
Affy7	0	0	78.6	7.1	95.9	1.9	99.5	0.3
Affy7-Y6A	0	0	79.2	5.8	93.5	3.3	98.7	0.4
Affy7-Y6I	0	0	67.0	3.9	83.6	2.4	93.2	0.7
Affy7-Y6L	0	0	66.9	2.2	84.5	2.8	93.5	0.3
Affy7-Y6V	0	0	68.1	8.2	81.9	5.4	93.2	2.1
Affy7-Y6M	0	0	61.2	3.0	77.5	1.8	90.6	0.4

TABLE 7

In Vitro Cytotoxicity of Parental- and Variant Affy7-Based MET CAR-T Cells Against Hs746T Human Stomach Carcinoma Cells								
Hs746T	0 hrs	stdev	24 hrs	stdev	48 hrs	stdev	72 hrs	stdev
Affy7	0	0	87.1	3.1	97.6	0.7	99.2	0.3
Affy7-Y6A	0	0	88.3	4.9	98.4	0.8	98.8	0.6
Affy7-Y6I	0	0	1.5	2.3	45.4	5.6	69.4	7.9
Affy7-Y6L	0	0	1.6	12.2	42.9	9.3	64.5	10.9
Affy7-Y6V	0	0	65.0	2.8	94.8	0.8	98.2	0.2
Affy7-Y6M	0	0	47.6	2	87	4.8	96.4	0.4

TABLE 8

In Vitro Cytotoxicity of Parental- and Variant Affy7-Based MET CAR-T Cells Against H1993 Non-Small Cell Lung Adenocarcinoma Cells								
H1993	0 hrs	stdev	24 hrs	stdev	48 hrs	stdev	72 hrs	stdev
Affy7	0	0	85.6	0.6	97.7	0.8	99.6	0.5
Affy7-Y6A	0	0	85.7	6.7	96.9	0.4	99.2	0.2
Affy7-Y6I	0	0	11.7	8.9	45.8	5.9	73.5	1.4

TABLE 8-continued

In Vitro Cytotoxicity of Parental- and Variant Affy7-Based MET CAR-T Cells Against H1993 Non-Small Cell Lung Adenocarcinoma Cells								
H1993	0 hrs	stdev	24 hrs	stdev	48 hrs	stdev	72 hrs	stdev
Affy7-Y6L	0	0	12.8	9.9	42.6	4.4	71.2	4.8
Affy7-Y6V	0	0	40.0	7.5	84.2	2.2	93.8	1.3
Affy7-Y6M	0	0	21.3	7	70.5	1.1	89.6	1.1

TABLE 9

In Vitro Cytotoxicity of Parental- and Variant Affy6-Based MET CAR-T Cells Against H1993 Non-Small Cell Lung Adenocarcinoma Cells								
H1993	0 hrs	stdev	24 hrs	stdev	48 hrs	stdev	72 hrs	stdev
Affy6	0	0						
Affy6-I3A	0	0	73.06	2.57	95.28	1.25	98.86	0.31
Affy6-I3F	0	0	64.67	2.71	91.32	1.73	96.99	0.21
Affy6-I3G	0	0	32.48	8.29	68.14	5.08	81.56	4.12
Affy6-I3W	0	0	74.88	5.65	93.5	0.88	98.20	0.24

Example 4. Cytotoxicity of Reduced Affinity c-MET CAR-T Cells in Primary Cells

[0141] As implicitly suggested above, an ideal CAR-T effector cell would be potent enough to kill c-MET overexpressing cancer cells while sparing normal c-MET expressing cells. Therefore, normal human cells were utilized as target cells for examining cytotoxicity in co-cultures with the reduced affinity c-MET CAR-T cells. More specifically, in vitro cytotoxicity assays were carried out by co-culturing the reduced affinity c-MET CAR-T cells with normal primary cells from human epithelial lung, liver, kidney, and colon cells.

[0142] Initially, the normal primary cells were evaluated for c-MET expression by flow cytometry. Briefly, 10 μL of human HGFR/c-met PE-conjugated antibody (R&D Systems, FAB3582P) was added to 1×10^6 cells in a 100 μL reaction volume containing 95 μL staining buffer (BioLegend, 420201). The resulting mixture was then incubated on ice for 30 min., washed with staining buffer and run on the flow cytometer. As shown in FIGS. **10**A-**10**D, the results of this analysis confirm expression of c-MET in normal human epithelial cells.

[0143] Next, cytotoxicity assays were carried out by co-culturing normal primary target cells (human epithelial lung, colon, kidney, and liver) with c-met CAR-T cells at an effector to target ratio of 2:1. The co-cultures were maintained in T cell culture medium with no cytokine supplementation over a period of 48 hours. Cytotoxicity was measured using label-free cellular impedance to continuously monitor and record cell killing using 96-well electronic microplates with the xCELLigence Real-Time Cell Analysis (RTCA) SP instrument (Agilent).

[0144] The results of this analysis are shown in Tables 10A-17B. The results show a correlation between the affinity of the binding between c-MET as shown in Tables 2-5 and cytotoxic activity. For example, among the Affy7 variants, Affy7-L9A and Affy7-Y6A CAR-T cells, which showed the least comparative reduction in binding affinity to c-MET (relative to the parental Affy7 CAR-T control cells) exhibited the greatest cytotoxicity against normal human lung (Tables 10A-10B; FIGS. **11**A-**11**B), liver (Tables 12A-12B; FIGS. **13**A-**13**B), kidney (Tables 14A-14B; FIGS. **15**A-**15**B), and colon cells (Tables 16A-16B; FIGS. **17**A-**17**B).

[0145] On the other hand, Affy7-Y6M CAR-T cells having a greater reduction in binding affinity to c-MET exhibited greatly reduced cytotoxicity toward normal human lung (Table 10B, FIG. **11**B) and colon cells (Table 16B, FIG. **17**B). Likewise, Affy7-Y6L CAR-T cells exhibited greatly reduced cytotoxicity toward normal human liver cells (Table 12B, FIG. **13**B), while Affy7-Y6V CAR-T cells exhibited greatly reduced cytotoxicity toward normal human lung (Table 10B, FIG. **11**B) and kidney cells (Table 14B, FIG. **15**B).

[0146] Among the Affy6 CAR-T cells variants, Affy6-I3F and I3G CAR-T cells were found to exhibit greatly reduced cytotoxicity toward normal human lung (Table 11B, FIG. **12**B), liver (Table 13B, FIG. **14**B), kidney (Table 15B, FIG. **16**B), and colon cells (Table 17B, FIG. **18**B).

TABLE 10A

Affy7 CAR-T Variants Co-cultured with Normal Human Lung Cells (FIG. 11A, Round 1)				
Time (hrs)	Affy7	Affy7_Y6A	Affy7_L9A	Negative
8	14.7	13.1	7.5	0
16	33.3	31.3	20.9	0
24	50.7	47.6	38.3	0
32	57.1	50.0	46.4	0
40	59.9	53.0	50.7	0
47	56.5	52.1	45.3	0

TABLE 10B

Affy7 CAR-T Variants Co-cultured with Normal Human Lung Cells (FIG. 11A, Round 2)

Time (hrs)	Affy7	Affy7_Y6A	Affy7_Y6V	Affy7_Y6M	Affy7_Y6L	Affy7_Y6I	Negative
8	31.8	30.6	18.2	9.9	21.1	23.1	6.7
16	35.7	32.6	18.0	9.7	23.6	27.5	5.4
24	43.2	40.1	15.4	6.3	22.8	26.9	4.1
32	55.8	48.3	17.9	6.9	25.4	29.7	11.6
40	67.5	56.5	16.7	9.5	25.9	29.9	9.7
48	76.0	64.0	21.5	13.8	30.1	30.9	16.5

TABLE 11A

Affy6 CAR-T Variants Co-cultured with Normal Human Lung Cells (FIG. 12A Round 1)

Time (hrs)	Affy6	Affy6-I3A	Negative
8	39.4	33.2	0
16	46.5	36.2	0
24	47.5	33.5	0
32	51.8	31.9	0
40	66.7	42.5	0
48	79.0	52.2	8.7

TABLE 11B

Affy6 CAR-T Variants Co-cultured with Normal Human Lung Cells (FIG. 12B, Round 2)

Time (hrs)	Affy6	I3W	I3F	I3G	Negative
8	38.4	14.3	17.1	14.8	4.3
16	47.1	20.9	16.4	17.3	2.6
24	54.7	30.6	19.1	19.5	8.9
32	63.1	34.0	18.0	18.4	8.8
40	76.4	41.2	29.4	28.0	4.3
48	85.2	42.7	34.8	33.1	4.2

TABLE 12A

Affy7 CAR-T Variants Co-Cultured with Normal Human Liver Cells (FIG. 13A, Round 1)

Time (hrs)	Affy7	Affy7_Y6A	Affy7_L9A	Negative
8	17.7	14.2	13.7	0
16	30.1	26.3	24.5	0
24	38.6	33.1	32.8	0
32	46.2	40.1	37.9	0
40	59.1	55.0	50.1	0
48	68.7	65.0	60.0	0

TABLE 12B

Affy7 CAR-T Variants Co-Cultured with Normal Human Liver Cells (FIG. 13B, Round 2)

Time (hrs)	Affy7	Affy7_Y6A	Affy7_Y6V	Affy7_Y6M	Affy7_Y6L	Affy7_Y6I	Negative
8	23.4	19.3	6.5	7.2	6.1	7.8	0
16	36.9	29.8	3.9	6.2	2.3	6.2	0
24	50.7	44.8	5.5	9.0	2.5	7.4	0
32	64.0	53.9	8.7	10.9	1.1	17.8	0
40	71.2	57.5	10.7	11.3	0	16.4	0
48	73.4	58.5	10.8	10.0	3.2	13.6	0

TABLE 13A

Affy6 CAR-T Variants Co-Cultured with Normal Human Liver Cells (FIG. 14A, Round 1)

Time (hrs)	Affy6	I3A	Negative
8	36.0	23.9	11.8
16	47.2	28.2	10.3
24	60.8	32.7	16.2
32	69.8	35.7	20.0
40	79.3	38.4	18.0
48	86.0	35.5	24.3

TABLE 13B

Affy6 CAR-T Variants Co-Cultured with Normal Human Liver Cells (FIG. 14B, Round 2)

Time (hrs)	Affy6	I3W	I3F	I3G	Negative
8	28.4	15.1	15.3	9.6	1.2
16	47.1	32.4	24.7	13.7	7.1
24	59.0	44.0	31.8	18.1	13.0
32	67.9	52.5	34.8	22.3	14.9
40	76.8	52.4	34.8	22.5	17.3
48	83.6	53.6	31.8	20.5	17.0

TABLE 14A

Affy7 CAR-T Variants Co-Cultured with Normal Human Kidney Cells (FIG. 15A, Round 1)

Time (hrs)	Affy7	Affy7_Y6A	Affy7_L9A	Negative
8	12.5	15.8	12.9	0
16	27.2	27.1	23.9	0
24	32.4	29.4	30.7	0
32	36.9	31.1	38.8	0
40	47.4	42.6	46.9	0
48	59.0	54.7	54.7	0

TABLE 14B

Affy7 CAR-T Variants Co-Cultured with Normal Human Kidney Cells (FIG. 15B, Round 2)							
Time (hrs)	Affy7	Affy7_Y6A	Affy7_Y6V	Affy7_Y6M	Affy7_Y6L	Affy7_Y6I	Negative
8	32.7	29.2	9.3	8.8	11.7	20.5	4.6
16	43.5	39.3	9.7	10.0	12.9	20.9	2.1
24	53.8	48.9	10.6	12.0	14.8	25.8	3.7
32	61.2	53.9	10.3	13.3	16.2	27.6	4.2
40	70.2	60.5	11.0	15.9	17.1	29.7	4.4
48	76.5	65.9	14.8	19.7	21.4	33.9	4.9

TABLE 15A

Affy6 CAR-T Variants Co-Cultured with Normal Human Kidney Cells (FIG. 16A, Round 1)			
Time (hrs)	Affy6	I3A	Negative
8	35.5	28.0	9.7
16	45.2	35.0	14.5
24	56.9	41.0	15.4
32	68.9	46.5	11.0
40	78.5	41.0	6.5
48	85.5	40.2	10.0

TABLE 15B

Affy6 CAR-T Variants Co-Cultured with Normal Human Kidney Cells (FIG. 16B, Round 2)					
Time (hrs)	Affy6	I3W	I3F	I3G	Negative
8	25.7	23.6	19.2	10.1	12.2
16	45.4	39.4	28.0	17.9	18.8
24	57.0	46.0	30.4	17.2	17.3
32	65.5	48.6	27.8	15.8	17.1
40	73.0	43.6	18.8	10.2	13.4
48	78.8	39.5	16.5	14.0	5.9

TABLE 16A

Affy7 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 17A, Round 1)				
Time (hrs)	Affy7	Affy7_Y6A	Affy7_L9A	Negative
8	13.8	13.1	12.1	0
16	29.3	25.0	25.0	0
24	34.9	25.6	27.2	0

TABLE 16A-continued

Affy7 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 17A, Round 1)				
Time (hrs)	Affy7	Affy7_Y6A	Affy7_L9A	Negative
32	42.8	35.8	38.2	0
40	54.1	45.4	48.5	0
48	60.1	45.6	49.6	0

TABLE 16B

Affy7 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 17B, Round 2)							
Time (hrs)	Affy7	Affy7_Y6A	Affy7_Y6V	Affy7_Y6M	Affy7_Y6L	Affy7_Y6I	Negative
8	28.5	26.5	9.4	1.4	12.1	15.1	0
16	33.9	36.3	13.5	1.7	16.4	18.8	0
24	43.2	42.5	14.0	1.4	13.6	20.8	0
32	57.0	51.5	18.2	7.1	19.0	28.9	0
40	69.7	62.7	20.3	1.8	20.0	32.5	5.4
48	79.4	70.7	22.2	3.6	17.2	36.4	4.3

TABLE 17A

Affy6 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 18A, Round 1)			
Time (hrs)	Affy6	I3A	Negative
8	31.9	19.4	2.7
16	39.2	22.5	4.4
24	50.3	30.9	7.8
32	57.8	31.8	6.2
40	65.7	31.1	5.2
48	76.8	30.0	2.9

TABLE 17B

Affy6 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 18B, Round 2)					
Time (hrs)	Affy6	I3W	I3F	I3G	Negative
8	24.6	18.8	13.2	8.2	8.9
16	37.7	22.7	19.2	14.4	13.7

TABLE 17B-continued

Affy6 CAR-T Variants Co-Cultured with Normal Human Colon Cells (FIG. 18B, Round 2)					
Time (hrs)	Affy6	I3W	I3F	I3G	Negative
24	46.4	27.0	22.4	16.4	16.6
32	57.3	29.9	13.3	11.4	13.6
40	72.3	30.9	1.6	2.0	10.0
48	83.8	38.6	8.3	10.7	14.1

LIST OF SEQUENCES

[0147] Exemplary amino acid sequences described herein are provided in Table 18 below:

TABLE 18

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
1	Affy7 HCDR1	SGYTFTSY
2	Affy7 HCDR2	LEWIGMIDPSNSDTRFN
3	Affy7-Y6A CDR3	AGSYVSPLDY
4	Affy7-L9A HCDR3	YGSYVSPADY
5	Affy7-Y6V HCDR3	VGSYVSPLDY
6	Affy7-Y6L HCDR3	LGSYVSPLDY
7	Affy7-Y6I HCDR3	IGSYVSPLDY
8	Affy7-Y6M HCDR3	MGSYVSPLDY
9	Affy7 LCDR1	QSLLYTSSQKN
10	Affy7 LCDR2	PKLLIYWASTRE
11	Affy7 LCDR3	YYAYP
12	Affy7 VH (wt)	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYCATYGSYVSPLDYWGQGTSVTVSS
13	Affy7-Y6A VH*	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYCATAGSYVSPLDYWGQGTSVTVSS
14	Affy7-L9A VH	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYCATYGSYVSPADYWGQGTSVTVSS
15	Affy7-Y6V VH	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYCATVGSYVSPLDYWGQGTSVTVSS
16	Affy7-Y6L VH	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYCATLGSYVSPLDYWGQGTSVTVSS
17	Affy7-Y6I VH	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYCATIGSYVSPLDYWGQGTSVTVSS
18	Affy7-Y6M VH	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTFNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYCATMGSYVSPLDYWGQGTSVTVSS

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
19	Affy7 VL (wt)	DIMMSQSPSSLTVSVGEKVTVSCKSSQSLLYTSSQKNYL AWYQQKPGQSPKLLIYWASTRESGVPDRFTGSGSGTDF TLTITSVKADDLAVYYCQQYYAYPWTFGGGTKLEIKR
20	Affy7 VH (wt)	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGGA TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG GACCGCTCGAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCCGTGTACTACTGT GCCACCTACGGCTCCTACGTGAGCCCTTTGGATTATTG GGGCCAGGGCACCTCCGTCACCGTGTCCTCC
21	Affy7-Y6A VH	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGGA TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG GACCGCTCGAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCCGTGTACTACTGT GCCACC**GCG**GGCTCCTACGTGAGCCCTTTGGATTATT GGGGCCAGGGCACCTCCGTCACCGTGTCCTCC
22	Affy7-L9A VH	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGGA TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG GACCGCTCGAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCCGTGTACTACTGT GCCACCTACGGCTCCTACGTGAGCCCT**GC**GGATTATT GGGGCCAGGGCACCTCCGTCACCGTGTCCTCC
23	Affy7-Y6V VH	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGGA TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG GACCGCTCGAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCCGTGTACTACTGT GCCACC**GT**CGGCTCCTACGTGAGCCCTTTGGATTATT GGGGCCAGGGCACCTCCGTCACCGTGTCCTCC
24	Affy7-Y6L VH	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGGA TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG GACCGCTCGAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCCGTGTACTACTGT GCCACC**CT**CGGCTCCTACGTGAGCCCTTTGGATTATTG GGGCCAGGGCACCTCCGTCACCGTGTCCTCC
25	Affy7-Y6I VH	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGGA TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG GACCGCTCGAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCCGTGTACTACTGT GCCACC**AT**CGGCTCCTACGTGAGCCCTTTGGATTATTG GGGCCAGGGCACCTCCGTCACCGTGTCCTCC

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
26	Affy7-Y6M VH	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACTG GGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGGA TCGGTATGATTGACCCGTCAAATAGTGACACCCGCTTC AACCCCAACTTCAAGGACAAGGCCACTCTCAACGTG GACCGCTCGAGCAACACGGCCTACATGCTGCTGTCC TCTCTGACCTCGGCGGACAGCGCCGTGTACTACTGT GCCACCATGGGCTCCTACGTGAGCCCTTTGGATTATT GGGGCCAGGGCACCTCCGTCACCGTGTCCTCC
27	Affy7 VL	GACATCATGATGTCTCAGAGTCCCTCTTCTCTGACC GTGTCCGTGGGCGAGAAGGTGACCGTGTCATGCAA GTCCTCACAGAGCCTTCTGTACACCAGCTCCCAGAAG AACTACCTGGCGTGGTACCAGCAGAAGCCCGGACA GAGCCCCAAGCTGCTGATCTATTGGGCTTCCACCCGC GAGAGCGGCGTCCCCGACCGCTTCACCGGCTCCGGC TCCGGGACCGACTTCACCCTGACCATCACCTCCGTG AAGGCCGATGACCTGGCCGTGTACTACTGTCAACA GTATTACGCCTACCCGTGGACCTTCGGAGGCGGCAC CAAGCTGGAGATCAAGCGC
28	Affy7 scFv (control)	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRENPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATYGSYVSPLDYWG QGTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR
29	Affy7-Y6A scFv	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATAGSYVSPLDYWG QGTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR
30	Affy7-L9A scFv	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATYGSYVSPADYWG QGTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR
31	Affy7-Y6V scFv	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRENPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATVGSYVSPLDYWG QGTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR
32	Affy7-Y6L scFv	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATLGSYVSPLDYWG QGTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR
33	Affy7-Y6I scFv	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCATIGSYVSPLDYWGQ GTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTVS VGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
34	Affy7-Y6M scFv	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRENPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCAT***M****GSYVSPLDY*WG QGTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKR
35	Affy7 scFv (control)	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC*TACGGCTCCTACGTGAGCCCTTTGGATT* *AT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCCG GTGGAGGCGGTTCAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGGACATCATGATGTCTCAGAGTCCCTCT TCTCTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCCTCACAGAGCCTTCTGTACACCAG CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCCTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCTACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC
36	Affy7-Y6A scFv	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC***GCG****GGCTCCTACGTGAGCCCTTTGGAT* *TAT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCC GGTGGAGGCGGTTCAGGCGGAGGTGGCTCTGGCGG TGGCGGATCGGACATCATGATGTCTCAGAGTCCCTC TTCTCTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCCTCACAGAGCCTTCTGTACACCAG CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCCTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCTACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC
37	Affy7-L9A scFv	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC*TACGGCTCCTACGTGAGCCCT****GC****GGAT* *TAT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCC GGTGGAGGCGGTTCAGGCGGAGGTGGCTCTGGCGG TGGCGGATCGGACATCATGATGTCTCAGAGTCCCTC TTCTCTGACCGTGTCCGTGGGCGAGAAGGTGACCGT

TABLE 18-continued

SEQ ID NO:	Name	Sequence
		GTCATGCAAGTCCTCACAGAGCCTTCTGTACACCAG CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCCTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCTACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC
38	Affy7-Y6V scFv	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC***GT****CGGCTCCTACGTGAGCCCTTTGGAT* *TAT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCC GGTGGAGGCGGTTCAGGCGGAGGTGGCTCTGGCGG TGGCGGATCGGACATCATGATGTCTCAGAGTCCCTC TTCTCTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCCTCACAGAGCCTTCTGTACACCAG CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCCTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCTACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC
39	Affy7-Y6L scFv	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC***CT****CGGCTCCTACGTGAGCCCTTTGGAT* *TAT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCC GGTGGAGGCGGTTCAGGCGGAGGTGGCTCTGGCGG TGGCGGATCGGACATCATGATGTCTCAGAGTCCCTC TTCTCTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCCTCACAGAGCCTTCTGTACACCAG CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCCTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCTACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC
40	Affy7-Y6I scFv	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC***AT****CGGCTCCTACGTGAGCCCTTTGGAT* *TAT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCC GGTGGAGGCGGTTCAGGCGGAGGTGGCTCTGGCGG TGGCGGATCGGACATCATGATGTCTCAGAGTCCCTC TTCTCTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCCTCACAGAGCCTTCTGTACACCAG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCCTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCTACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC
41	Affy7-Y6M scFv	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC***ATG****GGCTCCTACGTGAGCCCTTTGGAT* *TAT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCC GGTGGAGGCGGTTCAGGCGGAGGTGGCTCTGGCGG TGGCGGATCGGACATCATGATGTCTCAGAGTCCCTC TTCTCTGACCGTGTCCGTGGGCGAGAAGGTGACCGT GTCATGCAAGTCCTCACAGAGCCTTCTGTACACCAG CTCCCAGAAGAACTACCTGGCGTGGTACCAGCAGA AGCCCGGACAGAGCCCCAAGCTGCTGATCTATTGG GCTTCCACCCGCGAGAGCGGCGTCCCCGACCGCTTC ACCGGCTCCGGCTCCGGGACCGACTTCACCCTGACC ATCACCTCCGTGAAGGCCGATGACCTGGCCGTGTAC TACTGTCAACAGTATTACGCCTACCCGTGGACCTTC GGAGGCGGCACCAAGCTGGAGATCAAGCGC
42	Affy7 CAR (control)	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCAT*YGSYVSPLDY*WG QGTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKRASTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVVGGVLACYSLLVTVAFIIFWVRSKRSRLLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
43	Affy7-Y6A CAR	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCAT***A****GSYVSPLDY*WG QGTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKRASTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVVGGVLACYSLLVTVAFIIFWVRSKRSRLLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
44	Affy7-L9A CAR	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRENPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCAT*YGSYVSP****A****DY*WG QGTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKRASTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVVGGVLACYSLLVTVAFIIFWVRSKRSRLLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
45	Affy7-Y6V CAR	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCAT*VGSYVSPLDY*WG QGTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKRASTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVVGGVLACYSLLVTVAFIIFWVRSKRSRLLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
46	Affy7-Y6L CAR	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCAT*LGSYVSPLDY*WG QGTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKRASTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVVGGVLACYSLLVTVAFIIFWVRSKRSRLLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
47	Affy7-Y6I CAR	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRENPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCAT*IGSYVSPLDY*WGQ GTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTVS VGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKRASTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVVGGVLACYSLLVTVAFIIFWVRSKRSRLLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
48	Affy7-Y6M CAR	QVQLQQSGPELVRPGASVKMSCRASGYTFTSYWLHW VKQRPGQGLEWIGMIDPSNSDTRFNPNFKDKATLNVD RSSNTAYMLLSSLTSADSAVYYCAT*MGSYVSPLDY*WG QGTSVTVSSGGGGSGGGGSGGGGSDIMMSQSPSSLTV SVGEKVTVSCKSSQSLLYTSSQKNYLAWYQQKPGQSP KLLIYWASTRESGVPDRFTGSGSGTDFTLTITSVKADD LAVYYCQQYYAYPWTFGGGTKLEIKRASTTTPAPRPP TPAPTIASQPLSLRPEACRPAAGGAVHTRGLDFACDF WVLVVVGGVLACYSLLVTVAFIIFWVRSKRSRLLHSD YMNMTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFS RSADAPAYQQGQNQLYNELNLGRREEYDVLDKRRGR DPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMK GERRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
49	MycT-Affy7 CAR-SSTR2 (control)	GSEQKLISEEDLGSQVQLQQSGPELVRPGASVKMSCR ASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTR FNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYC AT*YGSYVSPLDY*WGQGTSVTVSSGGGGSGGGGSGGGG SDIMMSQSPSSLTVSVGEKVTVSCKSSQSLLYTSSQKN YLAWYQQKPGQSPKLLIYWASTRESGVPDRFTGSGSG TDFTLTITSVKADDLAVYYCQQYYAYPWTFGGGTKL EIKRASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGG AVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIF WVRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPR DFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQK DKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDT YDALHMQALPPRTSGSGATNFSLLKQAGDVEENPGPS

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		RMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEP YYDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKM KTITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGK AICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKS AKWRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICL CYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYA NSCANPILYAFLSDNFKKSFQNVLCLVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
50	MycT-Affy7-Y6A CAR-SSTR2	GSEQKLISEEDLGSQVQLQQSGPELVRPGASVKMSCR ASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTR FNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYC AT***A**GSYVSPLDY*WGQGTSVTVSSGGGGSGGGGSGGGG SDIMMSQSPSSLTVSVGEKVTVSCKSSQSLLYTSSQKN YLAWYQQKPGQSPKLLIYWASTRESGVPDRFTGSGSG TDFTLTITSVKADDLAVYYCQQYYAYPWTFGGGTKL EIKRASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGG AVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIF WVRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPR DFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQK DKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDT YDALHMQALPPRTSGSGATNFSLLKQAGDVEENPGPS RMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEP YYDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKM KTITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGK AICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKS AKWRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICL CYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYA NSCANPILYAFLSDNFKKSFQNVLCLVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
51	MycT-Affy7-L9A CAR-SSTR2	GSEQKLISEEDLGSQVQLQQSGPELVRPGASVKMSCR ASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTR FNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYC AT*YGSYVSP**A**DY*WGQGTSVTVSSGGGGSGGGGSGGGG SDIMMSQSPSSLTVSVGEKVTVSCKSSQSLLYTSSQKN YLAWYQQKPGQSPKLLIYWASTRESGVPDRFTGSGSG TDFTLTITSVKADDLAVYYCQQYYAYPWTFGGGTKL EIKRASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGG AVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIF WVRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPR DFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQK DKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDT YDALHMQALPPRTSGSGATNFSLLKQAGDVEENPGPS RMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEP YYDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKM KTITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGK AICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKS AKWRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICL CYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYA NSCANPILYAFLSDNFKKSFQNVLCLVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
52	MycT-Affy7-Y6V CAR-SSTR2	GSEQKLISEEDLGSQVQLQQSGPELVRPGASVKMSCR ASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTR FNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYC AT***V**GSYVSPLDY*WGQGTSVTVSSGGGGSGGGGSGGGG SDIMMSQSPSSLTVSVGEKVTVSCKSSQSLLYTSSQKN YLAWYQQKPGQSPKLLIYWASTRESGVPDRFTGSGSG TDFTLTITSVKADDLAVYYCQQYYAYPWTFGGGTKL EIKRASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGG AVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIF WVRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPR

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		DFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQK DKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDT YDALHMQALPPRTSGSGATNFSLLKQAGDVEENPGPS RMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEP YYDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKM KTITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGK AICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKS AKWRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICL CYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYA NSCANPILYAFLSDNFKKSFQNVLCLVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
53	MycT-Affy7-Y6L CAR-SSTR2	GSEQKLISEEDLGSQVQLQQSGPELVRPGASVKMSCR ASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTR FNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYC AT*LGSYVSPLDY*WGQGTSVTVSSGGGGSGGGGSGGGG SDIMMSQSPSSLTVSVGEKVTVSCKSSQSLLYTSSQKN YLAWYQQKPGQSPKLLIYWASTRESGVPDRFTGSGSG TDFTLTITSVKADDLAVYYCQQYYAYPWTFGGGTKL EIKRASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGG AVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIF WVRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPR DFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQK DKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDT YDALHMQALPPRTSGSGATNFSLLKQAGDVEENPGPS RMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEP YYDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKM KTITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGK AICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKS AKWRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICL CYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYA NSCANPILYAFLSDNFKKSFQNVLCLVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
54	MycT-Affy7-Y6I CAR-SSTR2	GSEQKLISEEDLGSQVQLQQSGPELVRPGASVKMSCR ASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTR FNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYC AT*IGSYVSPLDY*WGQGTSVTVSSGGGGSGGGGSGGGG SDIMMSQSPSSLTVSVGEKVTVSCKSSQSLLYTSSQKN YLAWYQQKPGQSPKLLIYWASTRESGVPDRFTGSGSG TDFTLTITSVKADDLAVYYCQQYYAYPWTFGGGTKL EIKRASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGG AVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIF WVRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPR DFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGR REEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQK DKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDT YDALHMQALPPRTSGSGATNFSLLKQAGDVEENPGPS RMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEP YYDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKM KTITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGK AICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKS AKWRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICL CYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYA NSCANPILYAFLSDNFKKSFQNVLCLVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
55	MycT-Affy7-Y6M CAR-SSTR2	GSEQKLISEEDLGSQVQLQQSGPELVRPGASVKMSCR ASGYTFTSYWLHWVKQRPGQGLEWIGMIDPSNSDTR FNPNFKDKATLNVDRSSNTAYMLLSSLTSADSAVYYC AT*MGSYVSPLDY*WGQGTSVTVSSGGGGSGGGGSGGG GSDIMMSQSPSSLTVSVGEKVTVSCKSSQSLLYTSSQK NYLAWYQQKPGQSPKLLIYWASTRESGVPDRFTGSGS

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		GTDFTLTITSVKADDLAVYYCQQYYAYPWTFGGGTK LEIKRASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAG GAVHTRGLDFACDFWVLVVVGGVLACYSLLVTVAFII FWVRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPP RDFAAYRSLRVKFSRSADAPAYQQGQNQLYNELNLG RREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQ KDKMAEAYSEIGMKGERRRGKGHDGLYQGLSTATK DTYDALHMQALPPRTSGSGATNFSLLKQAGDVEENP GPSRMDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQ TEPYYDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYA KMKTITNIYILNLAIADELFMLGLPFLAMQVALVHWP FGKAICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHP IKSAKWRRPRTAKMITMAVWGVSLLVILPIMIYAGLR SNQWGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTI ICLCYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVV AVFIFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVL TYANSCANPILYAFLSDNFKKSFQNVLCLVKVSGTDD GERSDSKQDKSRLNETTETQRTLLNGDLQTSI
56	Affy7 CAR (control)	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC*TACGGCTCCTACGTGAGCCCTTTGGATT* *AT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCCgg tggaggcggttcaggcggaggtggctctggcggtggcggatcgGACATCAT GATGTCTCAGAGTCCCTCTTCTCTGACCGTGTCCGT GGGCGAGAAGGTGACCGTGTCATGCAAGTCCTCAC AGAGCCTTCTGTACACCAGCTCCCAGAAGAACTAC CTGGCGTGGTACCAGCAGAAGCCCGGACAGAGCCC CAAGCTGCTGATCTATTGGGCTTCCACCCGCGAGAG CGGCGTCCCCGACCGCTTCACCGGCTCCGGCTCCGG GACCGACTTCACCCTGACCATCACCTCCGTGAAGGC CGATGACCTGGCCGTGTACTACTGTCAACAGTATTA CGCCTACCCGTGGACCTTCGGAGGCGGCACCAAGC TGGAGATCAAGCGCgctagcacgaccactccggcgccgcgcccaccg actccggccccaactatcgcgagccagcccctgtcgctgaggccggaagcatgccg ccctgccgccggaggtgctgtgcatacccggggattggacttcgcatgcgacTTT TGGGTGCTGGTGGTGGTTGGTGGAGTCCTGGCTTGC TATAGCTTGCTAGTAACAGTGGCCTTTATTATTTTCT GGGTGAGGAGTAAGAGGAGCAGGCTCCTGCACAGT GACTACATGAACATGACTCCCCGCCGCCCCGGGCC CACCCGCAAGCATTACCAGCCCTATGCCCCACCACG CGACTTCGCAGCCTATCGCTCCctgagagtgaagttcagcagga gcgcagacgcccccgcgtaccagcagggccagaaccagctctataacgagctcaa tctaggacgaagagaggagtacgatgttttggacaagagacgtggccgggaccctg agatgggggaaagccgagaaggaagaaccctcaggaaggcctgtacaatgaact gcagaaagataagatggcggaggcctacagtgagattgggatgaaaggcgagcgc cggaggggcaaggggcacgatggcctttaccagggtctcagtacagccaccaagg acacctacgacgcccttcacatgcaggccctgccccctcgc
57	Affy7-Y6A CAR	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC***GCG****GGCTCCTACGTGAGCCCTTTGGAT* *TAT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCC ggtggaggcggttcaggcggaggtggctctggcggtggcggatcgGACATC ATGATGTCTCAGAGTCCCTCTTCTCTGACCGTGTCC GTGGGCGAGAAGGTGACCGTGTCATGCAAGTCCTC ACAGAGCCTTCTGTACACCAGCTCCCAGAAGAACT ACCTGGCGTGGTACCAGCAGAAGCCCGGACAGAGC CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		AGCGGCGTCCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCTACCCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCgctagcacgaccactccggcgccgcgccca ccgactccggccccaactatcgcgagccagcccctgtcgctgaggccggaagcatg ccgccctgccgccggaggtgctgtgcatacccggggattggacttcgcatgcgacT TTTGGGTGCTGGTGGTGGTTGGTGGAGTCCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTTATTATTT TCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCCGCCGCCCCGG GCCCACCCGCAAGCATTACCAGCCCTATGCCCCACC ACGCGACTTCGCAGCCTATCGCTCCctgagagtgaagttcagc aggagcgcagacgcccccgcgtaccagcagggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgttttggacaagagacgtggccggga ccctgagatgggggaaagccgagaaggaagaaccctcaggaaggcctgtacaat gaactgcagaaagataagatggcggaggcctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggcctttaccagggtctcagtacagccac caaggacacctacgacgcccttcacatgcaggccctgccccctcgc
58	Affy7-L9A CAR	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC*TACGGCTCCTACGTGAGCCCT**GC**GGAT* *TAT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCC ggtggaggcggttcaggcggaggtggctctggcggtggcggatcgGACATC ATGATGTCTCAGAGTCCCTCTTCTCTGACCGTGTCC GTGGGCGAGAAGGTGACCGTGTCATGCAAGTCCTC ACAGAGCCTTCTGTACACCAGCTCCCAGAAGAACT ACCTGGCGTGGTACCAGCAGAAGCCCGGACAGAGC CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG AGCGGCGTCCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCTACCCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCgctagcacgaccactccggcgccgcgccca ccgactccggccccaactatcgcgagccagcccctgtcgctgaggccggaagcatg ccgccctgccgccggaggtgctgtgcatacccggggattggacttcgcatgcgacT TTTGGGTGCTGGTGGTGGTTGGTGGAGTCCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTTATTATTT TCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCCGCCGCCCCGG GCCCACCCGCAAGCATTACCAGCCCTATGCCCCACC ACGCGACTTCGCAGCCTATCGCTCCctgagagtgaagttcagc aggagcgcagacgcccccgcgtaccagcagggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgttttggacaagagacgtggccggga ccctgagatgggggaaagccgagaaggaagaaccctcaggaaggcctgtacaat gaactgcagaaagataagatggcggaggcctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggcctttaccagggtctcagtacagccac caaggacacctacgacgcccttcacatgcaggccctgccccctcgc
59	Affy7-Y6V CAR	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC***GT**CGGCTCCTACGTGAGCCCTTTGGAT* *TAT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCC ggtggaggcggttcaggcggaggtggctctggcggtggcggatcgGACATC ATGATGTCTCAGAGTCCCTCTTCTCTGACCGTGTCC GTGGGCGAGAAGGTGACCGTGTCATGCAAGTCCTC ACAGAGCCTTCTGTACACCAGCTCCCAGAAGAACT ACCTGGCGTGGTACCAGCAGAAGCCCGGACAGAGC

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG AGCGGCGTCCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCTACCCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCgctagcacgaccactccggcgccgcgccca ccgactccggccccaactatcgcgagccagcccctgtcgctgaggccggaagcatg ccgccctgccgccggaggtgctgtgcatacccggggattggacttcgcatgcgacT TTTGGGTGCTGGTGGTGGTTGGTGGAGTCCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTTATTATTT TCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCCGCCGCCCCGG GCCCACCCGCAAGCATTACCAGCCCTATGCCCCACC ACGCGACTTCGCAGCCTATCGCTCCctgagagtgaagttcagc aggagcgcagacgcccccgcgtaccagcagggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgttttggacaagagacgtggccggga ccctgagatgggggggaaagccgagaaggaagaaccctcaggaaggcctgtacaat gaactgcagaaagataagatggcggaggcctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggcctttaccagggtctcagtacagccac caaggacacctacgacgcccttcacatgcaggccctgccccctcgc
60	Affy7-Y6L CAR	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACCCT*CGGCTCCTACGTGAGCCCTTTGGAT* *TAT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCC ggtggaggcggttcaggcggaggtggctctggcggtggcggatcgGACATC ATGATGTCTCAGAGTCCCTCTTCTCTGACCGTGTCC GTGGGCGAGAAGGTGACCGTGTCATGCAAGTCCTC ACAGAGCCTTCTGTACACCAGCTCCCAGAAGAACT ACCTGGCGTGGTACCAGCAGAAGCCCGGACAGAGC CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG AGCGGCGTCCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCTACCCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCgctagcacgaccactccggcgccgcgccca ccgactccggccccaactatcgcgagccagcccctgtcgctgaggccggaagcatg ccgccctgccgccggaggtgctgtgcatacccggggattggacttcgcatgcgacT TTTGGGTGCTGGTGGTGGTTGGTGGAGTCCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTTATTATTT TCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCCGCCGCCCCGG GCCCACCCGCAAGCATTACCAGCCCTATGCCCCACC ACGCGACTTCGCAGCCTATCGCTCCctgagagtgaagttcagc aggagcgcagacgcccccgcgtaccagcagggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgttttggacaagagacgtggccggga ccctgagatgggggggaaagccgagaaggaagaaccctcaggaaggcctgtacaat gaactgcagaaagataagatggcggaggcctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggcctttaccagggtctcagtacagccac caaggacacctacgacgcccttcacatgcaggccctgccccctcgc
61	Affy7-Y6I CAR	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC***AT****CGGCTCCTACGTGAGCCCTTTGGAT* *TAT*TGGGGCCAGGGCACCTCCGTCACCGTGTCCTCC ggtggaggcggttcaggcggaggtggctctggcggtggcggatcgGACATC ATGATGTCTCAGAGTCCCTCTTCTCTGACCGTGTCC GTGGGCGAGAAGGTGACCGTGTCATGCAAGTCCTC ACAGAGCCTTCTGTACACCAGCTCCCAGAAGAACT

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		ACCTGGCGTGGTACCAGCAGAAGCCCGGACAGAGC CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG AGCGGCGTCCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCTACCCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCgctagcacgaccactccggcgccgcgccca ccgactccggccccaactatcgcgagccagcccctgtcgctgaggccggaagcatg ccgccctgccgccggaggtgctgtgcatacccggggattggacttcgcatgcgacT TTTGGGTGCTGGTGGTGGTTGGTGGAGTCCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTTATTATTT TCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCCGCCGCCCCGG GCCCACCCGCAAGCATTACCAGCCCTATGCCCCACC ACGCGACTTCGCAGCCTATCGCTCCctgagagtgaagttcagc aggagcgcagacgcccccgcgtaccagcagggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgttttggacaagagacgtggccggga ccctgagatggggggaaagccgagaaggaagaaccctcaggaaggcctgtacaat gaactgcagaaagataagatggcggaggcctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggcctttaccagggtctcagtacagccac caaggacacctacgacgcccttcacatgcaggccctgccccctcgc
62	Affy7-Y6M CAR	CAGGTGCAGCTCCAGCAGAGCGGCCCTGAGCTGGT GCGCCCCGGCGCTTCCGTGAAGATGTCATGCCGAG CTTCTGGCTACACGTTCACCAGCTATTGGCTACACT GGGTCAAGCAGAGGCCCGGACAGGGTCTGGAGTGG ATCGGTATGATTGACCCGTCAAATAGTGACACCCGC TTCAACCCCAACTTCAAGGACAAGGCCACTCTCAAC GTGGACCGCTCGAGCAACACGGCCTACATGCTGCT GTCCTCTCTGACCTCGGCGGACAGCGCCGTGTACTA CTGTGCCACC***AT***G*GGCTCCTACGTGAGCCCTTTGGAT* TATTGGGGCCAGGGCACCTCCGTCACCGTGTCCTCC ggtggaggcggttcaggcggaggtggctctggcggtggcggatcgGACATC ATGATGTCTCAGAGTCCCTCTTCTCTGACCGTGTCC GTGGGCGAGAAGGTGACCGTGTCATGCAAGTCCTC ACAGAGCCTTCTGTACACCAGCTCCCAGAAGAACT ACCTGGCGTGGTACCAGCAGAAGCCCGGACAGAGC CCCAAGCTGCTGATCTATTGGGCTTCCACCCGCGAG AGCGGCGTCCCCGACCGCTTCACCGGCTCCGGCTCC GGGACCGACTTCACCCTGACCATCACCTCCGTGAAG GCCGATGACCTGGCCGTGTACTACTGTCAACAGTAT TACGCCTACCCGTGGACCTTCGGAGGCGGCACCAA GCTGGAGATCAAGCGCgctagcacgaccactccggcgccgcgccca ccgactccggccccaactatcgcgagccagcccctgtcgctgaggccggaagcatg ccgccctgccgccggaggtgctgtgcatacccggggattggacttcgcatgcgacT TTTGGGTGCTGGTGGTGGTTGGTGGAGTCCTGGCTT GCTATAGCTTGCTAGTAACAGTGGCCTTTATTATTT TCTGGGTGAGGAGTAAGAGGAGCAGGCTCCTGCAC AGTGACTACATGAACATGACTCCCCGCCGCCCCGG GCCCACCCGCAAGCATTACCAGCCCTATGCCCCACC ACGCGACTTCGCAGCCTATCGCTCCctgagagtgaagttcagc aggagcgcagacgcccccgcgtaccagcagggccagaaccagctctataacgag ctcaatctaggacgaagagaggagtacgatgttttggacaagagacgtggccggga ccctgagatggggggaaagccgagaaggaagaaccctcaggaaggcctgtacaat gaactgcagaaagataagatggcggaggcctacagtgagattgggatgaaaggcg agcgccggaggggcaaggggcacgatggcctttaccagggtctcagtacagccac caaggacacctacgacgcccttcacatgcaggccctgccccctcgc
63	EF1α-MycT-Affy7 CAR-SSTR2 (control)	ggctccggtgcccgtcagtgggcagagcgcacatcgcccacagtccccgagaagtt ggggggaggggtcggcaattgaaccggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcgtgtactggctccgcctttttcccgagggtgggggagaacc gtatataagtgcagtagtcgccgtgaacgttctttttcgcaacgggtttgccgccagaa cacaggtaagtgccgtgtgtggttcccgcgggcctggcctctttacgggttatggccc ttgcgtgccttgaattacttccacctggctgcagtacgtgattcttgatcccgagcttcgg gttggaagtgggtgggagagttcgaggccttgcgcttaaggagccccttcgcctcgt gcttgagttgaggcctggcctgggcgctggggccgccgcgtgcgaatctggtggca ccttcgcgcctgtctcgctgctttcgataagtctctagccatttaaaatttttgatgacctg ctgcgacgctttttttctggcaagatagtcttgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggccgcgggcggcgacggggcccgtgcgtcccagcgcacat gttcggcgaggcggggcctgcgagcgcggccaccgagaatcggacgggggtagt ctcaagctggccggcctgctctggtgcctggtctcgcgccgccgtgtatcgccccgc

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		cctgggcggcaaggctggcccggtcggcaccagttgcgtgagcggaaagatggcc gcttcccggccctgctgcagggagctcaaaatggaggacgcggcgctcgggagag cgggcgggtgagtcacccacacaaaggaaaagggcctttccgtcctcagccgtcgc ttcatgtgactccacggagtaccgggcgccgtccaggcacctcgattagttctcgagc ttttggagtacgtcgtctttaggttggggggaggggttttatgcgatggagtttccccac actgagtgggtggagactgaagttaggccagcttggcacttgatgtaattctccttgga atttgccctttttgagtttggatcttggttcattctcaagcctcagacagtggttcaaagtt tttttcttccatttcaggtgtcgtgacaagtttgtacaaaaaagcaggctgccaccatggc cttaccagtgaccgccttgctcctgccgctggccttgctgctccacgccgccaggccg ggatccgaacaaaaactcatctcagaagaggatctgggatcgCAGGTGCAG CTCCAGCAGAGCGGCCCTGAGCTGGTGCGCCCCGG CGCTTCCGTGAAGATGTCATGCCGAGCTTCTGGCTA CACGTTCACCAGCTATTGGCTACACTGGGTCAAGCA GAGGCCCGGACAGGGTCTGGAGTGGATCGGTATGA TTGACCCGTCAAATAGTGACACCCGCTTCAACCCCA ACTTCAAGGACAAGGCCACTCTCAACGTGGACCGC TCGAGCAACACGGCCTACATGCTGCTGTCCTCTCTG ACCTCGGCGGACAGCGCCGTGTACTACTGTGCCACC *TACGGCTCCTACGTGAGCCCTTTGGATTAT*TGGGGCCA GGGCACCTCCGTCACCGTGTCCTCCggtggaggcggttcagg cggaggtggctctggcggtggcggatcgGACATCATGATGTCTCA GAGTCCCTCTTCTCTGACCGTGTCCGTGGGCGAGAA GGTGACCGTGTCATGCAAGTCCTCACAGAGCCTTCT GTACACCAGCTCCCAGAAGAACTACCTGGCGTGGT ACCAGCAGAAGCCCGGACAGAGCCCCAAGCTGCTG ATCTATTGGGCTTCCACCCGCGAGAGCGGCGTCCCC GACCGCTTCACCGGCTCCGGCTCCGGGACCGACTTC ACCCTGACCATCACCTCCGTGAAGGCCGATGACCTG GCCGTGTACTACTGTCAACAGTATTACGCCTACCCG TGGACCTTCGGAGGCGGCACCAAGCTGGAGATCAA GCGCgctagcacgaccactccggcgccgcgcccaccgactccggccccaacta tcgcgagccagcccctgtcgctgaggccggaagcatgccgccctgccgccggagg tgctgtgcatacccggggattggacttcgcatgcgacTTTTGGGTGCTGG TGGTGGTTGGTGGAGTCCTGGCTTGCTATAGCTTGC TAGTAACAGTGGCCTTTATTATTTTCTGGGTGAGGA GTAAGAGGAGCAGGCTCCTGCACAGTGACTACATG AACATGACTCCCCGCCGCCCCGGGCCCACCCGCAA GCATTACCAGCCCTATGCCCCACCACGCGACTTCGC AGCCTATCGCTCCctgagagtgaagttcagcaggagcgcagacgcccc cgcgtaccagcagggccagaaccagctctataacgagctcaatctaggacgaagag aggagtacgatgttttggacaagagacgtggccgggaccctgagatggggggaaa gccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaagataag atggcggaggcctacagtgagattgggatgaaaggcgagcgccggaggggcaag gggcacgatggcctttaccagggtctcagtacagccaccaaggacacctacgacgc ccttcacatgcaggccctgccccctcgcactagtggaagcggagctactaacttcag cctgctgaagcaggctggagacgtggaggagaaccctggaccttctagaatggaca tggcggatgagccactcaatggaagccacacatggctatccattccatttgacctcaat ggctctgtggtgtcaaccaacacctcaaaccagacagagccgtactatgacctgaca agcaatgcagtcctcacattcatctattttgtggtctgcatcattgggttgtgtggcaaca cacttgtcatttatgtcatcctccgctatgccaagatgaagaccatcaccaacatttacat cctcaacctggccatcgcagatgagctcttcatgctgggtctgcctttcttggctatgca ggtggctctggtccactggccctttggcaaggccatttgccgggtggtcatgactgtg gatggcatcaatcagttcaccagcatcttctgcctgacagtcatgagcatcgaccgata cctggctgtggtccaccccatcaagtcggccaagtggaggagaccccggacggcc aagatgatcaccatggctgtgtggggagtctctctgctggtcatcttgcccatcatgata tatgctgggctccggagcaaccagtgggggagaagcagctgcaccatcaactggcc aggtgaatctggggcttggtacacagggttcatcatctacactttcattctggggttcct ggtacccctcaccatcatctgtctttgctacctgttcattatcatcaaggtgaagtcctctg gaatccgagtgggctcctctaagaggaagaagtctgagaagaaggtcacccgaatg gtgtccatcgtggtggctgtcttcatcttctgctggcttcccttctacatattcaacgtttc ttccgtctccatggccatcagccccaccccagcccttaaaggcatgtttgactttgtggtg gtcctcacctatgctaacagctgtgccaaccctatcctatatgccttcttgtctgacaact tcaagaagagcttccagaatgtcctctgcttggtcaaggtgagcggcacagatgatgg ggagcggagtgacagtaagcaggacaaatcccggctgaatgagaccacggagac ccagaggaccctcctcaatggagacctccaaaccagtatctaa
64	EF1α-MycT-Affy7-Y6A CAR-SSTR2	ggctccggtgcccgtcagtgggcagagcgcacatcgcccacagtccccgagaagtt ggggggaggggtcggcaattgaaccggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcgtgtactggctccgcctttttcccgagggtgggggagaacc gtatataagtgcagtagtcgccgtgaacgttctttttcgcaacgggtttgccgccagaa cacaggtaagtgccgtgtgtggttcccgcgggcctggcctctttacgggttatggccc

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		ttgcgtgccttgaattacttccacctggctgcagtacgtgattcttgatcccgagcttcgg gttggaagtgggtgggagagttcgaggccttgcgcttaaggagccccttcgcctcgt gcttgagttgaggcctggcctgggcgctggggccgccgcgtgcgaatctggtggca ccttcgcgcctgtctcgctgctttcgataagtctctagccatttaaaatttttgatgacctg ctgcgacgctttttttctggcaagatagtcttgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggccgcgggcggcgacggggcccgtgcgtcccagcgcacat gttcggcgaggcggggcctgcgagcgcggccaccgagaatcggacgggggtagt ctcaagctggccggcctgctctggtgcctggtctcgcgccgccgtgtatcgccccgc cctgggcggcaaggctggcccggtcggcaccagttgcgtgagcggaaagatggcc gcttcccggccctgctgcagggagctcaaaatggaggacgcggcgctcgggagag cgggcgggtgagtcacccacacaaaggaaaagggcctttccgtcctcagccgtcgc ttcatgtgactccacggagtaccgggcgccgtccaggcacctcgattagttctcgagc ttttggagtacgtcgtctttaggttgggggaggggttttatgcgatggagtttccccac actgagtgggtggagactgaagttaggccagcttggcacttgatgtaattctccttgga atttgccctttttgagtttggatcttggttcattctcaagcctcagacagtggttcaaagtt tttttcttccatttcaggtgtcgtgacaagtttgtacaaaaaagcaggctgccaccatggc cttaccagtgaccgccttgctcctgccgctggccttgctgctccacgccgccaggccg ggatccgaacaaaaactcatctcagaagaggatctgggatcgCAGGTGCAG CTCCAGCAGAGCGGCCCTGAGCTGGTGCGCCCCGG CGCTTCCGTGAAGATGTCATGCCGAGCTTCTGGCTA CACGTTCACCAGCTATTGGCTACACTGGGTCAAGCA GAGGCCCGGACAGGGTCTGGAGTGGATCGGTATGA TTGACCCGTCAAATAGTGACACCCGCTTCAACCCCA ACTTCAAGGACAAGGCCACTCTCAACGTGGACCGC TCGAGCAACACGGCCTACATGCTGCTGTCCTCTCTG ACCTCGGCGGACAGCGCCGTGTACTACTGTGCCACC ***GCG***_GGCTCCTACGTGAGCCCTTTGGATTAT_TGGGGCC AGGGCACCTCCGTCACCGTGTCCTCCggtggaggcggttca ggcggaggtggctctggcggtggcggatcgGACATCATGATGTCTC AGAGTCCCTCTTCTCTGACCGTGTCCGTGGGCGAGA AGGTGACCGTGTCATGCAAGTCCTCACAGAGCCTTC TGTACACCAGCTCCCAGAAGAACTACCTGGCGTGG TACCAGCAGAAGCCCGGACAGAGCCCCAAGCTGCT GATCTATTGGGCTTCCACCCGCGAGAGCGGCGTCCC CGACCGCTTCACCGGCTCCGGCTCCGGGACCGACTT CACCCTGACCATCACCTCCGTGAAGGCCGATGACCT GGCCGTGTACTACTGTCAACAGTATTACGCCTACCC GTGGACCTTCGGAGGCGGCACCAAGCTGGAGATCA AGCGCgctagcacgaccactccggcgccgcgcccaccgactccggccccaac tatcgcgagccagcccctgtcgctgaggccggaagcatgccgccctgccgccgga ggtgctgtgcatacccggggattggacttcgcatgcgacTTTTGGGTGCTG GTGGTGGTTGGTGGAGTCCTGGCTTGCTATAGCTTG CTAGTAACAGTGGCCTTTATTATTTTCTGGGTGAGG AGTAAGAGGAGCAGGCTCCTGCACAGTGACTACAT GAACATGACTCCCCGCCGCCCCGGGCCCACCCGCA AGCATTACCAGCCCTATGCCCCACCACGCGACTTCG CAGCCTATCGCTCCctgagagtgaagttcagcaggagcgcagacgccc ccgcgtaccagcagggccagaaccagctctataacgagctcaatctaggacgaaga gaggagtacgatgttttggacaagagacgtggccgggaccctgagatggggggaa agccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaagataa gatggcggaggcctacagtgagattgggatgaaaggcgagcgccggaggggcaa ggggcacgatggcctttaccagggtctcagtacagccaccaaggacacctacgacg cccttcacatgcaggccctgccccctcgcactagtggaagcggagctactaacttca gcctgctgaagcaggctggagacgtggaggagaaccctggaccttctagaatggac atggcggatgagccactcaatggaagccacacatggctatccattccatttgacctca atggctctgtggtgtcaaccaacacctcaaaccagacagagccgtactatgacctgac aagcaatgcagtcctcacattcatctattttgtggtctgcatcattgggttgtgtggcaac acacttgtcatttatgtcatcctccgctatgccaagatgaagaccatcaccaacatttac atcctcaacctggccatcgcagatgagctcttcatgctgggtctgcctttcttggctatg caggtggctctggtccactggccctttggcaaggccatttgccgggtggtcatgactg tggatggcatcaatcagttcaccagcatcttctgcctgacagtcatgagcatcgaccga tacctggctgtggtccaccccatcaagtcggccaagtggaggagaccccggacggc caagatgatcaccatggctgtgtggggagtctctctgctggtcatcttgcccatcatgat atatgctgggctccggagcaaccagtgggggagaagcagctgcaccatcaactggc

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		caggtgaatctggggcttggtacacagggttcatcatctacactttcattctggggttcc tggtacccctcaccatcatctgtctttgctacctgttcattatcatcaaggtgaagtcctct ggaatccgagtgggctcctctaagaggaagaagtctgagaagaaggtcacccgaat ggtgtccatcgtggtggctgtcttcatcttctgctggcttcccttctacatattcaacgttt cttccgtctccatggccatcagccccaccccagcccttaaaggcatgtttgactttgtggt ggtcctcacctatgctaacagctgtgccaaccctatcctatatgccttcttgtctgacaac ttcaagaagagcttccagaatgtcctctgcttggtcaaggtgagcggcacagatgatg gggagcggagtgacagtaagcaggacaaatcccggctgaatgagaccacggaga cccagaggaccctcctcaatggagacctccaaaccagtatctaa
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TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		aagcaatgcagtcctcacattcatctattttgtggtctgcatcattgggttgtgtggcaac acacttgtcatttatgtcatcctccgctatgccaagatgaagaccatcaccaacatttac atcctcaacctggccatcgcagatgagctcttcatgctgggtctgcctttcttggctatg caggtggctctggtccactggccctttggcaaggccatttgccgggtggtcatgactg tggatggcatcaatcagttcaccagcatcttctgcctgacagtcatgagcatcgaccga tacctggctgtggtccaccccatcaagtcggccaagtggaggagaccccggacggc caagatgatcaccatggctgtgtggggagtctctctgctggtcatcttgcccatcatgat atatgctgggctccggagcaaccagtgggggagaagcagctgcaccatcaactggc caggtgaatctggggcttggtacacagggttcatcatctacactttcattctggggttcc tggtacccctcaccatcatctgtctttgctacctgttcattatcatcaaggtgaagtcctct ggaatccgagtgggctcctctaagaggaagaagtctgagaagaaggtcacccgaat ggtgtccatcgtggtggctgtcttcatcttctgctggcttcccttctacatattcaacgttt cttccgtctccatggccatcagccccaccccagcccttaaaggcatgtttgactttgtggt ggtcctcacctatgctaacagctgtgccaaccctatcctatatgccttcttgtctgacaac ttcaagaagagcttccagaatgtcctctgcttggtcaaggtgagcggcacagatgatg gggagcggagtgacagtaagcaggacaaatcccggctgaatgagaccacggaga cccagaggaccctcctcaatggagacctccaaaccagtatctaa
66	EF1α-MycT-Affy7-Y6V CAR-SSTR2	ggctccggtgcccgtcagtgggcagagcgcacatcgcccacagtccccgagaagtt ggggggaggggtcggcaattgaaccggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcgtgtactggctccgcctttttcccgagggtgggggagaacc gtatataagtgcagtagtcgccgtgaacgttctttttcgcaacgggtttgccgccagaa cacaggtaagtgccgtgtgtggttcccgcgggcctggcctctttacgggttatggccc ttgcgtgccttgaattacttccacctggctgcagtacgtgattcttgatcccgagcttcgg gttggaagtgggtgggagagttcgaggccttgcgcttaaggagccccttcgcctcgt gcttgagttgaggcctggcctgggcgctggggccgccgcgtgcgaatctggtggca ccttcgcgcctgtctcgctgctttcgataagtctctagccatttaaaatttttgatgacctg ctgcgacgctttttttctggcaagatagtcttgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggccgcgggcggcgacggggcccgtgcgtcccagcgcacat gttcggcgaggcggggcctgcgagcgcggccaccgagaatcggacgggggtagt ctcaagctggccggcctgctctggtgcctggtctcgcgccgccgtgtatcgccccgc cctgggcggcaaggctggcccggtcggcaccagttgcgtgagcggaaagatggcc gcttcccggccctgctgcagggagctcaaaatggaggacgcggcgctcgggagag cgggcgggtgagtcacccacacaaaggaaaagggcctttccgtcctcagccgtcgc ttcatgtgactccacggagtaccgggcgccgtccaggcacctcgattagttctcgagc ttttggagtacgtcgtctttaggttggggggaggggttttatgcgatggagtttccccac actgagtgggtggagactgaagttaggccagcttggcacttgatgtaattctccttgga atttgccctttttgagtttggatcttggttcattctcaagcctcagacagtggttcaaagtt tttttcttccatttcaggtgtcgtgacaagtttgtacaaaaaagcaggctgccaccatggc cttaccagtgaccgccttgctcctgccgctggccttgctgctccacgccgccaggccg ggatccgaacaaaaactcatctcagaagaggatctgggatcgCAGGTGCAG CTCCAGCAGAGCGGCCCTGAGCTGGTGCGCCCCGG CGCTTCCGTGAAGATGTCATGCCGAGCTTCTGGCTA CACGTTCACCAGCTATTGGCTACACTGGGTCAAGCA GAGGCCCGGACAGGGTCTGGAGTGGATCGGTATGA TTGACCCGTCAAATAGTGACACCCGCTTCAACCCCA ACTTCAAGGACAAGGCCACTCTCAACGTGGACCGC TCGAGCAACACGGCCTACATGCTGCTGTCCTCTCTG ACCTCGGCGGACAGCGCCGTGTACTACTGTGCCACC ***GT****CGGCTCCTACGTGAGCCCTTTGGATTAT*TGGGGCC AGGGCACCTCCGTCACCGTGTCCTCCggtggaggcggttca ggcggaggtggctctggcggtggcggatcgGACATCATGATGTCTC AGAGTCCCTCTTCTCTGACCGTGTCCGTGGGCGAGA AGGTGACCGTGTCATGCAAGTCCTCACAGAGCCTTC TGTACACCAGCTCCCAGAAGAACTACCTGGCGTGG TACCAGCAGAAGCCCGGACAGAGCCCCAAGCTGCT GATCTATTGGGCTTCCACCCGCGAGAGCGGCGTCCC CGACCGCTTCACCGGCTCCGGCTCCGGGACCGACTT CACCCTGACCATCACCTCCGTGAAGGCCGATGACCT GGCCGTGTACTACTGTCAACAGTATTACGCCTACCC GTGGACCTTCGGAGGCGGCACCAAGCTGGAGATCA AGCGCgctagcacgaccactccggcgccgcgcccaccgactccggccccaac tatcgcgagccagcccctgtcgctgaggccggaagcatgccgccctgccgccgga ggtgctgtgcatacccggggattggacttcgcatgcgacTTTTGGGTGCTG GTGGTGGTTGGTGGAGTCCTGGCTTGCTATAGCTTG CTAGTAACAGTGGCCTTTATTATTTTCTGGGTGAGG AGTAAGAGGAGCAGGCTCCTGCACAGTGACTACAT GAACATGACTCCCCGCCGCCCCGGGCCCACCCGCA AGCATTACCAGCCCTATGCCCCACCACGCGACTTCG CAGCCTATCGCTCCctgagagtgaagttcagcaggagcgcagacgccc ccgcgtaccagcagggccagaaccagctctataacgagctcaatctaggacgaaga

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
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TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		TACCAGCAGAAGCCCGGACAGAGCCCCAAGCTGCT GATCTATTGGGCTTCCACCCGCGAGAGCGGCGTCCC CGACCGCTTCACCGGCTCCGGCTCCGGGACCGACTT CACCCTGACCATCACCTCCGTGAAGGCCGATGACCT GGCCGTGTACTACTGTCAACAGTATTACGCCTACCC GTGGACCTTCGGAGGCGGCACCAAGCTGGAGATCA AGCGCgctagcacgaccactccggcgccgcgcccaccgactccggccccaac tatcgcgagccagcccctgtcgctgaggccggaagcatgccgccctgccgccgga ggtgctgtgcatacccggggattggacttcgcatgcgacTTTTGGGTGCTG GTGGTGGTTGGTGGAGTCCTGGCTTGCTATAGCTTG CTAGTAACAGTGGCCTTTATTATTTTCTGGGTGAGG AGTAAGAGGAGCAGGCTCCTGCACAGTGACTACAT GAACATGACTCCCCGCCGCCCCGGGCCCACCCGCA AGCATTACCAGCCCTATGCCCCACCACGCGACTTCG CAGCCTATCGCTCCctgagagtgaagttcagcaggagcgcagacgccc ccgcgtaccagcagggccagaaccagctctataacgagctcaatctaggacgaaga gaggagtacgatgttttggacaagagacgtggccgggaccctgagatggggggaa agccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaagataa gatggcggaggcctacagtgagattgggatgaaaggcgagcgccggaggggcaa ggggcacgatggcctttaccagggtctcagtacagccaccaaggacacctacgacg cccttcacatgcaggccctgccccctcgcactagtggaagcggagctactaacttca gcctgctgaagcaggctggagacgtggaggagaaccctggaccttctagaatggac atggcggatgagccactcaatggaagccacacatggctatccattccatttgacctca atggctctgtggtgtcaaccaacacctcaaaccagacagagccgtactatgacctgac aagcaatgcagtcctcacattcatctattttgtggtctgcatcattgggttgtgtggcaac acacttgtcatttatgtcatcctccgctatgccaagatgaagaccatcaccaacatttac atcctcaacctggccatcgcagatgagctcttcatgctgggtctgcctttcttggctatg caggtggctctggtccactggccctttggcaaggccatttgccgggtggtcatgactg tggatggcatcaatcagttcaccagcatcttctgcctgacagtcatgagcatcgaccga tacctggctgtggtccaccccatcaagtcggccaagtggaggagaccccggacggc caagatgatcaccatggctgtgtggggagtctctctgctggtcatcttgcccatcatgat atatgctgggctccggagcaaccagtgggggagaagcagctgcaccatcaactggc caggtgaatctggggcttggtacacagggttcatcatctacactttcattctggggttcc tggtacccctcaccatcatctgtctttgctacctgttcattatcatcaaggtgaagtcctct ggaatccgagtgggctcctctaagaggaagaagtctgagaagaaggtcacccgaat ggtgtccatcgtggtggctgtcttcatcttctgctggcttcccttctacatattcaacgtttc ttccgtctccatggccatcagccccaccccagcccttaaaggcatgtttgactttgtggt ggtcctcacctatgctaacagctgtgccaaccctatcctatatgccttcttgtctgacaac ttcaagaagagcttccagaatgtcctctgcttggtcaaggtgagcggcacagatgatg gggagcggagtgacagtaagcaggacaaatcccggctgaatgagaccacggaga cccagaggaccctcctcaatggagacctccaaaccagtatctaa
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TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		TCGAGCAACACGGCCTACATGCTGCTGTCCTCTCTG ACCTCGGCGGACAGCGCCGTGTACTACTGTGCCACC ***AT****CGGCTCCTACGTGAGCCCTTTGGATTAT*TGGGGCC AGGGCACCTCCGTCACCGTGTCCTCCggtggaggcggttca ggcggaggtggctctggcggtggcggatcgGACATCATGATGTCTC AGAGTCCCTCTTCTCTGACCGTGTCCGTGGGCGAGA AGGTGACCGTGTCATGCAAGTCCTCACAGAGCCTTC TGTACACCAGCTCCCAGAAGAACTACCTGGCGTGG TACCAGCAGAAGCCCGGACAGAGCCCCAAGCTGCT GATCTATTGGGCTTCCACCCGCGAGAGCGGCGTCCC CGACCGCTTCACCGGCTCCGGCTCCGGGACCGACTT CACCCTGACCATCACCTCCGTGAAGGCCGATGACCT GGCCGTGTACTACTGTCAACAGTATTACGCCTACCC GTGGACCTTCGGAGGCGGCACCAAGCTGGAGATCA AGCGCgctagcacgaccactccggcgccgcgcccaccgactccggccccaac tatcgcgagccagcccctgtcgctgaggccggaagcatgccgccctgccgccgga ggtgctgtgcatacccggggattggacttcgcatgcgacTTTTGGGTGCTG GTGGTGGTTGGTGGAGTCCTGGCTTGCTATAGCTTG CTAGTAACAGTGGCCTTTATTATTTTCTGGGTGAGG AGTAAGAGGAGCAGGCTCCTGCACAGTGACTACAT GAACATGACTCCCCGCCGCCCCGGGCCCACCCGCA AGCATTACCAGCCCTATGCCCCACCACGCGACTTCG CAGCCTATCGCTCCctgagagtgaagttcagcaggagcgcagacgccc ccgcgtaccagcagggccagaaccagctctataacgagctcaatctaggacgaaga gaggagtacgatgttttggacaagagacgtggccgggaccctgagatggggggaa agccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaagataa gatggcggaggcctacagtgagattgggatgaaaggcgagcgccggaggggcaa ggggcacgatggcctttaccagggtctcagtacagccaccaaggacacctacgacg cccttcacatgcaggccctgccccctcgcactagtggaagcggagctactaacttca gcctgctgaagcaggctggagacgtggaggagaaccctggaccttctagaatggac atggcggatgagccactcaatggaagccacacatggctatccattccatttgacctca atggctctgtggtgtcaaccaacacctcaaaccagacagagccgtactatgacctgac aagcaatgcagtcctcacattcatctattttgtggtctgcatcattgggttgtgtggcaac acacttgtcatttatgtcatcctccgctatgccaagatgaagaccatcaccaacatttac atcctcaacctggccatcgcagatgagctcttcatgctgggtctgcctttcttggctatg caggtggctctggtccactggccctttggcaaggccatttgccgggtggtcatgactg tggatggcatcaatcagttcaccagcatcttctgcctgacagtcatgagcatcgaccga tacctggctgtggtccaccccatcaagtcggccaagtggaggagaccccggacggc caagatgatcaccatggctgtgtggggagtctctctgctggtcatcttgcccatcatgat atatgctgggctccggagcaaccagtgggggagaagcagctgcaccatcaactggc caggtgaatctggggcttggtacacagggttcatcatctacactttcattctggggttcc tggtacccctcaccatcatctgtctttgctacctgttcattatcatcaaggtgaagtcctct ggaatccgagtgggctcctctaagaggaagaagtctgagaagaaggtcacccgaat ggtgtccatcgtggtggctgtcttcatcttctgctggcttcccttctacatattcaacgtttc ttccgtctccatggccatcagccccaccccagcccttaaaggcatgtttgactttgtggt ggtcctcacctatgctaacagctgtgccaaccctatcctatatgccttcttgtctgacaac ttcaagaagagcttccagaatgtcctctgcttggtcaaggtgagcggcacagatgatg gggagcggagtgacagtaagcaggacaaatcccggctgaatgagaccacggaga cccagaggaccctcctcaatggagacctccaaaccagtatctaa
69	EF1α-MycT-Affy7-Y6M CAR-SSTR2	ggctccggtgcccgtcagtgggcagagcgcacatcgcccacagtccccgagaagtt ggggggaggggtcggcaattgaaccggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcgtgtactggctccgcctttttcccgagggtgggggagaacc gtatataagtgcagtagtcgccgtgaacgttctttttcgcaacgggtttgccgccagaa cacaggtaagtgccgtgtgtggttcccgcgggcctggcctctttacgggttatggccc ttgcgtgccttgaattacttccacctggctgcagtacgtgattcttgatcccgagcttcgg gttggaagtgggtgggagagttcgaggccttgcgcttaaggagccccttcgcctcgt gcttgagttgaggcctggcctgggcgctggggccgccgcgtgcgaatctggtggca ccttcgcgcctgtctcgctgctttcgataagtctctagccatttaaaatttttgatgacctg ctgcgacgctttttttctggcaagatagtcttgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggccgcgggcggcgacggggcccgtgcgtcccagcgcacat gttcggcgaggcggggcctgcgagcgcggccaccgagaatcggacgggggtagt ctcaagctggccggcctgctctggtgcctggtctcgcgccgccgtgtatcgccccgc cctgggcggcaaggctggcccggtcggcaccagttgcgtgagcggaaagatggcc gcttcccggccctgctgcagggagctcaaaatggaggacgcggcgctcgggagag

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		cgggcgggtgagtcacccacacaaaggaaaagggcctttccgtcctcagccgtcgc ttcatgtgactccacggagtaccgggcgccgtccaggcacctcgattagttctcgagc ttttggagtacgtcgtctttaggttggggggaggggttttatgcgatggagtttccccac actgagtgggtggagactgaagttaggccagcttggcacttgatgtaattctccttgga atttgcccttttttgagtttggatcttggttcattctcaagcctcagacagtggttcaaagttt ttttcttccatttcaggtgtcgtgacaagtttgtacaaaaaagcaggctgccaccatggc cttaccagtgaccgccttgctcctgccgctggccttgctgctccacgccgccaggccg ggatccgaacaaaaactcatctcagaagaggatctgggatcgCAGGTGCAG CTCCAGCAGAGCGGCCCTGAGCTGGTGCGCCCCGG CGCTTCCGTGAAGATGTCATGCCGAGCTTCTGGCTA CACGTTCACCAGCTATTGGCTACACTGGGTCAAGCA GAGGCCCGGACAGGGTCTGGAGTGGATCGGTATGA TTGACCCGTCAAATAGTGACACCCGCTTCAACCCCA ACTTCAAGGACAAGGCCACTCTCAACGTGGACCGC TCGAGCAACACGGCCTACATGCTGCTGTCCTCTCTG ACCTCGGCGGACAGCGCCGTGTACTACTGTGCCACC ***ATG****GGCTCCTACGTGAGCCCTTTGGATTAT*TGGGGCC AGGGCACCTCCGTCACCGTGTCCTCCggtggaggcggttca ggcggaggtggctctggcggtggcggatcgGACATCATGATGTCTC AGAGTCCCTCTTCTCTGACCGTGTCCGTGGGCGAGA AGGTGACCGTGTCATGCAAGTCCTCACAGAGCCTTC TGTACACCAGCTCCCAGAAGAACTACCTGGCGTGG TACCAGCAGAAGCCCGGACAGAGCCCCAAGCTGCT GATCTATTGGGCTTCCACCCGCGAGAGCGGCGTCCC CGACCGCTTCACCGGCTCCGGCTCCGGGACCGACTT CACCCTGACCATCACCTCCGTGAAGGCCGATGACCT GGCCGTGTACTACTGTCAACAGTATTACGCCTACCC GTGGACCTTCGGAGGCGGCACCAAGCTGGAGATCA AGCGCgctagcacgaccactccggcgccgcgcccaccgactccggccccaac tatcgcgagccagcccctgtcgctgaggccggaagcatgccgccctgccgccgga ggtgctgtgcatacccggggattggacttcgcatgcgacTTTTGGGTGCTG GTGGTGGTTGGTGGAGTCCTGGCTTGCTATAGCTTG CTAGTAACAGTGGCCTTTATTATTTTCTGGGTGAGG AGTAAGAGGAGCAGGCTCCTGCACAGTGACTACAT GAACATGACTCCCCGCCGCCCCGGGCCCACCCGCA AGCATTACCAGCCCTATGCCCCACCACGCGACTTCG CAGCCTATCGCTCCctgagagtgaagttcagcaggagcgcagacgccc ccgcgtaccagcagggccagaaccagctctataacgagctcaatctaggacgaaga gaggagtacgatgttttggacaagagacgtggccgggaccctgagatggggggaa agccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaagataa gatggcggaggcctacagtgagattgggatgaaaggcgagcgccggaggggcaa ggggcacgatggcctttaccagggtctcagtacagccaccaaggacacctacgacg cccttcacatgcaggccctgccccctcgcactagtggaagcggagctactaacttca gcctgctgaagcaggctggagacgtggaggagaaccctggaccttctagaatggac atggcggatgagccactcaatggaagccacacatggctatccattccatttgacctca atggctctgtggtgtcaaccaacacctcaaaccagacagagccgtactatgacctgac aagcaatgcagtcctcacattcatctattttgtggtctgcatcattgggttgtgtggcaac acacttgtcatttatgtcatcctccgctatgccaagatgaagaccatcaccaacatttac atcctcaacctggccatcgcagatgagctcttcatgctgggtctgcctttcttggctatg caggtggctctggtccactggccctttggcaaggccatttgccgggtggtcatgactg tggatggcatcaatcagttcaccagcatcttctgcctgacagtcatgagcatcgaccga tacctggctgtggtccaccccatcaagtcggccaagtggaggagaccccggacggc caagatgatcaccatggctgtgtggggagtctctctgctggtcatcttgcccatcatgat atatgctgggctccggagcaaccagtgggggagaagcagctgcaccatcaactggc caggtgaatctggggcttggtacacagggttcatcatctacactttcattctggggttcc tggtacccctcaccatcatctgtctttgctacctgttcattatcatcaaggtgaagtcctct ggaatccgagtgggctcctctaagaggaagaagtctgagaagaaggtcacccgaat ggtgtccatcgtggtggctgtcttcatcttctgctggcttcccttctacatattcaacgtttc ttccgtctccatggccatcagccccaccccagcccttaaaggcatgtttgactttgtggt ggtcctcacctatgctaacagctgtgccaaccctatcctatatgccttcttgtctgacaac ttcaagaagagcttccagaatgtcctctgcttggtcaaggtgagcggcacagatgatg gggagcggagtgacagtaagcaggacaaatcccggctgaatgagaccacggaga cccagaggaccctcctcaatggagacctccaaaccagtatctaa
70	Affy6 HCDR1	GYIFTAY
71	Affy6 HCDR2	LEWMGWIKPNNGLANY
72	Affy6-I3A HCDR3	SE**A**TTEFDY
73	Affy6-I3W HCDR3	SE**W**TTEFDY

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
74	Affy6-I3F HCDR3	SE**F**TTEFDY
75	Affy6-I3G HCDR3	SE**G**TTEFDY
76	Affy6 LDR1	ESVDSY
77	Affy6 LDR2	PKLLIYRASTRE
78	Affy6 LDR3	SKEDPL
79	Affy6 VH (wt)	QVQLVQSGAEVKKPGASVKVSCKAS*GYIFTAY*TMHW VRQAPGQG*LEWMGWIKPNNGLANY*AQKFQGRVTMTR DTSISTAYMELSRLRSDDTAVYYCAR*SEITTEFDY*WGQ GTLVTVSS
80	Affy6-I3A VH	QVQLVQSGAEVKKPGASVKVSCKAS*GYIFTAY*TMHW VRQAPGQG*LEWMGWIKPNNGLANY*AQKFQGRVTMTR DTSISTAYMELSRLRSDDTAVYYCAR*SEATTEFDY*WG QGTLVTVSS
81	Affy6-I3W VH	QVQLVQSGAEVKKPGASVKVSCKAS*GYIFTAY*TMHW VRQAPGQG*LEWMGWIKPNNGLANY*AQKFQGRVTMTR DTSISTAYMELSRLRSDDTAVYYCAR*SEWTTEFDY*WG QGTLVTVSS
82	Affy6-I3F VH	QVQLVQSGAEVKKPGASVKVSCKAS*GYIFTAY*TMHW VRQAPGQG*LEWMGWIKPNNGLANY*AQKFQGRVTMTR DTSISTAYMELSRLRSDDTAVYYCAR*SEFTTEFDY*WG QGTLVTVSS
83	Affy6-I3G VH	QVQLVQSGAEVKKPGASVKVSCKAS*GYIFTAY*TMHW VRQAPGQG*LEWMGWIKPNNGLANY*AQKFQGRVTMTR DTSISTAYMELSRLRSDDTAVYYCAR*SEGTTEFDY*WG QGTLVTVSS
84	Affy6 VL (wt)	DIVLTQSPDSLAVSLGERATINCKSS*ESVDSY*ANSFMH WYQQKPGQPP*KLLIYRASTRE*SGVPDRFSGSGSRTDFTL TISSLQAEDVAVYYCQQ*SKEDPL*TFGGGTKVEIK
85	Affy6 VH (wt)	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCT*GGCTACATCTTCACCGCCTAC*ACCATGCACTG GGTGCGCCAGGCCCCGGGACAGGGG*CTGGAGTGGA* *TGGGCTGGATCAAGCCTAACAACGGTCTGGCCAACTAC* *GCG*CAGAAGTTCCAGGGCCGCGTGACCATGACTCG GGACACCAGCATCTCTACCGCGTACATGGAGCTGTC CCGCCTGCGCTCCGATGACACGGCCGTGTACTACTG CGCCCGC*TCGGAGATCACCACCGAGTTCGACTAT*TGG GGCCAGGGCACCCTGGTGACCGTGTCGTCC
86	Affy6-I3A VH	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCT*GGCTACATCTTCACCGCCTAC*ACCATGCACTG GGTGCGCCAGGCCCCGGGACAGGGG*CTGGAGTGGA* *TGGGCTGGATCAAGCCTAACAACGGTCTGGCCAACTAC* *GCG*CAGAAGTTCCAGGGCCGCGTGACCATGACTCG GGACACCAGCATCTCTACCGCGTACATGGAGCTGTC CCGCCTGCGCTCCGATGACACGGCCGTGTACTACTG CGCCCGC*TCGGAG**GC**CACCACCGAGTTCGACTAT*TGG GGCCAGGGCACCCTGGTGACCGTGTCGTCC

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
87	Affy6-I3W VH	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACTG GGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGGA TGGGCTGGATCAAGCCTAACAACGGTCTGGCCAACTAC GCGCAGAAGTTCCAGGGCCGCGTGACCATGACTCG GGACACCAGCATCTCTACCGCGTACATGGAGCTGTC CCGCCTGCGCTCCGATGACACGGCCGTGTACTACTG CGCCCGCTCGGAGTGGACCACCGAGTTCGACTATTGG GGCCAGGGCACCCTGGTGACCGTGTCGTCC
88	Affy6-I3F VH	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACTG GGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGGA TGGGCTGGATCAAGCCTAACAACGGTCTGGCCAACTAC GCGCAGAAGTTCCAGGGCCGCGTGACCATGACTCG GGACACCAGCATCTCTACCGCGTACATGGAGCTGTC CCGCCTGCGCTCCGATGACACGGCCGTGTACTACTG CGCCCGCTCGGAGTTCACCACCGAGTTCGACTATTGG GGCCAGGGCACCCTGGTGACCGTGTCGTCC
89	Affy6-I3G VH	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACTG GGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGGA TGGGCTGGATCAAGCCTAACAACGGTCTGGCCAACTAC GCGCAGAAGTTCCAGGGCCGCGTGACCATGACTCG GGACACCAGCATCTCTACCGCGTACATGGAGCTGTC CCGCCTGCGCTCCGATGACACGGCCGTGTACTACTG CGCCCGCTCGGAGGGCACCACCGAGTTCGACTATTGG GGCCAGGGCACCCTGGTGACCGTGTCGTCC
90	Affy6 VL (wt)	GACATCGTGCTGACCCAGAGCCCTGACAGCCTGGC CGTGTCCCTGGGCGAGCGCGCCACCATTAACTGCA AGAGCTCCGAGAGTGTCGATAGCTACGCCAACTCCTT CATGCACTGGTACCAGCAGAAGCCCGGCCAGCCAC CCAAGCTGCTGATCTACAGGGCTTCCACCCGCGAGAG CGGCGTCCCCGACAGGTTTTCAGGTTCTGGCTCCCG CACCGACTTCACCCTGACCATCTCTTCTCTGCAGGC CGAGGATGTGGCGGTGTACTACTGTCAACAGTCCAA GGAGGACCCGCTTACCTTCGGCGGCGGCACCAAGGT GGAGATCAAG
91	Affy6 scFv (control)	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMT RDTSISTAYMELSRLRSDDTAVYYCARSEITTEFDYWG QGTLVTVSSGGGGSGGGGSGGGGSDIVLTQSPDSLAV SLGERATINCKSSESVDSYANSFMHWYQQKPGQPPKL LIYRASTRESGVPDRFSGSGSRTDFTLTISSLQAEDVAV YYCQQSKEDPLTFGGGTKVEIK
92	Affy6-I3A scFv	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMT RDTSISTAYMELSRLRSDDTAVYYCARSEATTEFDYW GQGTLVTVSSGGGGSGGGGSGGGGSDIVLTQSPDSLA VSLGERATINCKSSESVDSYANSFMHWYQQKPGQPPK LLIYRASTRESGVPDRFSGSGSRTDFTLTISSLQAEDVA VYYCQQSKEDPLTFGGGTKVEIK
93	Affy6-I3W scFv	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMT RDTSISTAYMELSRLRSDDTAVYYCARSEWTTEFDYW GQGTLVTVSSGGGGSGGGGSGGGGSDIVLTQSPDSLA

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		VSLGERATINCKSSESVDSYANSFMHWYQQKPGQPPK LLIYRASTRESGVPDRFSGSGSRTDFTLTISSLQAEDVA VYYCQQSKEDPLTFGGGTKVEIK
94	Affy6-I3F scFv	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMT RDTSISTAYMELSRLRSDDTAVYYCAR*SE**F**TTEFDY*W GQGTLVTVSSGGGGSGGGGSGGGGSDIVLTQSPDSLA VSLGERATINCKSSESVDSYANSFMHWYQQKPGQPPK LLIYRASTRESGVPDRFSGSGSRTDFTLTISSLQAEDVA VYYCQQSKEDPLTFGGGTKVEIK
95	Affy6-I3G scFv	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMT RDTSISTAYMELSRLRSDDTAVYYCAR*SE**G**TTEFDY*W GQGTLVTVSSGGGGSGGGGSGGGGSDIVLTQSPDSLA VSLGERATINCKSSESVDSYANSFMHWYQQKPGQPPK LLIYRASTRESGVPDRFSGSGSRTDFTLTISSLQAEDVA VYYCQQSKEDPLTFGGGTKVEIK
96	Affy6 scFv (control)	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACAACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGC*TCGGAGATCACCACCGAGTTCGACT* *AT*TGGGGCCAGGGCACCCTGGTGACCGTGTCGTCCG GTGGAGGCGGTTCAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGGACATCGTGCTGACCCAGAGCCCTGA CAGCCTGGCCGTGTCCCTGGGCGAGCGCGCCACCA TTAACTGCAAGAGCTCCGAGAGTGTCGATAGCTAC GCCAACTCCTTCATGCACTGGTACCAGCAGAAGCCC GGCCAGCCACCCAAGCTGCTGATCTACAGGGCTTCC ACCCGCGAGAGCGGCGTCCCCGACAGGTTTTCAGG TTCTGGCTCCCGCACCGACTTCACCCTGACCATCTC TTCTCTGCAGGCCGAGGATGTGGCGGTGTACTACTG TCAACAGTCCAAGGAGGACCCGCTTACCTTCGGCG GCGGCACCAAGGTGGAGATCAAG
97	Affy6-I3A scFv	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACAACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGC*TCGGAG**GC**CACCACCGAGTTCGACT* *AT*TGGGGCCAGGGCACCCTGGTGACCGTGTCGTCCG GTGGAGGCGGTTCAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGGACATCGTGCTGACCCAGAGCCCTGA CAGCCTGGCCGTGTCCCTGGGCGAGCGCGCCACCA TTAACTGCAAGAGCTCCGAGAGTGTCGATAGCTAC GCCAACTCCTTCATGCACTGGTACCAGCAGAAGCCC GGCCAGCCACCCAAGCTGCTGATCTACAGGGCTTCC ACCCGCGAGAGCGGCGTCCCCGACAGGTTTTCAGG TTCTGGCTCCCGCACCGACTTCACCCTGACCATCTC TTCTCTGCAGGCCGAGGATGTGGCGGTGTACTACTG TCAACAGTCCAAGGAGGACCCGCTTACCTTCGGCG GCGGCACCAAGGTGGAGATCAAG
98	Affy6-I3W scFv	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACAACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGTGGACCACCGAGTTCGACT ATTGGGGCCAGGGCACCCTGGTGACCGTGTCGTCCG GTGGAGGCGGTTCAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGGACATCGTGCTGACCCAGAGCCCTGA CAGCCTGGCCGTGTCCCTGGGCGAGCGCGCCACCA TTAACTGCAAGAGCTCCGAGAGTGTCGATAGCTAC GCCAACTCCTTCATGCACTGGTACCAGCAGAAGCCC GGCCAGCCACCCAAGCTGCTGATCTACAGGGCTTCC ACCCGCGAGAGCGGCGTCCCCGACAGGTTTTCAGG TTCTGGCTCCCGCACCGACTTCACCCTGACCATCTC TTCTCTGCAGGCCGAGGATGTGGCGGTGTACTACTG TCAACAGTCCAAGGAGGACCCGCTTACCTTCGGCG GCGGCACCAAGGTGGAGATCAAG
99	Affy6-I3F scFv	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACAACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGTTCACCACCGAGTTCGACT ATTGGGGCCAGGGCACCCTGGTGACCGTGTCGTCCG GTGGAGGCGGTTCAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGGACATCGTGCTGACCCAGAGCCCTGA CAGCCTGGCCGTGTCCCTGGGCGAGCGCGCCACCA TTAACTGCAAGAGCTCCGAGAGTGTCGATAGCTAC GCCAACTCCTTCATGCACTGGTACCAGCAGAAGCCC GGCCAGCCACCCAAGCTGCTGATCTACAGGGCTTCC ACCCGCGAGAGCGGCGTCCCCGACAGGTTTTCAGG TTCTGGCTCCCGCACCGACTTCACCCTGACCATCTC TTCTCTGCAGGCCGAGGATGTGGCGGTGTACTACTG TCAACAGTCCAAGGAGGACCCGCTTACCTTCGGCG GCGGCACCAAGGTGGAGATCAAG
100	Affy6-I3G scFv	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACAACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGGGCACCACCGAGTTCGACT ATTGGGGCCAGGGCACCCTGGTGACCGTGTCGTCCG GTGGAGGCGGTTCAGGCGGAGGTGGCTCTGGCGGT GGCGGATCGGACATCGTGCTGACCCAGAGCCCTGA CAGCCTGGCCGTGTCCCTGGGCGAGCGCGCCACCA TTAACTGCAAGAGCTCCGAGAGTGTCGATAGCTAC GCCAACTCCTTCATGCACTGGTACCAGCAGAAGCCC GGCCAGCCACCCAAGCTGCTGATCTACAGGGCTTCC ACCCGCGAGAGCGGCGTCCCCGACAGGTTTTCAGG TTCTGGCTCCCGCACCGACTTCACCCTGACCATCTC TTCTCTGCAGGCCGAGGATGTGGCGGTGTACTACTG TCAACAGTCCAAGGAGGACCCGCTTACCTTCGGCG GCGGCACCAAGGTGGAGATCAAG

TABLE 18-continued

SEQ ID NO:	Name	Sequence
	Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein	
101	Affy6 CAR (control)	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMT RDTSISTAYMELSRLRSDDTAVYYCAR*SEITTEFDY*WG QGTLVTVSSGGGGSGGGGSGGGGSDIVLTQSPDSLAV SLGERATINCKSSESVDSYANSFMHWYQQKPGQPPKL LIYRASTRESGVPDRFSGSGSRTDFTLTISSLQAEDVAV YYCQQSKEDPLTFGGGTKVEIKASTTTPAPRPPTPAPTI ASQPLSLRPEACRPAAGGAVHTRGLDFACDFWVLVV VGGVLACYSLLVTVAFIIFWVRSKRSRLLHSDYMNMT PRRPGPTRKHYQPYAPPRDFAAYRSLRVKFSRSADAP AYQQGQNQLYNELNLGRREEYDVLDKRRGRDPEMG GKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRR GKGHDGLYQGLSTATKDTYDALHMQALPPR
102	Affy6-I3A CAR	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMT RDTSISTAYMELSRLRSDDTAVYYCAR*SE**A**TTEFDY*W GQGTLVTVSSGGGGSGGGGSGGGGSDIVLTQSPDSLA VSLGERATINCKSSESVDSYANSFMHWYQQKPGQPPK LLIYRASTRESGVPDRFSGSGSRTDFTLTISSLQAEDVA VYYCQQSKEDPLTFGGGTKVEIKASTTTPAPRPPTPAP TIASQPLSLRPEACRPAAGGAVHTRGLDFACDFWVLV VVGGVLACYSLLVTVAFIIFWVRSKRSRLLHSDYMN MTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFSRSAD APAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPE MGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGE RRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
103	Affy6-I3W CAR	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMT RDTSISTAYMELSRLRSDDTAVYYCAR*SE**W**TTEFDY*W GQGTLVTVSSGGGGSGGGGSGGGGSDIVLTQSPDSLA VSLGERATINCKSSESVDSYANSFMHWYQQKPGQPPK LLIYRASTRESGVPDRFSGSGSRTDFTLTISSLQAEDVA VYYCQQSKEDPLTFGGGTKVEIKASTTTPAPRPPTPAP TIASQPLSLRPEACRPAAGGAVHTRGLDFACDFWVLV VVGGVLACYSLLVTVAFIIFWVRSKRSRLLHSDYMN MTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFSRSAD APAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPE MGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGE RRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
104	Affy6-I3F CAR	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMT RDTSISTAYMELSRLRSDDTAVYYCAR*SE**F**TTEFDY*W GQGTLVTVSSGGGGSGGGGSGGGGSDIVLTQSPDSLA VSLGERATINCKSSESVDSYANSFMHWYQQKPGQPPK LLIYRASTRESGVPDRFSGSGSRTDFTLTISSLQAEDVA VYYCQQSKEDPLTFGGGTKVEIKASTTTPAPRPPTPAP TIASQPLSLRPEACRPAAGGAVHTRGLDFACDFWVLV VVGGVLACYSLLVTVAFIIFWVRSKRSRLLHSDYMN MTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFSRSAD APAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPE MGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGE RRRGKGHDGLYQGLSTATKDTYDALHMQALPPR
105	Affy6-I3G CAR	QVQLVQSGAEVKKPGASVKVSCKASGYIFTAYTMHW VRQAPGQGLEWMGWIKPNNGLANYAQKFQGRVTMT RDTSISTAYMELSRLRSDDTAVYYCAR*SE**G**TTEFDY*W GQGTLVTVSSGGGGSGGGGSGGGGSDIVLTQSPDSLA VSLGERATINCKSSESVDSYANSFMHWYQQKPGQPPK LLIYRASTRESGVPDRFSGSGSRTDFTLTISSLQAEDVA VYYCQQSKEDPLTFGGGTKVEIKASTTTPAPRPPTPAP TIASQPLSLRPEACRPAAGGAVHTRGLDFACDFWVLV VVGGVLACYSLLVTVAFIIFWVRSKRSRLLHSDYMN MTPRRPGPTRKHYQPYAPPRDFAAYRSLRVKFSRSAD APAYQQGQNQLYNELNLGRREEYDVLDKRRGRDPE MGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGE RRRGKGHDGLYQGLSTATKDTYDALHMQALPPR

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
106	MycT-Affy6 CAR-SSTR2 (control)	GSEQKLISEEDLGSQVQLVQSGAEVKKPGASVKVSCK ASGYIFTAYTMHWVRQAPGQGLEWMGWIKPNNGLA NYAQKFQGRVTMTRDTSISTAYMELSRLRSDDTAVY YCAR*SEITTEFDY*WGQGTLVTVSSGGGGSGGGGSGGG GSDIVLTQSPDSLAVSLGERATINCKSSESVDSYANSF MHWYQQKPGQPPKLLIYRASTRESGVPDRFSGSGSRT DFTLTISSLQAEDVAVYYCQQSKEDPLTFGGGTKVEIK ASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHT RGLDFACDFWVLVVVGGVLACYSLLVTVAFIIFWVRS KRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPRDFAA YRSLRVKFSRSADAPAYQQGQNQLYNELNLGRREEY DVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKM AEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTYDA LHMQALPPRTSGSGATNFSLLKQAGDVEENPGPSRMD MADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPYYD LTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKMKTIT NIYILNLAIADELFMLGLPFLAMQVALVHWPFGKAIC RVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKSAK WRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQW GRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICLCY LFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVFIF CWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYAN SCANPILYAFLSDNFKKSFQNVLCLVKVSGTDDGERS DSKQDKSRLNETTETQRTLLNGDLQTSI
107	MycTAffy6-I3A CAR-SSTR2	GSEQKLISEEDLGSQVQLVQSGAEVKKPGASVKVSCK ASGYIFTAYTMHWVRQAPGQGLEWMGWIKPNNGLA NYAQKFQGRVTMTRDTSISTAYMELSRLRSDDTAVY YCAR*SE**A**TTEFDY*WGQGTLVTVSSGGGGSGGGGSGG GGSDIVLTQSPDSLAVSLGERATINCKSSESVDSYANS FMHWYQQKPGQPPKLLIYRASTRESGVPDRFSGSGSR TDFTLTISSLQAEDVAVYYCQQSKEDPLTFGGGTKVEI KASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAV HTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIFW VRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPRDF AAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGRRE EYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKD KMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTY DALHMQALPPRTSGSGATNFSLLKQAGDVEENPGPSR MDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPY YDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKMK TITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGKA ICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKSA KWRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICL CYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYA NSCANPILYAFLSDNFKKSFQNVLCLVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
108	MycT-Affy6-I3W CAR-SSTR2	GSEQKLISEEDLGSQVQLVQSGAEVKKPGASVKVSCK ASGYIFTAYTMHWVRQAPGQGLEWMGWIKPNNGLA NYAQKFQGRVTMTRDTSISTAYMELSRLRSDDTAVY YCAR*SE**W**TTEFDY*WGQGTLVTVSSGGGGSGGGGSGG GGSDIVLTQSPDSLAVSLGERATINCKSSESVDSYANS FMHWYQQKPGQPPKLLIYRASTRESGVPDRFSGSGSR TDFTLTISSLQAEDVAVYYCQQSKEDPLTFGGGTKVEI KASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAV HTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIFW VRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPRDF AAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGRRE EYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKD KMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTY DALHMQALPPRTSGSGATNFSLLKQAGDVEENPGPSR MDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPY YDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKMK TITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGKA ICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKSA KWRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICL CYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		IFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYA NSCANPILYAFLSDNFKKSFQNVLCLVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
109	MycT-Affy6-I3F CAR-SSTR2	GSEQKLISEEDLGSQVQLVQSGAEVKKPGASVKVSCK ASGYIFTAYTMHWVRQAPGQGLEWMGWIKPNNGLA NYAQKFQGRVTMTRDTSISTAYMELSRLRSDDTAVY YCAR*SEFTTEFDY*WGQGTLVTVSSGGGGSGGGGSGG GGSDIVLTQSPDSLAVSLGERATINCKSSESVDSYANS FMHWYQQKPGQPPKLLIYRASTRESGVPDRFSGSGSR TDFTLTISSLQAEDVAVYYCQQSKEDPLTFGGGTKVEI KASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAV HTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIFW VRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPRDF AAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGRRE EYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKD KMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTY DALHMQALPPRTSGSGATNFSLLKQAGDVEENPGPSR MDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPY YDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKMK TITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGKA ICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKSA KWRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICL CYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYA NSCANPILYAFLSDNFKKSFQNVLCLVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
110	MycT-Affy6-I3G CAR-SSTR2	GSEQKLISEEDLGSQVQLVQSGAEVKKPGASVKVSCK ASGYIFTAYTMHWVRQAPGQGLEWMGWIKPNNGLA NYAQKFQGRVTMTRDTSISTAYMELSRLRSDDTAVY YCAR*SEGTTEFDY*WGQGTLVTVSSGGGGSGGGGSGG GGSDIVLTQSPDSLAVSLGERATINCKSSESVDSYANS FMHWYQQKPGQPPKLLIYRASTRESGVPDRFSGSGSR TDFTLTISSLQAEDVAVYYCQQSKEDPLTFGGGTKVEI KASTTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAV HTRGLDFACDFWVLVVVGGVLACYSLLVTVAFIIFW VRSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPRDF AAYRSLRVKFSRSADAPAYQQGQNQLYNELNLGRRE EYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKD KMAEAYSEIGMKGERRRGKGHDGLYQGLSTATKDTY DALHMQALPPRTSGSGATNFSLLKQAGDVEENPGPSR MDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPY YDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKMK TITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGKA ICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKSA KWRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICL CYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYA NSCANPILYAFLSDNFKKSFQNVLCLVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
111	Affy6 CAR (control)	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACAACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGC*TCGGAGATCACCACCGAGTTCGACT* *AT*TGGGGCCAGGGCACCCTGGTGACCGTGTCGTCCg gtggaggcggttcaggcggaggtggctctggcggtggcggatcgGACATCG TGCTGACCCAGAGCCCTGACAGCCTGGCCGTGTCCC TGGGCGAGCGCGCCACCATTAACTGCAAGAGCTCC GAGAGTGTCGATAGCTACGCCAACTCCTTCATGCAC TGGTACCAGCAGAAGCCCGGCCAGCCACCCAAGCT GCTGATCTACAGGGCTTCCACCCGCGAGAGCGGCG TCCCCGACAGGTTTTCAGGTTCTGGCTCCCGCACCG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		ACTTCACCCTGACCATCTCTTCTCTGCAGGCCGAGG ATGTGGCGGTGTACTACTGTCAACAGTCCAAGGAG GACCCGCTTACCTTCGGCGGCGGCACCAAGGTGGA GATCAAGgctagcacgaccactccggcgccgcgcccaccgactccggcccc aactatcgcgagccagcccctgtcgctgaggccggaagcatgccgccctgccgcc ggaggtgctgtgcatacccggggattggacttcgcatgcgacTTTTGGGTG CTGGTGGTGGTTGGTGGAGTCCTGGCTTGCTATAGC TTGCTAGTAACAGTGGCCTTTATTATTTTCTGGGTG AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTA CATGAACATGACTCCCCGCCGCCCCGGGCCCACCC GCAAGCATTACCAGCCCTATGCCCCACCACGCGACT TCGCAGCCTATCGCTCCctgagagtgaagttcagcaggagcgcaga cgcccccgcgtaccagcagggccagaaccagctctataacgagctcaatctaggac gaagagaggagtacgatgttttggacaagagacgtggccgggaccctgagatggg gggaaagccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaa gataagatggcggaggcctacagtgagattgggatgaaaggcgagcgccggagg ggcaaggggcacgatggcctttaccagggtctcagtacagccaccaaggacaccta cgacgcccttcacatgcaggccctgccccctcgc
112	Affy6-I3A CAR	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACAACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGGCCACCACCGAGTTCGACT ATTGGGGCCAGGGCACCCTGGTGACCGTGTCGTCCg gtggaggcggttcaggcggaggtggctctggcggtggcggatcgGACATCG TGCTGACCCAGAGCCCTGACAGCCTGGCCGTGTCCC TGGGCGAGCGCGCCACCATTAACTGCAAGAGCTCC GAGAGTGTCGATAGCTACGCCAACTCCTTCATGCAC TGGTACCAGCAGAAGCCCGGCCAGCCACCCAAGCT GCTGATCTACAGGGCTTCCACCCGCGAGAGCGGCG TCCCCGACAGGTTTTCAGGTTCTGGCTCCCGCACCG ACTTCACCCTGACCATCTCTTCTCTGCAGGCCGAGG ATGTGGCGGTGTACTACTGTCAACAGTCCAAGGAG GACCCGCTTACCTTCGGCGGCGGCACCAAGGTGGA GATCAAGgctagcacgaccactccggcgccgcgcccaccgactccggcccc aactatcgcgagccagcccctgtcgctgaggccggaagcatgccgccctgccgcc ggaggtgctgtgcatacccggggattggacttcgcatgcgacTTTTGGGTG CTGGTGGTGGTTGGTGGAGTCCTGGCTTGCTATAGC TTGCTAGTAACAGTGGCCTTTATTATTTTCTGGGTG AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTA CATGAACATGACTCCCCGCCGCCCCGGGCCCACCC GCAAGCATTACCAGCCCTATGCCCCACCACGCGACT TCGCAGCCTATCGCTCCctgagagtgaagttcagcaggagcgcaga cgcccccgcgtaccagcagggccagaaccagctctataacgagctcaatctaggac gaagagaggagtacgatgttttggacaagagacgtggccgggaccctgagatggg gggaaagccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaa gataagatggcggaggcctacagtgagattgggatgaaaggcgagcgccggagg ggcaaggggcacgatggcctttaccagggtctcagtacagccaccaaggacaccta cgacgcccttcacatgcaggccctgccccctcgc
113	Affy6-I3W CAR	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACAACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGCTCGGAGTGGACCACCGAGTTCGACT ATTGGGGCCAGGGCACCCTGGTGACCGTGTCGTCCg gtggaggcggttcaggcggaggtggctctggcggtggcggatcgGACATCG TGCTGACCCAGAGCCCTGACAGCCTGGCCGTGTCCC TGGGCGAGCGCGCCACCATTAACTGCAAGAGCTCC GAGAGTGTCGATAGCTACGCCAACTCCTTCATGCAC TGGTACCAGCAGAAGCCCGGCCAGCCACCCAAGCT GCTGATCTACAGGGCTTCCACCCGCGAGAGCGGCG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		TCCCCGACAGGTTTTCAGGTTCTGGCTCCCGCACCG ACTTCACCCTGACCATCTCTTCTCTGCAGGCCGAGG ATGTGGCGGTGTACTACTGTCAACAGTCCAAGGAG GACCCGCTTACCTTCGGCGGCGGCACCAAGGTGGA GATCAAGgctagcacgaccactccggcgccgcgcccaccgactccggcccc aactatcgcgagccagcccctgtcgctgaggccggaagcatgccgccctgccgcc ggaggtgctgtgcatacccggggattggacttcgcatgcgacTTTTGGGTG CTGGTGGTGGTTGGTGGAGTCCTGGCTTGCTATAGC TTGCTAGTAACAGTGGCCTTTATTATTTTCTGGGTG AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTA CATGAACATGACTCCCCGCCGCCCCGGGCCCACCC GCAAGCATTACCAGCCCTATGCCCCACCACGCGACT TCGCAGCCTATCGCTCCctgagagtgaagttcagcaggagcgcaga cgcccccgcgtaccagcagggccagaaccagctctataacgagctcaatctaggac gaagagaggagtacgatgttttggacaagagacgtggccgggaccctgagatggg gggaaagccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaa gataagatggcggaggcctacagtgagattgggatgaaaggcgagcgccggagg ggcaaggggcacgatggcctttaccagggtctcagtacagccaccaaggacaccta cgacgcccttcacatgcaggccctgccccctcgc
114	Affy6-I3F CAR	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACAACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGC<u>*TCGGAG***T***TCACCACCGAGTTCGACT*</u> <u>*AT*</u>TGGGGCCAGGGCACCCTGGTGACCGTGTCGTCCg gtggaggcggttcaggcggaggtggctctggcggtggcggatcgGACATCG TGCTGACCCAGAGCCCTGACAGCCTGGCCGTGTCCC TGGGCGAGCGCGCCACCATTAACTGCAAGAGCTCC GAGAGTGTCGATAGCTACGCCAACTCCTTCATGCAC TGGTACCAGCAGAAGCCCGGCCAGCCACCCAAGCT GCTGATCTACAGGGCTTCCACCCGCGAGAGCGGCG TCCCCGACAGGTTTTCAGGTTCTGGCTCCCGCACCG ACTTCACCCTGACCATCTCTTCTCTGCAGGCCGAGG ATGTGGCGGTGTACTACTGTCAACAGTCCAAGGAG GACCCGCTTACCTTCGGCGGCGGCACCAAGGTGGA GATCAAGgctagcacgaccactccggcgccgcgcccaccgactccggcccc aactatcgcgagccagcccctgtcgctgaggccggaagcatgccgccctgccgcc ggaggtgctgtgcatacccggggattggacttcgcatgcgacTTTTGGGTG CTGGTGGTGGTTGGTGGAGTCCTGGCTTGCTATAGC TTGCTAGTAACAGTGGCCTTTATTATTTTCTGGGTG AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTA CATGAACATGACTCCCCGCCGCCCCGGGCCCACCC GCAAGCATTACCAGCCCTATGCCCCACCACGCGACT TCGCAGCCTATCGCTCCctgagagtgaagttcagcaggagcgcaga cgcccccgcgtaccagcagggccagaaccagctctataacgagctcaatctaggac gaagagaggagtacgatgttttggacaagagacgtggccgggaccctgagatggg gggaaagccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaa gataagatggcggaggcctacagtgagattgggatgaaaggcgagcgccggagg ggcaaggggcacgatggcctttaccagggtctcagtacagccaccaaggacaccta cgacgcccttcacatgcaggccctgccccctcgc
115	Affy6-I3G CAR	CAGGTGCAGCTCGTGCAGAGCGGCGCCGAGGTGAA GAAGCCCGGCGCTTCCGTGAAGGTGTCCTGTAAGG CCTCTGGCTACATCTTCACCGCCTACACCATGCACT GGGTGCGCCAGGCCCCGGGACAGGGGCTGGAGTGG ATGGGCTGGATCAAGCCTAACAACGGTCTGGCCAA CTACGCGCAGAAGTTCCAGGGCCGCGTGACCATGA CTCGGGACACCAGCATCTCTACCGCGTACATGGAG CTGTCCCGCCTGCGCTCCGATGACACGGCCGTGTAC TACTGCGCCCGC<u>*TCGGAGGGCACCACCGAGTTCGACT*</u> <u>*AT*</u>TGGGGCCAGGGCACCCTGGTGACCGTGTCGTCCg gtggaggcggttcaggcggaggtggctctggcggtggcggatcgGACATCG TGCTGACCCAGAGCCCTGACAGCCTGGCCGTGTCCC TGGGCGAGCGCGCCACCATTAACTGCAAGAGCTCC GAGAGTGTCGATAGCTACGCCAACTCCTTCATGCAC TGGTACCAGCAGAAGCCCGGCCAGCCACCCAAGCT

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		GCTGATCTACAGGGCTTCCACCCGCGAGAGCGGCG TCCCCGACAGGTTTTCAGGTTCTGGCTCCCGCACCG ACTTCACCCTGACCATCTCTTCTCTGCAGGCCGAGG ATGTGGCGGTGTACTACTGTCAACAGTCCAAGGAG GACCCGCTTACCTTCGGCGGCGGCACCAAGGTGGA GATCAAGgctagcacgaccactccggcgccgcgcccaccgactccggcccc aactatcgcgagccagcccctgtcgctgaggccggaagcatgccgccctgccgcc ggaggtgctgtgcatacccggggattggacttcgcatgcgacTTTTGGGTG CTGGTGGTGGTTGGTGGAGTCCTGGCTTGCTATAGC TTGCTAGTAACAGTGGCCTTTATTATTTTCTGGGTG AGGAGTAAGAGGAGCAGGCTCCTGCACAGTGACTA CATGAACATGACTCCCCGCCGCCCCGGGCCCACCC GCAAGCATTACCAGCCCTATGCCCCACCACGCGACT TCGCAGCCTATCGCTCCctgagagtgaagttcagcaggagcgcaga cgcccccgcgtaccagcagggccagaaccagctctataacgagctcaatctaggac gaagagaggagtacgatgttttggacaagagacgtggccgggaccctgagatggg gggaaagccgagaaggaagaaccctcaggaaggcctgtacaatgaactgcagaaa gataagatggcggaggcctacagtgagattgggatgaaaggcgagcgccggagg ggcaaggggcacgatggcctttaccagggtctcagtacagccaccaaggacaccta cgacgcccttcacatgcaggccctgccccctcgc
116	EF1α-MycT-Affy6 CAR-SSTR2 (control)	ggctccggtgcccgtcagtgggcagagcgcacatcgcccacagtccccgagaagtt ggggggaggggtcggcaattgaaccggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcgtgtactggctccgcctttttcccgagggtgggggagaacc gtatataagtgcagtagtcgccgtgaacgttctttttcgcaacgggtttgccgccagaa cacaggtaagtgccgtgtgtggttcccgcgggcctggcctctttacgggttatggccc ttgcgtgccttgaattacttccacctggctgcagtacgtgattcttgatcccgagcttcgg gttggaagtgggtgggagagttcgaggccttgcgcttaaggagccccttcgcctcgt gcttgagttgaggcctggcctgggcgctggggccgccgcgtgcgaatctggtggca ccttcgcgcctgtctcgctgctttcgataagtctctagccatttaaaatttttgatgacctg ctgcgacgctttttttctggcaagatagtcttgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggccgcgggcggcgacggggcccgtgcgtcccagcgcacat gttcggcgaggcggggcctgcgagcgcggccaccgagaatcggacgggggtagt ctcaagctggccggcctgctctggtgcctggtctcgcgccgccgtgtatcgccccgc cctgggcggcaaggctggcccggtcggcaccagttgcgtgagcggaaagatggcc gcttcccggccctgctgcagggagctcaaaatggaggacgcggcgctcgggagag cgggcgggtgagtcacccacacaaaggaaaagggcctttccgtcctcagccgtcgc ttcatgtgactccacggagtaccgggcgccgtccaggcacctcgattagttctcgagc ttttggagtacgtcgtctttaggttggggggaggggttttatgcgatggagtttccccac actgagtgggtggagactgaagttaggccagcttggcacttgatgtaattctccttgga atttgccctttttgagtttggatcttggttcattctcaagcctcagacagtggttcaaagttt ttttcttccatttcaggtgtcgtgacaagtttgtacaaaaaagcaggctgccaccatggc cttaccagtgaccgccttgctcctgccgctggccttgctgctccacgccgccaggccg ggatccgaacaaaaactcatctcagaagaggatctgggatcgCAGGTGCAG CTCGTGCAGAGCGGCGCCGAGGTGAAGAAGCCCGG CGCTTCCGTGAAGGTGTCCTGTAAGGCCTCTGGCTA CATCTTCACCGCCTACACCATGCACTGGGTGCGCCA GGCCCCGGGACAGGGGCTGGAGTGGATGGGCTGGA TCAAGCCTAACAACGGTCTGGCCAACTACGCGCAG AAGTTCCAGGGCCGCGTGACCATGACTCGGGACAC CAGCATCTCTACCGCGTACATGGAGCTGTCCCGCCT GCGCTCCGATGACACGGCCGTGTACTACTGCGCCCG C*TCGGAGATCACCACCGAGTTCGACTAT*TGGGGCCAG GGCACCCTGGTGACCGTGTCGTCCggtggaggcggttcaggc ggaggtggctctggcggtggcggatcgGACATCGTGCTGACCCAG AGCCCTGACAGCCTGGCCGTGTCCCTGGGCGAGCG CGCCACCATTAACTGCAAGAGCTCCGAGAGTGTCG ATAGCTACGCCAACTCCTTCATGCACTGGTACCAGC AGAAGCCCGGCCAGCCACCCAAGCTGCTGATCTAC AGGGCTTCCACCCGCGAGAGCGGCGTCCCCGACAG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		GTTTTCAGGTTCTGGCTCCCGCACCGACTTCACCCT GACCATCTCTTCTCTGCAGGCCGAGGATGTGGCGGT GTACTACTGTCAACAGTCCAAGGAGGACCCGCTTA CCTTCGGCGGCGGCACCAAGGTGGAGATCAAGgctag cacgaccactccggcgccgcgcccaccgactccggccccaactatcgcgagccag cccctgtcgctgaggccggaagcatgccgccctgccgccggaggtgctgtgcatac ccggggattggacttcgcatgcgacTTTTGGGTGCTGGTGGTGGT TGGTGGAGTCCTGGCTTGCTATAGCTTGCTAGTAAC AGTGGCCTTTATTATTTTCTGGGTGAGGAGTAAGAG GAGCAGGCTCCTGCACAGTGACTACATGAACATGA CTCCCCGCCGCCCCGGGCCCACCCGCAAGCATTACC AGCCCTATGCCCCACCACGCGACTTCGCAGCCTATC GCTCCctgagagtgaagttcagcaggagcgcagacgcccccgcgtaccagca gggccagaaccagctctataacgagctcaatctaggacgaagagaggagtacgatg ttttggacaagagacgtggccgggaccctgagatggggggaaagccgagaaggaa gaaccctcaggaaggcctgtacaatgaactgcagaaagataagatggcggaggcct acagtgagattgggatgaaaggcgagcgccggaggggcaaggggcacgatggcc tttaccagggtctcagtacagccaccaaggacacctacgacgcccttcacatgcagg ccctgccccctcgcactagtggaagcggagctactaacttcagcctgctgaagcagg ctggagacgtggaggagaaccctggaccttctagaatggacatggcggatgagcca ctcaatggaagccacacatggctatccattccatttgacctcaatggctctgtggtgtca accaacacctcaaaccagacagagccgtactatgacctgacaagcaatgcagtcctc acattcatctatttgtggtctgcatcattgggttgtgtggcaacacacttgtcatttatgtc atcctccgctatgccaagatgaagaccatcaccaacatttacatcctcaacctggccat cgcagatgagctcttcatgctgggtctgcctttcttggctatgcaggtggctctggtcca ctggccctttggcaaggccatttgccgggggtcatgactgtggatggcatcaatcagt tcaccagcatcttctgcctgacagtcatgagcatcgaccgatacctggctgtggtccac cccatcaagtcggccaagtggaggagaccccggacggccaagatgatcaccatgg ctgtgtggggagtctctctgctggtcatcttgcccatcatgatatatgctgggctccgga gcaaccagtgggggagaagcagctgcaccatcaactggccaggtgaatctggggc ttggtacacagggttcatcatctacactttcattctggggttcctggtacccctcaccatc atctgtctttgctacctgttcattatcatcaaggtgaagtcctctggaatccgagtgggct cctctaagaggaagaagtctgagaagaaggtcacccgaatggtgtccatcgtggtgg ctgtcttcatcttctgctggcttcccttctacatattcaacgtttcttccgtctccatggcca tcagccccaccccagcccttaaaggcatgtttgactttgtggtggtcctcacctatgcta acagctgtgccaaccctatcctatatgccttcttgtctgacaacttcaagaagagcttcc agaatgtcctctgcttggtcaaggtgagcggcacagatgatggggagcggagtgac agtaagcaggacaaatcccggctgaatgagaccacggagacccagaggaccctcc tcaatggagacctccaaaccagtatctaa
117	EF1α-MycT-Affy6-I3A CAR-SSTR2	ggctccggtgcccgtcagtgggcagagcgcacatcgcccacagtccccgagaagtt ggggggaggggtcggcaattgaaccggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcgtgtactggctccgcctttttcccgagggtgggggagaacc gtatataagtgcagtagtcgccgtgaacgttctttttcgcaacgggtttgccgccagaa cacaggtaagtgccgtgtgtggttcccgcgggcctggcctctttacgggttatggccc ttgcgtgccttgaattacttccacctggctgcagtacgtgattcttgatcccgagcttcgg gttggaagtgggtgggagagttcgaggccttgcgcttaaggagccccttcgcctcgt gcttgagttgaggcctggcctgggcgctggggccgccgcgtgcgaatctggtggca ccttcgcgcctgtctcgctgctttcgataagtctctagccatttaaaatttttgatgacctg ctgcgacgctttttttctggcaagatagtcttgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggccgcgggcggcgacggggcccgtgcgtcccagcgcacat gttcggcgaggcggggcctgcgagcgcggccaccgagaatcggacgggggtagt ctcaagctggccggcctgctctggtgcctggtctcgcgccgccgtgtatcgccccgc cctgggcggcaaggctggcccggtcggcaccagttgcgtgagcggaaagatggcc gcttcccggccctgctgcagggagctcaaaatggaggacgcggcgctcgggagag cgggcgggtgagtcacccacacaaaggaaaagggcctttccgtcctcagccgtcgc ttcatgtgactccacggagtaccgggcgccgtccaggcacctcgattagttctcgagc ttttggagtacgtcgtctttaggttggggggaggggttttatgcgatggagtttccccac actgagtgggtggagactgaagttaggccagcttggcacttgatgtaattctccttgga atttgcccttttttgagtttggatcttggttcattctcaagcctcagacagtggttcaaagttt ttttcttccatttcaggtgtcgtgacaagtttgtacaaaaaagcaggctgccaccatggc cttaccagtgaccgccttgctcctgccgctggccttgctgctccacgccgccaggccg ggatccgaacaaaaactcatctcagaagaggatctgggatcgCAGGTGCAG CTCGTGCAGAGCGGCGCCGAGGTGAAGAAGCCCGG CGCTTCCGTGAAGGTGTCCTGTAAGGCCTCTGGCTA CATCTTCACCGCCTACACCATGCACTGGGTGCGCCA GGCCCCGGGACAGGGGCTGGAGTGGATGGGCTGGA TCAAGCCTAACAACGGTCTGGCCAACTACGCGCAG AAGTTCCAGGGCCGCGTGACCATGACTCGGGACAC CAGCATCTCTACCGCGTACATGGAGCTGTCCCGCCT GCGCTCCGATGACACGGCCGTGTACTACTGCGCCCG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein		
SEQ ID NO:	Name	Sequence
		C*TCGGAG**GC**CACCACCGAGTTCGACTAT*TGGGGCCAG GGCACCCTGGTGACCGTGTCGTCCggtggaggcggttcaggc ggaggtggctctggcggtggcggatcgGACATCGTGCTGACCCAG AGCCCTGACAGCCTGGCCGTGTCCCTGGGCGAGCG CGCCACCATTAACTGCAAGAGCTCCGAGAGTGTCG ATAGCTACGCCAACTCCTTCATGCACTGGTACCAGC AGAAGCCCGGCCAGCCACCCAAGCTGCTGATCTAC AGGGCTTCCACCCGCGAGAGCGGCGTCCCCGACAG GTTTTCAGGTTCTGGCTCCCGCACCGACTTCACCCT GACCATCTCTTCTCTGCAGGCCGAGGATGTGGCGGT GTACTACTGTCAACAGTCCAAGGAGGACCCGCTTA CCTTCGGCGGCGGCACCAAGGTGGAGATCAAGgctag cacgaccactccggcgccgcgcccaccgactccggccccaactatcgcgagccag cccctgtcgctgaggccggaagcatgccgccctgccgccggaggtgctgtgcatac ccggggattggacttcgcatgcgacTTTTGGGTGCTGGTGGTGGT TGGTGGAGTCCTGGCTTGCTATAGCTTGCTAGTAAC AGTGGCCTTTATTATTTTCTGGGTGAGGAGTAAGAG GAGCAGGCTCCTGCACAGTGACTACATGAACATGA CTCCCCGCCGCCCCGGGCCCACCCGCAAGCATTACC AGCCCTATGCCCCACCACGCGACTTCGCAGCCTATC GCTCCctgagagtgaagttcagcaggagcgcagacgcccccgcgtaccagca gggccagaaccagctctataacgagctcaatctaggacgaagagaggagtacgatg ttttggacaagagacgtggccgggaccctgagatggggggaaagccgagaaggaa gaaccctcaggaaggcctgtacaatgaactgcagaaagataagatggcggaggcct acagtgagattgggatgaaaggcgagcgccggaggggcaaggggcacgatggcc tttaccagggtctcagtacagccaccaaggacacctacgacgcccttcacatgcagg ccctgccccctcgcactagtggaagcggagctactaacttcagcctgctgaagcagg ctggagacgtggaggagaaccctggaccttctagaatggacatggcggatgagcca ctcaatggaagccacacatggctatccattccatttgacctcaatggctctgtggtgtca accaacacctcaaaccagacagagccgtactatgacctgacaagcaatgcagtcctc acattcatctattttgtggtctgcatcattgggttgtgtggcaacacacttgtcatttatgtc atcctccgctatgccaagatgaagaccatcaccaacatttacatcctcaacctggccat cgcagatgagctcttcatgctgggtctgcctttcttggctatgcaggtggctctggtcca ctggccctttggcaaggccatttgccgggtggtcatgactgtggatggcatcaatcagt tcaccagcatcttctgcctgacagtcatgagcatcgaccgatacctggctgtggtccac cccatcaagtcggccaagtggaggagaccccggacggccaagatgatcaccatgg ctgtgtggggagtctctctgctggtcatcttgcccatcatgatatatgctgggctccgga gcaaccagtgggggagaagcagctgcaccatcaactggccaggtgaatctggggc ttggtacacagggttcatcatctacactttcattctggggttcctggtacccctcaccatc atctgtctttgctacctgttcattatcatcaaggtgaagtcctctggaatccgagtgggct cctctaagaggaagaagtctgagaagaaggtcacccgaatggtgtccatcgtggtgg ctgtcttcatcttctgctggcttcccttctacatattcaacgtttcttccgtctccatggcca tcagccccaccccagcccttaaaggcatgtttgactttgtggtggtcctcacctatgcta acagctgtgccaaccctatcctatatgccttcttgtctgacaacttcaagaagagcttcc agaatgtcctctgcttggtcaaggtgagcggcacagatgatggggagcggagtgac agtaagcaggacaaatcccggctgaatgagaccacggagacccagaggaccctcc tcaatggagacctccaaaccagtatctaa
118	EF1α-MycT-Affy6-I3W CAR-SSTR2	ggctccggtgcccgtcagtgggcagagcgcacatcgcccacagtccccgagaagtt ggggggaggggtcggcaattgaaccggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcgtgtactggctccgcctttttcccgagggtgggggagaacc gtatataagtgcagtagtcgccgtgaacgttctttttcgcaacgggtttgccgccagaa cacaggtaagtgccgtgtgtggttcccgcgggcctggcctctttacgggttatggccc ttgcgtgccttgaattacttccacctggctgcagtacgtgattcttgatcccgagcttcgg gttggaagtgggtgggagagttcgaggccttgcgcttaaggagccccttcgcctcgt gcttgagttgaggcctggcctgggcgctggggccgccgcgtgcgaatctggtggca ccttcgcgcctgtctcgctgctttcgataagtctctagccatttaaaatttttgatgacctg ctgcgacgctttttttctggcaagatagtcttgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggccgcgggcggcgacggggcccgtgcgtcccagcgcacat gttcggcgaggcggggcctgcgagcgcggccaccgagaatcggacgggggtagt ctcaagctggccggcctgctctggtgcctggtctcgcgccgccgtgtatcgccccgc cctgggcggcaaggctggcccggtcggcaccagttgcgtgagcggaaagatggcc gcttcccggccctgctgcagggagctcaaaatggaggacgcggcgctcgggagag cgggcgggtgagtcacccacacaaaggaaaagggcctttccgtcctcagccgtcgc ttcatgtgactccacggagtaccgggcgccgtccaggcacctcgattagttctcgagc ttttggagtacgtcgtctttaggttggggggaggggttttatgcgatggagtttccccac actgagtgggtggagactgaagttaggccagcttggcacttgatgtaattctccttgga atttgccctttttgagtttggatcttggttcattctcaagcctcagacagtggttcaaagttt ttttcttccatttcaggtgtcgtgacaagtttgtacaaaaaagcaggctgccaccatggc cttaccagtgaccgccttgctcctgccgctggccttgctgctccacgccgccaggccg ggatccgaacaaaaactcatctcagaagaggatctgggatcgCAGGTGCAG

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		CTCGTGCAGAGCGGCGCCGAGGTGAAGAAGCCCGG CGCTTCCGTGAAGGTGTCCTGTAAGGCCTCTGGCTA CATCTTCACCGCCTACACCATGCACTGGGTGCGCCA GGCCCCGGGACAGGGGCTGGAGTGGATGGGCTGGA TCAAGCCTAACAACGGTCTGGCCAACTACGCGCAG AAGTTCCAGGGCCGCGTGACCATGACTCGGGACAC CAGCATCTCTACCGCGTACATGGAGCTGTCCCGCCT GCGCTCCGATGACACGGCCGTGTACTACTGCGCCCG C*TCGGAG*___TGG___*ACCACCGAGTTCGACTAT*TGGGGCCAG GGCACCCTGGTGACCGTGTCGTCCggtggaggcggttcaggc ggaggtggctctggcggtggcggatcgGACATCGTGCTGACCCAG AGCCCTGACAGCCTGGCCGTGTCCCTGGGCGAGCG CGCCACCATTAACTGCAAGAGCTCCGAGAGTGTCG ATAGCTACGCCAACTCCTTCATGCACTGGTACCAGC AGAAGCCCGGCCAGCCACCCAAGCTGCTGATCTAC AGGGCTTCCACCCGCGAGAGCGGCGTCCCCGACAG GTTTTCAGGTTCTGGCTCCCGCACCGACTTCACCCT GACCATCTCTTCTCTGCAGGCCGAGGATGTGGCGGT GTACTACTGTCAACAGTCCAAGGAGGACCCGCTTA CCTTCGGCGGCGGCACCAAGGTGGAGATCAAGgctag cacgaccactccggcgccgcgcccaccgactccggccccaactatcgcgagccag cccctgtcgctgaggccggaagcatgccgccctgccgccggaggtgctgtgcatac ccggggattggacttcgcatgcgacTTTTGGGTGCTGGTGGTGGT TGGTGGAGTCCTGGCTTGCTATAGCTTGCTAGTAAC AGTGGCCTTTATTATTTTCTGGGTGAGGAGTAAGAG GAGCAGGCTCCTGCACAGTGACTACATGAACATGA CTCCCCGCCGCCCCGGGCCCACCCGCAAGCATTACC AGCCCTATGCCCCACCACGCGACTTCGCAGCCTATC GCTCCctgagagtgaagttcagcaggagcgcagacgcccccgcgtaccagca gggccagaaccagctctataacgagctcaatctaggacgaagagaggagtacgatg ttttggacaagagacgtggccgggaccctgagatggggggaaagccgagaaggaa gaaccctcaggaaggcctgtacaatgaactgcagaaagataagatggcggaggcct acagtgagattgggatgaaaggcgagcgccggaggggcaaggggcacgatggcc tttaccagggtctcagtacagccaccaaggacacctacgacgcccttcacatgcagg ccctgccccctcgcactagtggaagcggagctactaacttcagcctgctgaagcagg ctggagacgtggaggagaaccctggaccttctagaatggacatggcggatgagcca ctcaatggaagccacacatggctatccattccatttgacctcaatggctctgtggtgtca accaacacctcaaaccagacagagccgtactatgacctgacaagcaatgcagtcctc acattcatctattttgtggtctgcatcattgggttgtgtggcaacacacttgtcatttatgtc atcctccgctatgccaagatgaagaccatcaccaacatttacatcctcaacctggccat cgcagatgagctcttcatgctgggtctgcctttcttggctatgcaggtggctctggtcca ctggccctttggcaaggccatttgccgggtggtcatgactgtggatggcatcaatcagt tcaccagcatcttctgcctgacagtcatgagcatcgaccgatacctggctgtggtccac cccatcaagtcggccaagtggaggagaccccggacggccaagatgatcaccatgg ctgtgtggggagtctctctgctggtcatcttgcccatcatgatatatgctgggctccgga gcaaccagtgggggagaagcagctgcaccatcaactggccaggtgaatctggggc ttggtacacagggttcatcatctacactttcattctggggttcctggtacccctcaccatc atctgtctttgctacctgttcattatcatcaaggtgaagtcctctggaatccgagtgggct cctctaagaggaagaagtctgagaagaaggtcacccgaatggtgtccatcgtggtgg ctgtcttcatcttctgctggcttcccttctacatattcaacgtttcttccgtctccatggcca tcagccccaccccagcccttaaaggcatgtttgactttgtggtggtcctcacctatgcta acagctgtgccaaccctatcctatatgccttcttgtctgacaacttcaagaagagcttcc agaatgtcctctgcttggtcaaggtgagcggcacagatgatggggagcggagtgac agtaagcaggacaaatcccggctgaatgagaccacggagacccagaggaccctcc tcaatggagacctccaaaccagtatctaa
119	EF1α-MycT-Affy6-I3F CAR-SSTR2	ggctccggtgcccgtcagtgggcagagcgcacatcgcccacagtccccgagaagtt ggggggaggggtcggcaattgaaccggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcgtgtactggctccgcctttttcccgagggtgggggagaacc gtatataagtgcagtagtcgccgtgaacgttctttttcgcaacgggtttgccgccagaa cacaggtaagtgccgtgtgtggttcccgcgggcctggcctctttacgggttatggccc ttgcgtgccttgaattacttccacctggctgcagtacgtgattcttgatcccgagcttcgg gttggaagtgggtgggagagttcgaggccttgcgcttaaggagccccttcgcctcgt gcttgagttgaggcctggcctgggcgctggggccgccgcgtgcgaatctggtggca ccttcgcgcctgtctcgctgctttcgataagtctctagccatttaaaatttttgatgacctg ctgcgacgctttttttctggcaagatagtcttgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggccgcgggcggcgacggggcccgtgcgtcccagcgcacat gttcggcgaggcggggcctgcgagcgcggccaccgagaatcggacgggggtagt ctcaagctggccggcctgctctggtgcctggtctcgcgccgccgtgtatcgccccgc cctgggcggcaaggctggcccggtcggcaccagttgcgtgagcggaaagatggcc gcttcccggccctgctgcagggagctcaaaatggaggacgcggcgctcgggagag

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		cgggcgggtgagtcacccacacaaaggaaaagggcctttccgtcctcagccgtcgc ttcatgtgactccacggagtaccgggcgccgtccaggcacctcgattagttctcgagc ttttggagtacgtcgtctttaggttggggggaggggttttatgcgatggagtttccccac actgagtgggtggagactgaagttaggccagcttggcacttgatgtaattctccttgga atttgccctttttgagtttggatcttggttcattctcaagcctcagacagtggttcaaagttt ttttcttccatttcaggtgtcgtgacaagtttgtacaaaaaagcaggctgccaccatggc cttaccagtgaccgccttgctcctgccgctggccttgctgctccacgccgccaggccg ggatccgaacaaaaactcatctcagaagaggatctgggatcgCAGGTGCAG CTCGTGCAGAGCGGCGCCGAGGTGAAGAAGCCCGG CGCTTCCGTGAAGGTGTCCTGTAAGGCCTCTGGCTA CATCTTCACCGCCTACACCATGCACTGGGTGCGCCA GGCCCCGGGACAGGGGCTGGAGTGGATGGGCTGGA TCAAGCCTAACAACGGTCTGGCCAACTACGCGCAG AAGTTCCAGGGCCGCGTGACCATGACTCGGGACAC CAGCATCTCTACCGCGTACATGGAGCTGTCCCGCCT GCGCTCCGATGACACGGCCGTGTACTACTGCGCCCG C*TCGGAG**T**TCACCACCGAGTTCGACTAT*TGGGGCCAG GGCACCCTGGTGACCGTGTCGTCCggtggaggcggttcaggc ggaggtggctctggcggtggcggatcgGACATCGTGCTGACCCAG AGCCCTGACAGCCTGGCCGTGTCCCTGGGCGAGCG CGCCACCATTAACTGCAAGAGCTCCGAGAGTGTCG ATAGCTACGCCAACTCCTTCATGCACTGGTACCAGC AGAAGCCCGGCCAGCCACCCAAGCTGCTGATCTAC AGGGCTTCCACCCGCGAGAGCGGCGTCCCCGACAG GTTTTCAGGTTCTGGCTCCCGCACCGACTTCACCCT GACCATCTCTTCTCTGCAGGCCGAGGATGTGGCGGT GTACTACTGTCAACAGTCCAAGGAGGACCCGCTTA CCTTCGGCGGCGGCACCAAGGTGGAGATCAAGgctag cacgaccactccggcgccgcgcccaccgactccggccccaactatcgcgagccag cccctgtcgctgaggccggaagcatgccgccctgccgccggaggtgctgtgcatac ccggggattggacttcgcatgcgacTTTTGGGTGCTGGTGGTGGT TGGTGGAGTCCTGGCTTGCTATAGCTTGCTAGTAAC AGTGGCCTTTATTATTTTCTGGGTGAGGAGTAAGAG GAGCAGGCTCCTGCACAGTGACTACATGAACATGA CTCCCCGCCGCCCCGGGCCCACCCGCAAGCATTACC AGCCCTATGCCCCACCACGCGACTTCGCAGCCTATC GCTCCctgagagtgaagttcagcaggagcgcagacgcccccgcgtaccagca gggccagaaccagctctataacgagctcaatctaggacgaagagaggagtacgatg ttttggacaagagacgtggccgggaccctgagatggggggaaagccgagaaggaa gaaccctcaggaaggcctgtacaatgaactgcagaaagataagatggcggaggcct acagtgagattgggatgaaaggcgagcgccggaggggcaaggggcacgatggcc tttaccagggtctcagtacagccaccaaggacacctacgacgcccttcacatgcagg ccctgccccctcgcactagtggaagcggagctactaacttcagcctgctgaagcagg ctggagacgtggaggagaaccctggaccttctagaatggacatggcggatgagcca ctcaatggaagccacacatggctatccattccatttgacctcaatggctctgtggtgtca accaacacctcaaaccagacagagccgtactatgacctgacaagcaatgcagtcctc acattcatctattttgtggtctgcatcattgggttgtgtggcaacacacttgtcatttatgtc atcctccgctatgccaagatgaagaccatcaccaacatttacatcctcaacctggccat cgcagatgagctcttcatgctgggtctgcctttcttggctatgcaggtggctctggtcca ctggccctttggcaaggccatttgccgggtggtcatgactgtggatggcatcaatcagt tcaccagcatcttctgcctgacagtcatgagcatcgaccgatacctggctgtggtccac cccatcaagtcggccaagtggaggagaccccggacggccaagatgatcaccatgg ctgtgtggggagtctctctgctggtcatcttgcccatcatgatatatgctgggctccgga gcaaccagtgggggagaagcagctgcaccatcaactggccaggtgaatctggggc ttggtacacagggttcatcatctacactttcattctggggttcctggtacccctcaccatc atctgtctttgctacctgttcattatcatcaaggtgaagtcctctggaatccgagtgggct cctctaagaggaagaagtctgagaagaaggtcacccgaatggtgtccatcgtggtgg ctgtcttcatcttctgctggcttcccttctacatattcaacgtttcttccgtctccatggcca tcagccccaccccagcccttaaaggcatgtttgactttgtggtggtcctcacctatgcta acagctgtgccaaccctatcctatatgccttcttgtctgacaacttcaagaagagcttcc agaatgtcctctgcttggtcaaggtgagcggcacagatgatggggagcggagtgac agtaagcaggacaaatcccggctgaatgagaccacggagacccagaggaccctcc tcaatggagacctccaaaccagtatctaa
120	EF1α-MycT-Affy6-I3G CAR-SSTR2	ggctccggtgcccgtcagtgggcagagcgcacatcgcccacagtccccgagaagtt ggggggaggggtcggcaattgaaccggtgcctagagaaggtggcgcggggtaaa ctgggaaagtgatgtcgtgtactggctccgcctttttcccgagggtgggggagaacc gtatataagtgcagtagtcgccgtgaacgttctttttcgcaacgggtttgccgccagaa cacaggtaagtgccgtgtgtggttcccgcgggcctggcctctttacgggttatggccc ttgcgtgccttgaattacttccacctggctgcagtacgtgattcttgatcccgagcttcgg gttggaagtgggggagagttcgaggccttgcgcttaaggagccccttcgcctcgt

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		gcttgagttgaggcctggcctgggcgctggggccgccgcgtgcgaatctggtggca ccttcgcgcctgtctcgctgctttcgataagtctctagccatttaaaatttttgatgacctg ctgcgacgctttttttctggcaagatagtcttgtaaatgcgggccaagatctgcacactg gtatttcggtttttggggccgcgggcggcgacggggcccgtgcgtcccagcgcacat gttcggcgaggcggggcctgcgagcgcggccaccgagaatcggacgggggtagt ctcaagctggccggcctgctctggtgcctggtctcgcgccgccgtgtatcgccccgc cctgggcggcaaggctggcccggtcggcaccagttgcgtgagcggaaagatggcc gcttcccggccctgctgcagggagctcaaaatggaggacgcggcgctcgggagag cgggcgggtgagtcacccacacaaaggaaaagggcctttccgtcctcagccgtcgc ttcatgtgactccacggagtaccgggcgccgtccaggcacctcgattagttctcgagc ttttggagtacgtcgtctttaggttggggggaggggttttatgcgatggagtttccccac actgagtgggtggagactgaagttaggccagcttggcacttgatgtaattctccttgga atttgcccttttttgagtttggatcttggttcattctcaagcctcagacagtggttcaaagttt ttttcttccatttcaggtgtcgtgacaagtttgtacaaaaaagcaggctgccaccatggc cttaccagtgaccgccttgctcctgccgctggccttgctgctccacgccgccaggccg ggatccgaacaaaaactcatctcagaagaggatctgggatcgCAGGTGCAG CTCGTGCAGAGCGGCGCCGAGGTGAAGAAGCCCGG CGCTTCCGTGAAGGTGTCCTGTAAGGCCTCTGGCTA CATCTTCACCGCCTACACCATGCACTGGGTGCGCCA GGCCCCGGGACAGGGGCTGGAGTGGATGGGCTGGA TCAAGCCTAACAACGGTCTGGCCAACTACGCGCAG AAGTTCCAGGGCCGCGTGACCATGACTCGGGACAC CAGCATCTCTACCGCGTACATGGAGCTGTCCCGCCT GCGCTCCGATGACACGGCCGTGTACTACTGCGCCCG C*TCGGAG**GG**CACCACCGAGTTCGACTAT*TGGGGCCAG GGCACCCTGGTGACCGTGTCGTCCggtggaggcggttcaggc ggaggtggctctggcggtggcggatcgGACATCGTGCTGACCCAG AGCCCTGACAGCCTGGCCGTGTCCCTGGGCGAGCG CGCCACCATTAACTGCAAGAGCTCCGAGAGTGTCG ATAGCTACGCCAACTCCTTCATGCACTGGTACCAGC AGAAGCCCGGCCAGCCACCCAAGCTGCTGATCTAC AGGGCTTCCACCCGCGAGAGCGGCGTCCCCGACAG GTTTTCAGGTTCTGGCTCCCGCACCGACTTCACCCT GACCATCTCTTCTCTGCAGGCCGAGGATGTGGCGGT GTACTACTGTCAACAGTCCAAGGAGGACCCGCTTA CCTTCGGCGGCGGCACCAAGGTGGAGATCAAGgctag cacgaccactccggcgccgcgcccaccgactccggccccaactatcgcgagccag cccctgtcgctgaggccggaagcatgccgccctgccgccggaggtgctgtgcatac ccggggattggacttcgcatgcgacTTTTGGGTGCTGGTGGTGGT TGGTGGAGTCCTGGCTTGCTATAGCTTGCTAGTAAC AGTGGCCTTTATTATTTTCTGGGTGAGGAGTAAGAG GAGCAGGCTCCTGCACAGTGACTACATGAACATGA CTCCCCGCCGCCCCGGGCCCACCCGCAAGCATTACC AGCCCTATGCCCCACCACGCGACTTCGCAGCCTATC GCTCCctgagagtgaagttcagcaggagcgcagacgcccccgcgtaccagca gggccagaaccagctctataacgagctcaatctaggacgaagagaggagtacgatg ttttggacaagagacgtggccgggaccctgagatggggggaaagccgagaaggaa gaaccctcaggaaggcctgtacaatgaactgcagaaagataagatggcggaggcct acagtgagattgggatgaaaggcgagcgccggaggggcaaggggcacgatggcc tttaccagggtctcagtacagccaccaaggacacctacgacgcccttcacatgcagg ccctgccccctcgcactagtggaagcggagctactaacttcagcctgctgaagcagg ctggagacgtggaggagaaccctggaccttctagaatggacatggcggatgagcca ctcaatggaagccacacatggctatccattccatttgacctcaatggctctgtggtgtca accaacacctcaaaccagacagagccgtactatgacctgacaagcaatgcagtcctc acattcatctattttgtggtctgcatcattgggttgtgtggcaacacacttgtcatttatgtc atcctccgctatgccaagatgaagaccatcaccaacatttacatcctcaacctggccat cgcagatgagctcttcatgctgggtctgcctttcttggctatgcaggtggctctggtcca ctggccctttggcaaggccatttgccgggtggtcatgactgtggatggcatcaatcagt tcaccagcatcttctgcctgacagtcatgagcatcgaccgatacctggctgtggtccac cccatcaagtcggccaagtggaggagaccccggacggccaagatgatcaccatgg ctgtgtggggagtctctctgctggtcatcttgcccatcatgatatatgctgggctccgga gcaaccagtgggggagaagcagctgcaccatcaactggccaggtgaatctggggc ttggtacacagggttcatcatctacactttcattctggggttcctggtacccctcaccatc atctgtctttgctacctgttcattatcatcaaggtgaagtcctctggaatccgagtgggct cctctaagaggaagaagtctgagaagaaggtcacccgaatggtgtccatcgtggtgg ctgtcttcatcttctgctggcttcccttctacatattcaacgtttcttccgtctccatggcca tcagccccaccccagcccttaaaggcatgtttgactttgtggtggtcctcacctatgcta acagctgtgccaaccctatcctatatgccttcttgtctgacaacttcaagaagagcttcc agaatgtcctctgcttggtcaaggtgagcggcacagatgatggggagcggagtgac agtaagcaggacaaatcccggctgaatgagaccacggagacccagaggaccctcc tcaatggagacctccaaaccagtatctaa

TABLE 18-continued

SEQ ID NO:	Name	Sequence
	Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein	
121	(GGGGS)x3 linker	GGGGSGGGGSGGGGS
122	4-1BB costimulatory domain	KRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEE GGCEL
123	CD28 costimulatory domain	RSKRSRLLHSDYMNMTPRRPGPTRKHYQPYAPPRDFA AYRS
124	OX-40 costimulatory domain	ALYLLRRDQRLPPDAHKPPGGGSFRTPIQEEQADAHS TLAKI
125	CD3 zeta signaling domain	RVKFSRSADAPAYKQGQNQLYNELNLGRREEYDVLD KRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAY SEIGMKGERRRGKGHDGLYQGLSTATKDTYDALHMQ ALPPR
126	CD8 alpha hinge	TTTPAPRPPTPAPTIASQPLSLRPEACRPAAGGAVHTR GLDFACD
127	CD28 hinge	RAAAIEVMYPPPYLDNEKSNGTIIHVKGKHLCPSPLFP GPSKPKDPK
128	IgG4 hinge	SKYGPPCPSCP
129	CD8 alpha transmembrane domain	IYIWAPLAGTCGVLLLSLVITLYC
130	CD28 transmembrane domain	FWVLVVVGGVLACYSLLVTVAFII
131	ICOS transmembrane domain	FWLPIGCAAFVVVCILGCILICWL
132	GITR transmembrane domain	LGWLTVVLLAVAACVLLLTSAQLGL
133	αL subunit of LFA-1	YNLDVRGARSFSPPRAGRHFGYRVLQVGNGVIVGAP GEGNSTGSLYQCQSGTGHCLPVTLRGSNYTSKYLGM TLATDPTDGSILACDPGLSRTCDQNTYLSGLCYLFRQN LQGPMLQGRPGFQECIKGNVDLVFLFDGSMSLQPDEF QKILDFMKDVMKKLSNTSYQFAAVQFSTSYKTEFDFS DYVKWKDPDALLKHVKHMLLLTNTFGAINYVATEVF REELGARPDATKVLIIITDGEATDSGNIDAAKDIIR YIIG IGKHFQTKESQETLHKFASKPASEFVKILDTFEKLKDL FTELQKKIYVIEGTSKQDLTSFNMELSSSGISADLSRGH AVVGAVGAKDWAGGFLDLKADLQDDTFIGNEPLTPE VRAGYLGYTVTWLPSRQKTSLLASGAPRYQHMGRVL LFQEPQGGGHWSQVQTIHGTQIGSYFGGELCGVDVD QDGETELLLIGAPLFYGEQRGGRVFIYQRRQLGFEEVS ELQGDPGYPLGRFGEAITALTDINGDGLVDVAVGAPL EEQGAVYIFNGRHGGLSPQPSQRIEGTQVLSGIQWFGR SIHGVKDLEGDGLADVAVGAESQMIVLSSRPVVDMV TLMSFSPAEIPVHEVECSYSTSNKMKEGVNITICFQIKS LYPQFQGRLVANLTYTLQLDGHRTRRRGLFPGGRHEL RRNIAVTTSMSCTDFSFHFPVCVQDLISPINVSLNFSLW EEEGTPRDQRAQGKDIPPILRPSLHSETWEIPFEKNCGE DKKCEANLRVSFSPARSRALRLTAFASLSVELSLSNLE

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
		EDAYWVQLDLHFPPGLSFRKVEMLKPHSQIPVSCEEL PEESRLLSRALSCNVSSPIFKAGHSVALQMMENTLVNS SWGDSVELHANVTCNNEDSDLLEDNSATTIIPILYPINI LIQDQEDSTLYVSFTPKGPKIHQVKHMYQVRIQPSIHD HNIPTLEAVVGVPQPPSEGPITHQWSVQMEPPVPCHYE DLERLPDAAEPCLPGALFRCPVVFRQEILVQVIGTLEL VGEIEASSMFSLCSSLSISFNSSKHFHLYGSNASLAQVV MKVDVVYEKQMLYLYVLSGIGGLLLLLLIFIVLYKVG FFKRNLKEKMEAGRGVPNGIPAEDSEQLASGQEAGDP GCLKPLHEKDSESGGGKD
134	I domain of αL subunit of LFA-1	VDLVFLFDGSMSLQPDEFQKILDFMKDVMKKLSNTSY QFAAVQFSTSYKTEFDFSDYVKWKDPDALLKHVKHM LLLTNTFGAINYVATEVFREELGARPDATKVLIIITDGE ATDSGNIDAAKDIIRYIIGIGKHFQTKESQETLHKFASK PASEFVKILDTFEKLKDLFTELQKKIYVIE
135	SSTR2 (aa1-381)	MDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPY YDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKMK TITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGKA ICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKSA KWRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICL CYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYA NSCANPILYAFLSDNFKKSFQNVLCLVKVSGTDDGER SDSKQDKSRLNETTETQRTLLNGDLQTSI
136	SSTR2 (aa 1-314)	MDMADEPLNGSHTWLSIPFDLNGSVVSTNTSNQTEPY YDLTSNAVLTFIYFVVCIIGLCGNTLVIYVILRYAKMK TITNIYILNLAIADELFMLGLPFLAMQVALVHWPFGKA ICRVVMTVDGINQFTSIFCLTVMSIDRYLAVVHPIKSA KWRRPRTAKMITMAVWGVSLLVILPIMIYAGLRSNQ WGRSSCTINWPGESGAWYTGFIIYTFILGFLVPLTIICL CYLFIIIKVKSSGIRVGSSKRKKSEKKVTRMVSIVVAVF IFCWLPFYIFNVSSVSMAISPTPALKGMFDFVVVLTYA NSCANPILYAF
137	SSTR2 (aa 1-381) nucleotide sequence	ATGGACATGGCGGATGAGCCACTCAATGGAAGCCA CACATGGCTATCCATTCCATTTGACCTCAATGGCTC TGTGGTGTCAACCAACACCTCAAACCAGACAGAGC CGTACTATGACCTGACAAGCAATGCAGTCCTCACAT TCATCTATTTTGTGGTCTGCATCATTGGGTTGTGTGG CAACACACTTGTCATTTATGTCATCCTCCGCTATGC CAAGATGAAGACCATCACCAACATTTACATCCTCA ACCTGGCCATCGCAGATGAGCTCTTCATGCTGGGTC TGCCTTTCTTGGCTATGCAGGTGGCTCTGGTCCACT GGCCCTTTGGCAAGGCCATTTGCCGGGTGGTCATGA CTGTGGATGGCATCAATCAGTTCACCAGCATCTTCT GCCTGACAGTCATGAGCATCGACCGATACCTGGCT GTGGTCCACCCCATCAAGTCGGCCAAGTGGAGGAG ACCCCGGACGGCCAAGATGATCACCATGGCTGTGT GGGGAGTCTCTCTGCTGGTCATCTTGCCCATCATGA TATATGCTGGGCTCCGGAGCAACCAGTGGGGGAGA AGCAGCTGCACCATCAACTGGCCAGGTGAATCTGG GGCTTGGTACACAGGGTTCATCATCTACACTTTCAT TCTGGGGTTCCTGGTACCCCTCACCATCATCTGTCTT TGCTACCTGTTCATTATCATCAAGGTGAAGTCCTCT GGAATCCGAGTGGGCTCCTCTAAGAGGAAGAAGTC TGAGAAGAAGGTCACCCGAATGGTGTCCATCGTGG TGGCTGTCTTCATCTTCTGCTGGCTTCCCTTCTACAT ATTCAACGTTTCTTCCGTCTCCATGGCCATCAGCCC CACCCCAGCCCTTAAAGGCATGTTTGACTTTGTGGT GGTCCTCACCTATGCTAACAGCTGTGCCAACCCTAT CCTATATGCCTTCTTGTCTGACAACTTCAAGAAGAG CTTCCAGAATGTCCTCTGCTTGGTCAAGGTGAGCGG CACAGATGATGGGGAGCGGAGTGACAGTAAGCAGG ACAAATCCCGGCTGAATGAGACCACGGAGACCCAG AGGACCCTCCTCAATGGAGACCTCCAAACCAGTAT CTAA

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
138	SSTR2 (aa1-314) nucleotide sequence	ATGGACATGGCGGATGAGCCACTCAATGGAAGCCA CACATGGCTATCCATTCCATTTGACCTCAATGGCTC TGTGGTGTCAACCAACACCTCAAACCAGACAGAGC CGTACTATGACCTGACAAGCAATGCAGTCCTCACAT TCATCTATTTTGTGGTCTGCATCATTGGGTTGTGTGG CAACACACTTGTCATTTATGTCATCCTCCGCTATGC CAAGATGAAGACCATCACCAACATTTACATCCTCA ACCTGGCCATCGCAGATGAGCTCTTCATGCTGGGTC TGCCTTTCTTGGCTATGCAGGTGGCTCTGGTCCACT GGCCCTTTGGCAAGGCCATTTGCCGGGTGGTCATGA CTGTGGATGGCATCAATCAGTTCACCAGCATCTTCT GCCTGACAGTCATGAGCATCGACCGATACCTGGCT GTGGTCCACCCCATCAAGTCGGCCAAGTGGAGGAG ACCCCGGACGGCCAAGATGATCACCATGGCTGTGT GGGGAGTCTCTCTGCTGGTCATCTTGCCCATCATGA TATATGCTGGGCTCCGGAGCAACCAGTGGGGGAGA AGCAGCTGCACCATCAACTGGCCAGGTGAATCTGG GGCTTGGTACACAGGGTTCATCATCTACACTTTCAT TCTGGGGTTCCTGGTACCCCTCACCATCATCTGTCTT TGCTACCTGTTCATTATCATCAAGGTGAAGTCCTCT GGAATCCGAGTGGGCTCCTCTAAGAGGAAGAAGTC TGAGAAGAAGGTCACCCGAATGGTGTCCATCGTGG TGGCTGTCTTCATCTTCTGCTGGCTTCCCTTCTACAT ATTCAACGTTTCTTCCGTCTCCATGGCCATCAGCCC CACCCCAGCCCTTAAAGGCATGTTTGACTTTGTGGT GGTCCTCACCTATGCTAACAGCTGTGCCAACCCTAT CCTATAT
139	P2A peptide	GSGATNFSLLKQAGDVEENPGP
140	E2A peptide	GSGQCTNYALLKLAGDVESNPGP
141	T2A peptide	GSGEGRGSLLTCGDVEENPGP
142	F2A peptide	GSGVKQTLNFDLLKLAGDVESNPGP
143	Linker	(GGGS)n
144	AB_001 primer	CTGTGCCACCgcgGGCTCCTACGTGAGC
145	AB_002 primer	TAGTACACGGCGCTGTCC
146	AB_015 primer	CGTGAGCCCTgcgGATTATTGGGGC
147	AB_016 primer	TAGGAGCCGTAGGTGGCA
148	AB_037 primer	CTGTGCCACCgtCGGCTCCTACG
149	AB_038 primer	TAGTACACGGCGCTGTCC
150	AB_039 primer	CTGTGCCACCctCGGCTCCTAC
151	AB_040 primer	CTGTGCCACCatCGGCTCCTAC
152	AB_042 primer	CTGTGCCACCatgGGCTCCTACGTGAGCCC
153	AB_086 primer	CCGCTCGGAGgcCACCACCGAG
154	AB_087 primer	GCGCAGTAGTACACGGCC
155	AB_088 primer	CTCGGAGATCgCCACCGAGTTC
156	AB_089 primer	CGGGCGCAGTAGTACACG
157	AB_117 primer	GCGCAGTAGTACACGGCC
158	AB_120 primer	CCGCTCGGAGggcACCACCGAGTTC
159	AB_123 primer	CCGCTCGGAGttcACCACCGAGT
160	AB_125 primer	CCGCTCGGAGtggACCACCGAGTTC

TABLE 18-continued

Amino Acid and Nucleotide Sequences for Exemplary Polypeptides Described Herein

SEQ ID NO:	Name	Sequence
161	EF1α promoter	GGCTCCGGTGCCCGTCAGTGGGCAGAGCGCACATC GCCCACAGTCCCCGAGAAGTTGGGGGGAGGGGTCG GCAATTGAACCGGTGCCTAGAGAAGGTGGCGCGGG GTAAACTGGGAAAGTGATGTCGTGTACTGGCTCCG CCTTTTTCCCGAGGGTGGGGGAGAACCGTATATAA GTGCAGTAGTCGCCGTGAACGTTCTTTTTCGCAACG GGTTTGCCGCCAGAACACAGGTAAGTGCCGTGTGT GGTTCCCGCGGGCCTGGCCTCTTTACGGGTTATGGC CCTTGCGTGCCTTGAATTACTTCCACCTGGCTGCAG TACGTGATTCTTGATCCCGAGCTTCGGGTTGGAAGT GGGTGGGAGAGTTCGAGGCCTTGCGCTTAAGGAGC CCCTTCGCCTCGTGCTTGAGTTGAGGCCTGGCCTGG GCGCTGGGGCCGCCGCGTGCGAATCTGGTGGCACC TTCGCGCCTGTCTCGCTGCTTTCGATAAGTCTCTAG CCATTTAAAATTTTTGATGACCTGCTGCGACGCTTT TTTTCTGGCAAGATAGTCTTGTAAATGCGGGCCAAG ATCTGCACACTGGTATTTCGGTTTTTGGGGCCGCGG GCGGCGACGGGGCCCGTGCGTCCCAGCGCACATGT TCGGCGAGGCGGGGCCTGCGAGCGCGGCCACCGAG AATCGGACGGGGGTAGTCTCAAGCTGGCCGGCCTG CTCTGGTGCCTGGTCTCGCGCCGCCGTGTATCGCCC CGCCCTGGGCGGCAAGGCTGGCCCGGTCGGCACCA GTTGCGTGAGCGGAAAGATGGCCGCTTCCCGGCCC TGCTGCAGGGAGCTCAAAATGGAGGACGCGGCGCT CGGGAGAGCGGGCGGGTGAGTCACCCACACAAAGG AAAAGGGCCTTTCCGTCCTCAGCCGTCGCTTCATGT GACTCCACGGAGTACCGGGCGCCGTCCAGGCACCT CGATTAGTTCTCGAGCTTTTGGAGTACGTCGTCTTT AGGTTGGGGGGAGGGGTTTTATGCGATGGAGTTTC CCCACACTGAGTGGGTGGAGACTGAAGTTAGGCCA GCTTGGCACTTGATGTAATTCTCCTTGGAATTTGCC CTTTTTGAGTTTGGATCTTGGTTCATTCTCAAGCCTC AGACAGTGGTTCAAAGTTTTTTTCTTCCATTTCAGG TGTCGTGA
162	CD8 alpha chain signal peptide	MALPVTALLLPLALLLHAARP
163	CD8 alpha chain signal peptide	ATGGCCTTACCAGTGACCGCCTTGCTCCTGCCGCTG GCCTTGCTGCTCCACGCCGCCAGGCCG

*CDR regions underlined and italicized. CDR1, CD2, and CD3 regions are underlined and italicized in the VH and VL sequences. Only HCDR3 regions are underlined and italicized in the scFv and CAR sequences. Mutated amino acids of nucleotides in the HCDR3 regions are bolded.

Other Embodiments

[0148] All of the features disclosed in this specification may be combined in any combination. Each feature disclosed in this specification may be replaced by an alternative feature serving the same, equivalent, or similar purpose. Thus, unless expressly stated otherwise, each feature disclosed is only an example of a generic series of equivalent or similar features.

[0149] From the above description, one skilled in the art can easily ascertain the essential characteristics of the present invention, and without departing from the spirit and scope thereof, can make various changes and modifications of the invention to adapt it to various usages and conditions. Thus, other embodiments are also within the claims.

EQUIVALENTS

[0150] While several inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

[0151] All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

[0152] All references, patents and patent applications disclosed herein are incorporated by reference with respect to the subject matter for which each is cited, which in some cases may encompass the entirety of the document.

[0153] The indefinite articles "a" and "an," as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean "at least one."

[0154] The phrase "and/or," as used herein in the specification and in the claims, should be understood to mean "either or both" of the elements so conjoined, e.g., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with "and/or" should be construed in the same fashion, e.g., "one or more" of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the "and/or" clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to "A and/or B", when used in conjunction with open-ended language such as "comprising" can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0155] As used herein in the specification and in the claims, "or" should be understood to have the same meaning as "and/or" as defined above. For example, when separating items in a list, "or" or "and/or" shall be interpreted as being inclusive, e.g., the inclusion of at least one, but also including more than one of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as "only one of" or "exactly one of," or, when used in the claims, "consisting of," will refer to the inclusion of exactly one element of a number or list of elements. In general, the term "or" as used herein shall only be interpreted as indicating exclusive alternatives (e.g., "one or the other but not both") when preceded by terms of exclusivity, such as "either," "one of," "only one of," or "exactly one of." "Consisting essentially of," when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0156] As used herein in the specification and in the claims, the phrase "at least one," in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase "at least one" refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, "at least one of A and B" (or, equivalently, "at least one of A or B," or, equivalently "at least one of A and/or B") can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0157] It should also be understood that, unless clearly indicated to the contrary, in any methods claimed herein that include more than one step or act, the order of the steps or acts of the method is not necessarily limited to the order in which the steps or acts of the method are recited.

SEQUENCE LISTING

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Sequence total quantity: 163
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                       organism = synthetic construct
SEQUENCE: 1
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SEQ ID NO: 2           moltype = AA  length = 17
FEATURE                Location/Qualifiers
source                 1..17
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 2
LEWIGMIDPS NSDTRFN                                                 17

SEQ ID NO: 3           moltype = AA  length = 10
FEATURE                Location/Qualifiers
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 3
AGSYVSPLDY                                                         10
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source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 4
YGSYVSPADY                                                          10

SEQ ID NO: 5            moltype = AA  length = 10
FEATURE                 Location/Qualifiers
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 5
VGSYVSPLDY                                                          10

SEQ ID NO: 6            moltype = AA  length = 10
FEATURE                 Location/Qualifiers
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 6
LGSYVSPLDY                                                          10

SEQ ID NO: 7            moltype = AA  length = 10
FEATURE                 Location/Qualifiers
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 7
IGSYVSPLDY                                                          10

SEQ ID NO: 8            moltype = AA  length = 10
FEATURE                 Location/Qualifiers
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 8
MGSYVSPLDY                                                          10

SEQ ID NO: 9            moltype = AA  length = 11
FEATURE                 Location/Qualifiers
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 9
QSLLYTSSQK N                                                        11

SEQ ID NO: 10           moltype = AA  length = 12
FEATURE                 Location/Qualifiers
source                  1..12
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 10
PKLLIYWAST RE                                                       12

SEQ ID NO: 11           moltype = AA  length = 5
FEATURE                 Location/Qualifiers
source                  1..5
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 11
YYAYP                                                               5

SEQ ID NO: 12           moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 12
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NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPLDYWG QGTSVTVSS    119

SEQ ID NO: 13           moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
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                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 13
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NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATAG SYVSPLDYWG QGTSVTVSS   119

SEQ ID NO: 14           moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 14
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPADYWG QGTSVTVSS   119

SEQ ID NO: 15           moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 15
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATVG SYVSPLDYWG QGTSVTVSS   119

SEQ ID NO: 16           moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 16
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATLG SYVSPLDYWG QGTSVTVSS   119

SEQ ID NO: 17           moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 17
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATIG SYVSPLDYWG QGTSVTVSS   119

SEQ ID NO: 18           moltype = AA  length = 119
FEATURE                 Location/Qualifiers
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 18
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATMG SYVSPLDYWG QGTSVTVSS   119

SEQ ID NO: 19           moltype = AA  length = 114
FEATURE                 Location/Qualifiers
source                  1..114
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 19
DIMMSQSPSS LTVSVGEKVT VSCKSSQSLL YTSSQKNYLA WYQQKPGQSP KLLIYWASTR  60
ESGVPDRFTG SGSGTDFTLT ITSVKADDLA VYYCQQYYAY PWTFGGGTKL EIKR        114

SEQ ID NO: 20           moltype = DNA  length = 357
FEATURE                 Location/Qualifiers
source                  1..357
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 20
caggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg  60
tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc cacctacggc  300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcaccgt gtcctcc     357

SEQ ID NO: 21           moltype = DNA  length = 357
FEATURE                 Location/Qualifiers
source                  1..357
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                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 21
caggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg  60
tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc caccgcgggc  300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcaccgt gtcctcc     357

SEQ ID NO: 22           moltype = DNA  length = 357
FEATURE                 Location/Qualifiers
source                  1..357
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 22
caggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg  60
tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc cacctacggc  300
tcctacgtga gccctgcgga ttattggggc cagggcacct ccgtcaccgt gtcctcc     357

SEQ ID NO: 23           moltype = DNA  length = 357
FEATURE                 Location/Qualifiers
source                  1..357
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 23
caggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg  60
tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc caccgtcggc  300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcaccgt gtcctcc     357

SEQ ID NO: 24           moltype = DNA  length = 357
FEATURE                 Location/Qualifiers
source                  1..357
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 24
caggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg  60
tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc caccctcggc  300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcaccgt gtcctcc     357

SEQ ID NO: 25           moltype = DNA  length = 357
FEATURE                 Location/Qualifiers
source                  1..357
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 25
caggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg  60
tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc caccatcggc  300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcaccgt gtcctcc     357

SEQ ID NO: 26           moltype = DNA  length = 357
FEATURE                 Location/Qualifiers
source                  1..357
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 26
caggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg  60
tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc caccatgggc  300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcaccgt gtcctcc     357

SEQ ID NO: 27           moltype = DNA  length = 342
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FEATURE                Location/Qualifiers
source                 1..342
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 27
gacatcatga tgtctcagag tccctcttct ctgaccgtgt ccgtgggcga gaaggtgacc  60
gtgtcatgca agtcctcaca gagccttctg tacaccagct cccagaagaa ctacctggcg  120
tggtaccagc agaagcccgg acagagcccc aagctgctga tctattgggc ttccacccgc  180
gagagcggcg tccccgaccg cttcaccggc tccggctccg ggaccgactt caccctgacc  240
atcacctccg tgaaggccga tgacctggcc gtgtactact gtcaacagta ttacgcctac  300
ccgtggacct tcggaggcgg caccaagctg gagatcaagc gc                     342

SEQ ID NO: 28          moltype = AA  length = 248
FEATURE                Location/Qualifiers
source                 1..248
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 28
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPLDYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKR                                                           248

SEQ ID NO: 29          moltype = AA  length = 248
FEATURE                Location/Qualifiers
source                 1..248
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 29
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATAG SYVSPLDYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKR                                                           248

SEQ ID NO: 30          moltype = AA  length = 248
FEATURE                Location/Qualifiers
source                 1..248
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 30
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPADYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKR                                                           248

SEQ ID NO: 31          moltype = AA  length = 248
FEATURE                Location/Qualifiers
source                 1..248
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 31
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATVG SYVSPLDYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKR                                                           248

SEQ ID NO: 32          moltype = AA  length = 248
FEATURE                Location/Qualifiers
source                 1..248
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 32
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATLG SYVSPLDYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKR                                                           248

SEQ ID NO: 33          moltype = AA  length = 248
FEATURE                Location/Qualifiers
source                 1..248
                       mol_type = protein
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SEQUENCE: 33
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NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATIG SYVSPLDYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKR                                                           248

SEQ ID NO: 34            moltype = AA  length = 248
FEATURE                  Location/Qualifiers
source                   1..248
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 34
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NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATMG SYVSPLDYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKR                                                           248

SEQ ID NO: 35            moltype = DNA  length = 744
FEATURE                  Location/Qualifiers
source                   1..744
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 35
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tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc cacctacggc  300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcaccgt gtcctccggt  360
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cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc  540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac  600
cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccatcacctc cgtgaaggcc  660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc  720
ggcaccaagc tggagatcaa gcgc                                         744

SEQ ID NO: 36            moltype = DNA  length = 744
FEATURE                  Location/Qualifiers
source                   1..744
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 36
caggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg  60
tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc caccgcgggc  300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcaccgt gtcctccggt  360
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agtccctctt ctctgaccgt gtccgtgggc gagaaggtga ccgtgtcatg caagtcctca  480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc  540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac  600
cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccatcacctc cgtgaaggcc  660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc  720
ggcaccaagc tggagatcaa gcgc                                         744

SEQ ID NO: 37            moltype = DNA  length = 744
FEATURE                  Location/Qualifiers
source                   1..744
                         mol_type = other DNA
                         organism = synthetic construct
SEQUENCE: 37
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tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc cacctacggc  300
tcctacgtga gccctgcgga ttattggggc cagggcacct ccgtcaccgt gtcctccggt  360
ggaggcggtt caggcggagg tggctctggc ggtggcggat cggacatcat gatgtctcag  420
agtccctctt ctctgaccgt gtccgtgggc gagaaggtga ccgtgtcatg caagtcctca  480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc  540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac  600
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cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccatcacctc cgtgaaggcc  660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc  720
ggcaccaagc tggagatcaa gcgc                                         744

SEQ ID NO: 38           moltype = DNA  length = 744
FEATURE                 Location/Qualifiers
source                  1..744
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 38
caggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg  60
tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc caccgtcggc  300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcaccgt gtcctccggt  360
ggaggcggtt caggcggagg tggctctggc ggtggcggat cggacatcat gatgtctcag  420
agtccctctt ctctgaccgt gtccgtgggc gagaaggtga ccgtgtcatg caagtcctca  480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc  540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac  600
cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccatcacctc cgtgaaggcc  660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc  720
ggcaccaagc tggagatcaa gcgc                                         744

SEQ ID NO: 39           moltype = DNA  length = 744
FEATURE                 Location/Qualifiers
source                  1..744
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 39
caggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg  60
tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc caccctcggc  300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcaccgt gtcctccggt  360
ggaggcggtt caggcggagg tggctctggc ggtggcggat cggacatcat gatgtctcag  420
agtccctctt ctctgaccgt gtccgtgggc gagaaggtga ccgtgtcatg caagtcctca  480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc  540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac  600
cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccatcacctc cgtgaaggcc  660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc  720
ggcaccaagc tggagatcaa gcgc                                         744

SEQ ID NO: 40           moltype = DNA  length = 744
FEATURE                 Location/Qualifiers
source                  1..744
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 40
caggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg  60
tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc caccatcggc  300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcaccgt gtcctccggt  360
ggaggcggtt caggcggagg tggctctggc ggtggcggat cggacatcat gatgtctcag  420
agtccctctt ctctgaccgt gtccgtgggc gagaaggtga ccgtgtcatg caagtcctca  480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc  540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac  600
cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccatcacctc cgtgaaggcc  660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc  720
ggcaccaagc tggagatcaa gcgc                                         744

SEQ ID NO: 41           moltype = DNA  length = 744
FEATURE                 Location/Qualifiers
source                  1..744
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 41
caggtgcagc tccagcagag cggccctgag ctggtgcgcc ccggcgcttc cgtgaagatg  60
tcatgccgag cttctggcta cacgttcacc agctattggc tacactgggt caagcagagg  120
cccggacagg gtctggagtg gatcggtatg attgacccgt caaatagtga cacccgcttc  180
aaccccaact tcaaggacaa ggccactctc aacgtggacc gctcgagcaa cacggcctac  240
atgctgctgt cctctctgac ctcggcggac agcgccgtgt actactgtgc caccatgggc  300
tcctacgtga gccctttgga ttattggggc cagggcacct ccgtcaccgt gtcctccggt  360
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ggaggcggtt caggcggagg tggctctggc ggtggcggat cggacatcat gatgtctcag  420
agtccctctt ctctgaccgt gtccgtgggc gagaaggtga ccgtgtcatg caagtcctca  480
cagagccttc tgtacaccag ctcccagaag aactacctgg cgtggtacca gcagaagccc  540
ggacagagcc ccaagctgct gatctattgg gcttccaccc gcgagagcgg cgtccccgac  600
cgcttcaccg gctccggctc cgggaccgac ttcaccctga ccatcacctc cgtgaaggcc  660
gatgacctgg ccgtgtacta ctgtcaacag tattacgcct acccgtggac cttcggaggc  720
ggcaccaagc tggagatcaa gcgc                                          744

SEQ ID NO: 42           moltype = AA  length = 476
FEATURE                 Location/Qualifiers
source                  1..476
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 42
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPLDYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKRAS TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDFWVLV  300
VVGGVLACYS LLVTVAFIIF WVRSKRSRLL HSDYMNMTPR RPGPTRKHYQ PYAPPRDFAA  360
YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVLD KRRGRDPEMG GKPRRKNPQE  420
GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM QALPPR       476

SEQ ID NO: 43           moltype = AA  length = 476
FEATURE                 Location/Qualifiers
source                  1..476
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 43
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATAG SYVSPLDYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKRAS TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDFWVLV  300
VVGGVLACYS LLVTVAFIIF WVRSKRSRLL HSDYMNMTPR RPGPTRKHYQ PYAPPRDFAA  360
YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVLD KRRGRDPEMG GKPRRKNPQE  420
GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM QALPPR       476

SEQ ID NO: 44           moltype = AA  length = 476
FEATURE                 Location/Qualifiers
source                  1..476
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 44
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATYG SYVSPADYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKRAS TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDFWVLV  300
VVGGVLACYS LLVTVAFIIF WVRSKRSRLL HSDYMNMTPR RPGPTRKHYQ PYAPPRDFAA  360
YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVLD KRRGRDPEMG GKPRRKNPQE  420
GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM QALPPR       476

SEQ ID NO: 45           moltype = AA  length = 476
FEATURE                 Location/Qualifiers
source                  1..476
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 45
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATVG SYVSPLDYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKRAS TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDFWVLV  300
VVGGVLACYS LLVTVAFIIF WVRSKRSRLL HSDYMNMTPR RPGPTRKHYQ PYAPPRDFAA  360
YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVLD KRRGRDPEMG GKPRRKNPQE  420
GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM QALPPR       476

SEQ ID NO: 46           moltype = AA  length = 476
FEATURE                 Location/Qualifiers
source                  1..476
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 46
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATLG SYVSPLDYWG QGTSVTVSSG  120
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GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKRAS TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDFWVLV  300
VVGGVLACYS LLVTVAFIIF WVRSKRSRLL HSDYMNMTPR RPGPTRKHYQ PYAPPRDFAA  360
YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVLD KRRGRDPEMG GKPRRKNPQE  420
GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM QALPPR      476

SEQ ID NO: 47           moltype = AA  length = 476
FEATURE                 Location/Qualifiers
source                  1..476
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 47
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATIG SYVSPLDYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKRAS TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDFWVLV  300
VVGGVLACYS LLVTVAFIIF WVRSKRSRLL HSDYMNMTPR RPGPTRKHYQ PYAPPRDFAA  360
YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVLD KRRGRDPEMG GKPRRKNPQE  420
GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM QALPPR      476

SEQ ID NO: 48           moltype = AA  length = 476
FEATURE                 Location/Qualifiers
source                  1..476
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 48
QVQLQQSGPE LVRPGASVKM SCRASGYTFT SYWLHWVKQR PGQGLEWIGM IDPSNSDTRF  60
NPNFKDKATL NVDRSSNTAY MLLSSLTSAD SAVYYCATMG SYVSPLDYWG QGTSVTVSSG  120
GGGSGGGGSG GGGSDIMMSQ SPSSLTVSVG EKVTVSCKSS QSLLYTSSQK NYLAWYQQKP  180
GQSPKLLIYW ASTRESGVPD RFTGSGSGTD FTLTITSVKA DDLAVYYCQQ YYAYPWTFGG  240
GTKLEIKRAS TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDFWVLV  300
VVGGVLACYS LLVTVAFIIF WVRSKRSRLL HSDYMNMTPR RPGPTRKHYQ PYAPPRDFAA  360
YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVLD KRRGRDPEMG GKPRRKNPQE  420
GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM QALPPR      476

SEQ ID NO: 49           moltype = AA  length = 885
FEATURE                 Location/Qualifiers
source                  1..885
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 49
GSEQKLISEE DLGSQVQLQQ SGPELVRPGA SVKMSCRASG YTFTSYWLHW VKQRPGQGLE  60
WIGMIDPSNS DTRFNPNFKD KATLNVDRSS NTAYMLLSSL TSADSAVYYC ATYGSYVSPL  120
DYWGQGTSVT VSSGGGGSGG GGSGGGGSDI MMSQSPSSLT VSVGEKVTVS CKSSQSLLYT  180
SSQKNYLAWY QQKPGQSPKL LIYWASTRES GVPDRFTGSG SGTDFTLTIT SVKADDLAVY  240
YCQQYYAYPW TFGGGTKLEI KRASTTTPAP RPPTPAPTIA SQPLSLRPEA CRPAAGGAVH  300
TRGLDFACDF WVLVVVGGVL ACYSLLVTVA FIIFWVRSKR SRLLHSDYMN MTPRRPGPTR  360
KHYQPYAPPR DFAAYRSLRV KFSRSADAPA YQQGQNQLYN ELNLGRREEY DVLDKRRGRD  420
PEMGGKPRRK NPQEGLYNEL QKDKMAEAYS EIGMKGERRR GKGHDGLYQG LSTATKDTYD  480
ALHMQALPPR TSGSGATNFS LLKQAGDVEE NPGPSRMDMA DEPLNGSHTW LSIPFDLNGS  540
VVSTNTSNQT EPYYDLTSNA VLTFIYFVVC IIGLCGNTLV IYVILRYAKM KTITNIYILN  600
LAIADELFML GLPFLAMQVA LVHWPFGKAI CRVVMTVDGI NQFTSIFCLT VMSIDRYLAV  660
VHPIKSAKWR RPRTAKMITM AVWGVSLLVI LPIMIYAGLR SNQWGRSSCT INWPGESGAW  720
YTGFIIYTFI LGFLVPLTII CLCYLFIIIK VKSSGIRVGS SKRKKSEKKV TRMVSIVVAV  780
FIFCWLPFYI FNVSSVSMAI SPTPALKGMF DFVVVLTYAN SCANPILYAF LSDNFKKSFQ  840
NVLCLVKVSG TDDGERSDSK QDKSRLNETT ETQRTLLNGD LQTSI                  885

SEQ ID NO: 50           moltype = AA  length = 885
FEATURE                 Location/Qualifiers
source                  1..885
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 50
GSEQKLISEE DLGSQVQLQQ SGPELVRPGA SVKMSCRASG YTFTSYWLHW VKQRPGQGLE  60
WIGMIDPSNS DTRFNPNFKD KATLNVDRSS NTAYMLLSSL TSADSAVYYC ATAGSYVSPL  120
DYWGQGTSVT VSSGGGGSGG GGSGGGGSDI MMSQSPSSLT VSVGEKVTVS CKSSQSLLYT  180
SSQKNYLAWY QQKPGQSPKL LIYWASTRES GVPDRFTGSG SGTDFTLTIT SVKADDLAVY  240
YCQQYYAYPW TFGGGTKLEI KRASTTTPAP RPPTPAPTIA SQPLSLRPEA CRPAAGGAVH  300
TRGLDFACDF WVLVVVGGVL ACYSLLVTVA FIIFWVRSKR SRLLHSDYMN MTPRRPGPTR  360
KHYQPYAPPR DFAAYRSLRV KFSRSADAPA YQQGQNQLYN ELNLGRREEY DVLDKRRGRD  420
PEMGGKPRRK NPQEGLYNEL QKDKMAEAYS EIGMKGERRR GKGHDGLYQG LSTATKDTYD  480
ALHMQALPPR TSGSGATNFS LLKQAGDVEE NPGPSRMDMA DEPLNGSHTW LSIPFDLNGS  540
VVSTNTSNQT EPYYDLTSNA VLTFIYFVVC IIGLCGNTLV IYVILRYAKM KTITNIYILN  600
LAIADELFML GLPFLAMQVA LVHWPFGKAI CRVVMTVDGI NQFTSIFCLT VMSIDRYLAV  660
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VHPIKSAKWR RPRTAKMITM AVWGVSLLVI LPIMIYAGLR SNQWGRSSCT INWPGESGAW  720
YTGFIIYTFI LGFLVPLTII CLCYLFIIIK VKSSGIRVGS SKRKKSEKKV TRMVSIVVAV  780
FIFCWLPFYI FNVSSVSMAI SPTPALKGMF DFVVVLTYAN SCANPILYAF LSDNFKKSFQ  840
NVLCLVKVSG TDDGERSDSK QDKSRLNETT ETQRTLLNGD LQTSI                  885

SEQ ID NO: 51           moltype = AA  length = 885
FEATURE                 Location/Qualifiers
source                  1..885
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 51
GSEQKLISEE DLGSQVQLQQ SGPELVRPGA SVKMSCRASG YTFTSYWLHW VKQRPGQGLE  60
WIGMIDPSNS DTRFNPNFKD KATLNVDRSS NTAYMLLSSL TSADSAVYYC ATYGSYVSPA  120
DYWGQGTSVT VSSGGGGSGG GGSGGGGSDI MMSQSPSSLT VSVGEKVTVS CKSSQSLLYT  180
SSQKNYLAWY QQKPGQSPKL LIYWASTRES GVPDRFTGSG SGTDFTLTIT SVKADDLAVY  240
YCQQYYAYPW TFGGGTKLEI KRASTTTPAP RPPTPAPTIA SQPLSLRPEA CRPAAGGAVH  300
TRGLDFACDF WVLVVVGGVL ACYSLLVTVA FIIFWVRSKR SRLLHSDYMN MTPRRPGPTR  360
KHYQPYAPPR DFAAYRSLRV KFSRSADAPA YQQGQNQLYN ELNLGRREEY DVLDKRRGRD  420
PEMGGKPRRK NPQEGLYNEL QKDKMAEAYS EIGMKGERRR GKGHDGLYQG LSTATKDTYD  480
ALHMQALPPR TSGSGATNFS LLKQAGDVEE NPGPSRMDMA DEPLNGSHTW LSIPFDLNGS  540
VVSTNTSNQT EPYYDLTSNA VLTFIYFVVC IIGLCGNTLV IYVILRYAKM KTITNIYILN  600
LAIADELFML GLPFLAMQVA LVHWPFGKAI CRVVMTVDGI NQFTSIFCLT VMSIDRYLAV  660
VHPIKSAKWR RPRTAKMITM AVWGVSLLVI LPIMIYAGLR SNQWGRSSCT INWPGESGAW  720
YTGFIIYTFI LGFLVPLTII CLCYLFIIIK VKSSGIRVGS SKRKKSEKKV TRMVSIVVAV  780
FIFCWLPFYI FNVSSVSMAI SPTPALKGMF DFVVVLTYAN SCANPILYAF LSDNFKKSFQ  840
NVLCLVKVSG TDDGERSDSK QDKSRLNETT ETQRTLLNGD LQTSI                  885

SEQ ID NO: 52           moltype = AA  length = 885
FEATURE                 Location/Qualifiers
source                  1..885
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 52
GSEQKLISEE DLGSQVQLQQ SGPELVRPGA SVKMSCRASG YTFTSYWLHW VKQRPGQGLE  60
WIGMIDPSNS DTRFNPNFKD KATLNVDRSS NTAYMLLSSL TSADSAVYYC ATVGSYVSPL  120
DYWGQGTSVT VSSGGGGSGG GGSGGGGSDI MMSQSPSSLT VSVGEKVTVS CKSSQSLLYT  180
SSQKNYLAWY QQKPGQSPKL LIYWASTRES GVPDRFTGSG SGTDFTLTIT SVKADDLAVY  240
YCQQYYAYPW TFGGGTKLEI KRASTTTPAP RPPTPAPTIA SQPLSLRPEA CRPAAGGAVH  300
TRGLDFACDF WVLVVVGGVL ACYSLLVTVA FIIFWVRSKR SRLLHSDYMN MTPRRPGPTR  360
KHYQPYAPPR DFAAYRSLRV KFSRSADAPA YQQGQNQLYN ELNLGRREEY DVLDKRRGRD  420
PEMGGKPRRK NPQEGLYNEL QKDKMAEAYS EIGMKGERRR GKGHDGLYQG LSTATKDTYD  480
ALHMQALPPR TSGSGATNFS LLKQAGDVEE NPGPSRMDMA DEPLNGSHTW LSIPFDLNGS  540
VVSTNTSNQT EPYYDLTSNA VLTFIYFVVC IIGLCGNTLV IYVILRYAKM KTITNIYILN  600
LAIADELFML GLPFLAMQVA LVHWPFGKAI CRVVMTVDGI NQFTSIFCLT VMSIDRYLAV  660
VHPIKSAKWR RPRTAKMITM AVWGVSLLVI LPIMIYAGLR SNQWGRSSCT INWPGESGAW  720
YTGFIIYTFI LGFLVPLTII CLCYLFIIIK VKSSGIRVGS SKRKKSEKKV TRMVSIVVAV  780
FIFCWLPFYI FNVSSVSMAI SPTPALKGMF DFVVVLTYAN SCANPILYAF LSDNFKKSFQ  840
NVLCLVKVSG TDDGERSDSK QDKSRLNETT ETQRTLLNGD LQTSI                  885

SEQ ID NO: 53           moltype = AA  length = 885
FEATURE                 Location/Qualifiers
source                  1..885
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 53
GSEQKLISEE DLGSQVQLQQ SGPELVRPGA SVKMSCRASG YTFTSYWLHW VKQRPGQGLE  60
WIGMIDPSNS DTRFNPNFKD KATLNVDRSS NTAYMLLSSL TSADSAVYYC ATLGSYVSPL  120
DYWGQGTSVT VSSGGGGSGG GGSGGGGSDI MMSQSPSSLT VSVGEKVTVS CKSSQSLLYT  180
SSQKNYLAWY QQKPGQSPKL LIYWASTRES GVPDRFTGSG SGTDFTLTIT SVKADDLAVY  240
YCQQYYAYPW TFGGGTKLEI KRASTTTPAP RPPTPAPTIA SQPLSLRPEA CRPAAGGAVH  300
TRGLDFACDF WVLVVVGGVL ACYSLLVTVA FIIFWVRSKR SRLLHSDYMN MTPRRPGPTR  360
KHYQPYAPPR DFAAYRSLRV KFSRSADAPA YQQGQNQLYN ELNLGRREEY DVLDKRRGRD  420
PEMGGKPRRK NPQEGLYNEL QKDKMAEAYS EIGMKGERRR GKGHDGLYQG LSTATKDTYD  480
ALHMQALPPR TSGSGATNFS LLKQAGDVEE NPGPSRMDMA DEPLNGSHTW LSIPFDLNGS  540
VVSTNTSNQT EPYYDLTSNA VLTFIYFVVC IIGLCGNTLV IYVILRYAKM KTITNIYILN  600
LAIADELFML GLPFLAMQVA LVHWPFGKAI CRVVMTVDGI NQFTSIFCLT VMSIDRYLAV  660
VHPIKSAKWR RPRTAKMITM AVWGVSLLVI LPIMIYAGLR SNQWGRSSCT INWPGESGAW  720
YTGFIIYTFI LGFLVPLTII CLCYLFIIIK VKSSGIRVGS SKRKKSEKKV TRMVSIVVAV  780
FIFCWLPFYI FNVSSVSMAI SPTPALKGMF DFVVVLTYAN SCANPILYAF LSDNFKKSFQ  840
NVLCLVKVSG TDDGERSDSK QDKSRLNETT ETQRTLLNGD LQTSI                  885

SEQ ID NO: 54           moltype = AA  length = 885
FEATURE                 Location/Qualifiers
source                  1..885
                        mol_type = protein
                        organism = synthetic construct
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SEQUENCE: 54
GSEQKLISEE DLGSQVQLQQ SGPELVRPGA SVKMSCRASG YTFTSYWLHW VKQRPGQGLE  60
WIGMIDPSNS DTRFNPNFKD KATLNVDRSS NTAYMLLSSL TSADSAVYYC ATIGSYVSPL  120
DYWGQGTSVT VSSGGGGSGG GGSGGGGSDI MMSQSPSSLT VSVGEKVTVS CKSSQSLLYT  180
SSQKNYLAWY QQKPGQSPKL LIYWASTRES GVPDRFTGSG SGTDFTLTIT SVKADDLAVY  240
YCQQYYAYPW TFGGGTKLEI KRASTTTPAP RPPTPAPTIA SQPLSLRPEA CRPAAGGAVH  300
TRGLDFACDF WVLVVVGGVL ACYSLLVTVA FIIFWVRSKR SRLLHSDYMN MTPRRPGPTR  360
KHYQPYAPPR DFAAYRSLRV KFSRSADAPA YQQGQNQLYN ELNLGRREEY DVLDKRRGRD  420
PEMGGKPRRK NPQEGLYNEL QKDKMAEAYS EIGMKGERRR GKGHDGLYQG LSTATKDTYD  480
ALHMQALPPR TSGSGATNFS LLKQAGDVEE NPGPSRMDMA DEPLNGSHTW LSIPFDLNGS  540
VVSTNTSNQT EPYYDLTSNA VLTFIYFVVC IIGLCGNTLV IYVILRYAKM KTITNIYILN  600
LAIADELFML GLPFLAMQVA LVHWPFGKAI CRVVMTVDGI NQFTSIFCLT VMSIDRYLAV  660
VHPIKSAKWR RPRTAKMITM AVWGVSLLVI LPIMIYAGLR SNQWGRSSCT INWPGESGAW  720
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FIFCWLPFYI FNVSSVSMAI SPTPALKGMF DFVVVLTYAN SCANPILYAF LSDNFKKSFQ  840
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SEQ ID NO: 55          moltype = AA  length = 885
FEATURE                Location/Qualifiers
source                 1..885
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SSQKNYLAWY QQKPGQSPKL LIYWASTRES GVPDRFTGSG SGTDFTLTIT SVKADDLAVY  240
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VHPIKSAKWR RPRTAKMITM AVWGVSLLVI LPIMIYAGLR SNQWGRSSCT INWPGESGAW  720
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FIFCWLPFYI FNVSSVSMAI SPTPALKGMF DFVVVLTYAN SCANPILYAF LSDNFKKSFQ  840
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SEQ ID NO: 56          moltype = DNA  length = 1428
FEATURE                Location/Qualifiers
source                 1..1428
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                       organism = synthetic construct
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SEQ ID NO: 57          moltype = DNA  length = 1428
FEATURE                Location/Qualifiers
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                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 57
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aagagacgtg gccgggaccc tgagatgggg ggaaagccga gaaggaagaa ccctcaggaa  1260
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aaaggcgagc gccggagggg caaggggcac gatggccttt accagggtct cagtacagcc  1380
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SEQ ID NO: 58          moltype = DNA  length = 1428
FEATURE                Location/Qualifiers
source                 1..1428
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                       organism = synthetic construct
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SEQ ID NO: 59          moltype = DNA  length = 1428
FEATURE                Location/Qualifiers
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                       mol_type = other DNA
                       organism = synthetic construct
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                        organism = synthetic construct
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                         organism = synthetic construct
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SEQ ID NO: 63            moltype = DNA  length = 3930
FEATURE                  Location/Qualifiers
source                   1..3930
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                         organism = synthetic construct
SEQUENCE: 63
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ctcaatggct ctgtggtgtc aaccaacacc tcaaaccaga cagagccgta ctatgacctg  2940
acaagcaatg cagtcctcac attcatctat tttgtggtct gcatcattgg gttgtgtggc  3000
aacacacttg tcatttatgt catcctccgc tatgccaaga tgaagaccat caccaacatt  3060
tacatcctca acctggccat cgcagatgag ctcttcatgc tgggtctgcc tttcttggct  3120
atgcaggtgg ctctggtcca ctggcccttt ggcaaggcca tttgccgggt ggtcatgact  3180
gtggatggca tcaatcagtt caccagcatc ttctgcctga cagtcatgag catcgaccga  3240
tacctggctg tggtccaccc catcaagtcg gccaagtgga ggagaccccg gacggccaag  3300
atgatcacca tggctgtgtg ggaggtctct ctgctggtca tcttgcccat catgatatat  3360
gctgggctcc ggagcaacca gtgggggaga agcagctgca ccatcaactg gccaggtgaa  3420
tctggggctt ggtacacagg gttcatcatc tacactttca ttctggggtt cctggtaccc  3480
ctcaccatca tctgtctttg ctacctgttc attatcatca aggtgaagtc ctctggaatc  3540
cgagtgggct cctctaagag gaagaagtct gagaagaagg tcacccgaat ggtgtccatc  3600
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gtggtggctg tcttcatctt ctgctggctt cccttctaca tattcaacgt ttcttccgtc  3660
tccatggcca tcagccccac cccagccctt aaaggcatgt ttgactttgt ggtggtcctc  3720
acctatgcta acagctgtgc caaccctatc ctatatgcct tcttgtctga caacttcaag  3780
aagagcttcc agaatgtcct ctgcttggtc aaggtgagcg gcacagatga tggggagcgg  3840
agtgacagta agcaggacaa atcccggctg aatgagacca cggagaccca gaggaccctc  3900
ctcaatggag acctccaaac cagtatctaa                                   3930

SEQ ID NO: 70          moltype = AA  length = 7
FEATURE                Location/Qualifiers
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 70
GYIFTAY                                                            7

SEQ ID NO: 71          moltype = AA  length = 16
FEATURE                Location/Qualifiers
source                 1..16
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 71
LEWMGWIKPN NGLANY                                                  16

SEQ ID NO: 72          moltype = AA  length = 9
FEATURE                Location/Qualifiers
source                 1..9
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 72
SEATTEFDY                                                          9

SEQ ID NO: 73          moltype = AA  length = 9
FEATURE                Location/Qualifiers
source                 1..9
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 73
SEWTTEFDY                                                          9

SEQ ID NO: 74          moltype = AA  length = 9
FEATURE                Location/Qualifiers
source                 1..9
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 74
SEFTTEFDY                                                          9

SEQ ID NO: 75          moltype = AA  length = 9
FEATURE                Location/Qualifiers
source                 1..9
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 75
SEGTTEFDY                                                          9

SEQ ID NO: 76          moltype = AA  length = 6
FEATURE                Location/Qualifiers
source                 1..6
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 76
ESVDSY                                                             6

SEQ ID NO: 77          moltype = AA  length = 12
FEATURE                Location/Qualifiers
source                 1..12
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 77
PKLLIYRAST RE                                                      12

SEQ ID NO: 78          moltype = AA  length = 6
FEATURE                Location/Qualifiers
source                 1..6
                       mol_type = protein
                       organism = synthetic construct
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SEQUENCE: 78
SKEDPL                                                        6

SEQ ID NO: 79          moltype = AA  length = 118
FEATURE                Location/Qualifiers
source                 1..118
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 79
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE ITTEFDYWGQ GTLVTVSS    118

SEQ ID NO: 80          moltype = AA  length = 118
FEATURE                Location/Qualifiers
source                 1..118
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 80
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE ATTEFDYWGQ GTLVTVSS    118

SEQ ID NO: 81          moltype = AA  length = 118
FEATURE                Location/Qualifiers
source                 1..118
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 81
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE WTTEFDYWGQ GTLVTVSS    118

SEQ ID NO: 82          moltype = AA  length = 118
FEATURE                Location/Qualifiers
source                 1..118
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 82
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE FTTEFDYWGQ GTLVTVSS    118

SEQ ID NO: 83          moltype = AA  length = 118
FEATURE                Location/Qualifiers
source                 1..118
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 83
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE GTTEFDYWGQ GTLVTVSS    118

SEQ ID NO: 84          moltype = AA  length = 111
FEATURE                Location/Qualifiers
source                 1..111
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 84
DIVLTQSPDS LAVSLGERAT INCKSSESVD SYANSFMHWY QQKPGQPPKL LIYRASTRES  60
GVPDRFSGSG SRTDFTLTIS SLQAEDVAVY YCQQSKEDPL TFGGGTKVEI K           111

SEQ ID NO: 85          moltype = DNA  length = 354
FEATURE                Location/Qualifiers
source                 1..354
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 85
caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc ccggcgcttc cgtgaaggtg  60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc  120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag  300
atcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtcc        354

SEQ ID NO: 86          moltype = DNA  length = 354
FEATURE                Location/Qualifiers
source                 1..354
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 86
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caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc ccggcgcttc cgtgaaggtg  60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc  120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag  300
gccaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtcc        354

SEQ ID NO: 87           moltype = DNA  length = 354
FEATURE                 Location/Qualifiers
source                  1..354
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 87
caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc ccggcgcttc cgtgaaggtg  60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc  120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag  300
tggaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtcc        354

SEQ ID NO: 88           moltype = DNA  length = 354
FEATURE                 Location/Qualifiers
source                  1..354
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 88
caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc ccggcgcttc cgtgaaggtg  60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc  120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag  300
ttcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtcc        354

SEQ ID NO: 89           moltype = DNA  length = 354
FEATURE                 Location/Qualifiers
source                  1..354
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 89
caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc ccggcgcttc cgtgaaggtg  60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc  120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag  300
ggcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtcc        354

SEQ ID NO: 90           moltype = DNA  length = 333
FEATURE                 Location/Qualifiers
source                  1..333
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 90
gacatcgtgc tgacccagag ccctgacagc ctggccgtgt ccctgggcga gcgcgccacc  60
attaactgca agagctccga gagtgtcgat agctacgcca actccttcat gcactggtac  120
cagcagaagc ccggccagcc acccaagctg ctgatctaca gggcttccac ccgcgagagc  180
ggcgtccccg acaggtttc aggttctggc tcccgcaccg acttcaccct gaccatctct  240
tctctgcagg ccgaggatgt ggcggtgtac tactgtcaac agtccaagga ggacccgctt  300
accttcggcg gcggcaccaa ggtggagatc aag                                333

SEQ ID NO: 91           moltype = AA  length = 244
FEATURE                 Location/Qualifiers
source                  1..244
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 91
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE ITTEFDYWGQ GTLVTVSSGG  120
GGSGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP  180
PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK  240
VEIK                                                                244

SEQ ID NO: 92           moltype = AA  length = 244
FEATURE                 Location/Qualifiers
source                  1..244
                        mol_type = protein
                        organism = synthetic construct
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SEQUENCE: 92
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE ATTEFDYWGQ GTLVTVSSGG  120
GGSGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP  180
PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK  240
VEIK                                                              244

SEQ ID NO: 93           moltype = AA  length = 244
FEATURE                 Location/Qualifiers
source                  1..244
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 93
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE WTTEFDYWGQ GTLVTVSSGG  120
GGSGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP  180
PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK  240
VEIK                                                              244

SEQ ID NO: 94           moltype = AA  length = 244
FEATURE                 Location/Qualifiers
source                  1..244
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 94
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE FTTEFDYWGQ GTLVTVSSGG  120
GGSGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP  180
PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK  240
VEIK                                                              244

SEQ ID NO: 95           moltype = AA  length = 244
FEATURE                 Location/Qualifiers
source                  1..244
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 95
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE GTTEFDYWGQ GTLVTVSSGG  120
GGSGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP  180
PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK  240
VEIK                                                              244

SEQ ID NO: 96           moltype = DNA  length = 732
FEATURE                 Location/Qualifiers
source                  1..732
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 96
caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc ccggcgcttc cgtgaaggtg  60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc  120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag  300
atcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtccggtgga  360
ggcggttcag gcggaggtgg ctctggcggt ggcggatcgg acatcgtgct gacccagagc  420
cctgacagcc tggccgtgtc cctgggcgag cgcgccacca ttaactgcaa gagctccgag  480
agtgtcgata gctacgccaa ctccttcatg cactggtacc agcagaagcc cggccagcca  540
cccaagctgc tgatctacag ggcttccacc cgcgagagcg gcgtccccga caggttttca  600
ggttctggct cccgcaccga cttcaccctg accatctctt ctctgcaggc cgaggatgtg  660
gcggtgtact actgtcaaca gtccaaggag gacccgctta ccttcggcgg cggcaccaag  720
gtggagatca ag                                                     732

SEQ ID NO: 97           moltype = DNA  length = 732
FEATURE                 Location/Qualifiers
source                  1..732
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 97
caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc ccggcgcttc cgtgaaggtg  60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc  120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag  300
gccaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtccggtgga  360
ggcggttcag gcggaggtgg ctctggcggt ggcggatcgg acatcgtgct gacccagagc  420
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cctgacagcc tggccgtgtc cctgggcgag cgcgccacca ttaactgcaa gagctccgag  480
agtgtcgata gctacgccaa ctccttcatg cactggtacc agcagaagcc cggccagcca  540
cccaagctgc tgatctacag ggcttccacc cgcgagagcg gcgtccccga caggttttca  600
ggttctggct cccgcaccga cttcaccctg accatctctt ctctgcaggc cgaggatgtg  660
gcggtgtact actgtcaaca gtccaaggag gacccgctta ccttcggcgg cggcaccaag  720
gtggagatca ag                                                      732

SEQ ID NO: 98          moltype = DNA  length = 732
FEATURE                Location/Qualifiers
source                 1..732
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 98
caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc ccggcgcttc cgtgaaggtg  60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc  120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag  300
tggaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtccggtgga  360
ggcggttcag gcggaggtgg ctctggcggt ggcggatcgg acatcgtgct gacccagagc  420
cctgacagcc tggccgtgtc cctgggcgag cgcgccacca ttaactgcaa gagctccgag  480
agtgtcgata gctacgccaa ctccttcatg cactggtacc agcagaagcc cggccagcca  540
cccaagctgc tgatctacag ggcttccacc cgcgagagcg gcgtccccga caggttttca  600
ggttctggct cccgcaccga cttcaccctg accatctctt ctctgcaggc cgaggatgtg  660
gcggtgtact actgtcaaca gtccaaggag gacccgctta ccttcggcgg cggcaccaag  720
gtggagatca ag                                                      732

SEQ ID NO: 99          moltype = DNA  length = 732
FEATURE                Location/Qualifiers
source                 1..732
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 99
caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc ccggcgcttc cgtgaaggtg  60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc  120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag  300
ttcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtccggtgga  360
ggcggttcag gcggaggtgg ctctggcggt ggcggatcgg acatcgtgct gacccagagc  420
cctgacagcc tggccgtgtc cctgggcgag cgcgccacca ttaactgcaa gagctccgag  480
agtgtcgata gctacgccaa ctccttcatg cactggtacc agcagaagcc cggccagcca  540
cccaagctgc tgatctacag ggcttccacc cgcgagagcg gcgtccccga caggttttca  600
ggttctggct cccgcaccga cttcaccctg accatctctt ctctgcaggc cgaggatgtg  660
gcggtgtact actgtcaaca gtccaaggag gacccgctta ccttcggcgg cggcaccaag  720
gtggagatca ag                                                      732

SEQ ID NO: 100         moltype = DNA  length = 732
FEATURE                Location/Qualifiers
source                 1..732
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 100
caggtgcagc tcgtgcagag cggcgccgag gtgaagaagc ccggcgcttc cgtgaaggtg  60
tcctgtaagg cctctggcta catcttcacc gcctacacca tgcactgggt gcgccaggcc  120
ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag  300
ggcaccaccg agttcgacta ttggggccag ggcaccctgg tgaccgtgtc gtccggtgga  360
ggcggttcag gcggaggtgg ctctggcggt ggcggatcgg acatcgtgct gacccagagc  420
cctgacagcc tggccgtgtc cctgggcgag cgcgccacca ttaactgcaa gagctccgag  480
agtgtcgata gctacgccaa ctccttcatg cactggtacc agcagaagcc cggccagcca  540
cccaagctgc tgatctacag ggcttccacc cgcgagagcg gcgtccccga caggttttca  600
ggttctggct cccgcaccga cttcaccctg accatctctt ctctgcaggc cgaggatgtg  660
gcggtgtact actgtcaaca gtccaaggag gacccgctta ccttcggcgg cggcaccaag  720
gtggagatca ag                                                      732

SEQ ID NO: 101         moltype = AA  length = 472
FEATURE                Location/Qualifiers
source                 1..472
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 101
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE ITTEFDYWGQ GTLVTVSSGG  120
GGSGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP  180
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PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK  240
VEIKASTTTP APRPPTPAPT IASQPLSLRP EACRPAAGGA VHTRGLDFAC DFWVLVVVGG  300
VLACYSLLVT VAFIIFWVRS KRSRLLHSDY MNMTPRRPGP TRKHYQPYAP PRDFAAYRSL  360
RVKFSRSADA PAYQQGQNQL YNELNLGRRE EYDVLDKRRG RDPEMGGKPR RKNPQEGLYN  420
ELQKDKMAEA YSEIGMKGER RRGKGHDGLY QGLSTATKDT YDALHMQALP PR          472

SEQ ID NO: 102          moltype = AA  length = 472
FEATURE                 Location/Qualifiers
source                  1..472
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 102
QVQLVQSGAE VKKPGASVKV SCKASGYIFT AYTMHWVRQA PGQGLEWMGW IKPNNGLANY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE ATTEFDYWGQ GTLVTVSSGG  120
GGSGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP  180
PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK  240
VEIKASTTTP APRPPTPAPT IASQPLSLRP EACRPAAGGA VHTRGLDFAC DFWVLVVVGG  300
VLACYSLLVT VAFIIFWVRS KRSRLLHSDY MNMTPRRPGP TRKHYQPYAP PRDFAAYRSL  360
RVKFSRSADA PAYQQGQNQL YNELNLGRRE EYDVLDKRRG RDPEMGGKPR RKNPQEGLYN  420
ELQKDKMAEA YSEIGMKGER RRGKGHDGLY QGLSTATKDT YDALHMQALP PR          472

SEQ ID NO: 103          moltype = AA  length = 472
FEATURE                 Location/Qualifiers
source                  1..472
                        mol_type = protein
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AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE WTTEFDYWGQ GTLVTVSSGG  120
GGSGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP  180
PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK  240
VEIKASTTTP APRPPTPAPT IASQPLSLRP EACRPAAGGA VHTRGLDFAC DFWVLVVVGG  300
VLACYSLLVT VAFIIFWVRS KRSRLLHSDY MNMTPRRPGP TRKHYQPYAP PRDFAAYRSL  360
RVKFSRSADA PAYQQGQNQL YNELNLGRRE EYDVLDKRRG RDPEMGGKPR RKNPQEGLYN  420
ELQKDKMAEA YSEIGMKGER RRGKGHDGLY QGLSTATKDT YDALHMQALP PR          472

SEQ ID NO: 104          moltype = AA  length = 472
FEATURE                 Location/Qualifiers
source                  1..472
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 104
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AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE FTTEFDYWGQ GTLVTVSSGG  120
GGSGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP  180
PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK  240
VEIKASTTTP APRPPTPAPT IASQPLSLRP EACRPAAGGA VHTRGLDFAC DFWVLVVVGG  300
VLACYSLLVT VAFIIFWVRS KRSRLLHSDY MNMTPRRPGP TRKHYQPYAP PRDFAAYRSL  360
RVKFSRSADA PAYQQGQNQL YNELNLGRRE EYDVLDKRRG RDPEMGGKPR RKNPQEGLYN  420
ELQKDKMAEA YSEIGMKGER RRGKGHDGLY QGLSTATKDT YDALHMQALP PR          472

SEQ ID NO: 105          moltype = AA  length = 472
FEATURE                 Location/Qualifiers
source                  1..472
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 105
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AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCARSE GTTEFDYWGQ GTLVTVSSGG  120
GGSGGGGSGG GGSDIVLTQS PDSLAVSLGE RATINCKSSE SVDSYANSFM HWYQQKPGQP  180
PKLLIYRAST RESGVPDRFS GSGSRTDFTL TISSLQAEDV AVYYCQQSKE DPLTFGGGTK  240
VEIKASTTTP APRPPTPAPT IASQPLSLRP EACRPAAGGA VHTRGLDFAC DFWVLVVVGG  300
VLACYSLLVT VAFIIFWVRS KRSRLLHSDY MNMTPRRPGP TRKHYQPYAP PRDFAAYRSL  360
RVKFSRSADA PAYQQGQNQL YNELNLGRRE EYDVLDKRRG RDPEMGGKPR RKNPQEGLYN  420
ELQKDKMAEA YSEIGMKGER RRGKGHDGLY QGLSTATKDT YDALHMQALP PR          472

SEQ ID NO: 106          moltype = AA  length = 881
FEATURE                 Location/Qualifiers
source                  1..881
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 106
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WMGWIKPNNG LANYAQKFQG RVTMTRDTSI STAYMELSRL RSDDTAVYYC ARSEITTEFD  120
YWGQGTLVTV SSGGGGSGGG GSGGGGSDIV LTQSPDSLAV SLGERATINC KSSESVDSYA  180
NSFMHWYQQK PGQPPKLLIY RASTRESGVP DRFSGSGSRT DFTLTISSLQ AEDVAVYYCQ  240
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QSKEDPLTFG GGTKVEIKAS TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL  300
DFACDFWVLV VVGGVLACYS LLVTVAFIIF WVRSKRSRLL HSDYMNMTPR RPGPTRKHYQ  360
PYAPPRDFAA YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVLD KRRGRDPEMG  420
GKPRRKNPQE GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM  480
QALPPRTSGS GATNFSLLKQ AGDVEENPGP SRMDMADEPL NGSHTWLSIP FDLNGSVVST  540
NTSNQTEPYY DLTSNAVLTF IYFVVCIIGL CGNTLVIYVI LRYAKMKTIT NIYILNLAIA  600
DELFMLGLPF LAMQVALVHW PFGKAICRVV MTVDGINQFT SIFCLTVMSI DRYLAVVHPI  660
KSAKWRRPRT AKMITMAVWG VSLLVILPIM IYAGLRSNQW GRSSCTINWP GESGAWYTGF  720
IIYTFILGFL VPLTIICLCY LFIIIKVKSS GIRVGSSKRK KSEKKVTRMV SIVVAVFIFC  780
WLPFYIFNVS SVSMAISPTP ALKGMFDFVV VLTYANSCAN PILYAFLSDN FKKSFQNVLC  840
LVKVSGTDDG ERSDSKQDKS RLNETTETQR TLLNGDLQTS I                    881

SEQ ID NO: 107         moltype = AA  length = 881
FEATURE                Location/Qualifiers
source                 1..881
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 107
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WMGWIKPNNG LANYAQKFQG RVTMTRDTSI STAYMELSRL RSDDTAVYYC ARSEATTEFD  120
YWGQGTLVTV SSGGGGSGGG GSGGGGSDIV LTQSPDSLAV SLGERATINC KSSESVDSYA  180
NSFMHWYQQK PGQPPKLLIY RASTRESGVP DRFSGSGSRT DFTLTISSLQ AEDVAVYYCQ  240
QSKEDPLTFG GGTKVEIKAS TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL  300
DFACDFWVLV VVGGVLACYS LLVTVAFIIF WVRSKRSRLL HSDYMNMTPR RPGPTRKHYQ  360
PYAPPRDFAA YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVLD KRRGRDPEMG  420
GKPRRKNPQE GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM  480
QALPPRTSGS GATNFSLLKQ AGDVEENPGP SRMDMADEPL NGSHTWLSIP FDLNGSVVST  540
NTSNQTEPYY DLTSNAVLTF IYFVVCIIGL CGNTLVIYVI LRYAKMKTIT NIYILNLAIA  600
DELFMLGLPF LAMQVALVHW PFGKAICRVV MTVDGINQFT SIFCLTVMSI DRYLAVVHPI  660
KSAKWRRPRT AKMITMAVWG VSLLVILPIM IYAGLRSNQW GRSSCTINWP GESGAWYTGF  720
IIYTFILGFL VPLTIICLCY LFIIIKVKSS GIRVGSSKRK KSEKKVTRMV SIVVAVFIFC  780
WLPFYIFNVS SVSMAISPTP ALKGMFDFVV VLTYANSCAN PILYAFLSDN FKKSFQNVLC  840
LVKVSGTDDG ERSDSKQDKS RLNETTETQR TLLNGDLQTS I                    881

SEQ ID NO: 108         moltype = AA  length = 881
FEATURE                Location/Qualifiers
source                 1..881
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 108
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WMGWIKPNNG LANYAQKFQG RVTMTRDTSI STAYMELSRL RSDDTAVYYC ARSEWTTEFD  120
YWGQGTLVTV SSGGGGSGGG GSGGGGSDIV LTQSPDSLAV SLGERATINC KSSESVDSYA  180
NSFMHWYQQK PGQPPKLLIY RASTRESGVP DRFSGSGSRT DFTLTISSLQ AEDVAVYYCQ  240
QSKEDPLTFG GGTKVEIKAS TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL  300
DFACDFWVLV VVGGVLACYS LLVTVAFIIF WVRSKRSRLL HSDYMNMTPR RPGPTRKHYQ  360
PYAPPRDFAA YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVLD KRRGRDPEMG  420
GKPRRKNPQE GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM  480
QALPPRTSGS GATNFSLLKQ AGDVEENPGP SRMDMADEPL NGSHTWLSIP FDLNGSVVST  540
NTSNQTEPYY DLTSNAVLTF IYFVVCIIGL CGNTLVIYVI LRYAKMKTIT NIYILNLAIA  600
DELFMLGLPF LAMQVALVHW PFGKAICRVV MTVDGINQFT SIFCLTVMSI DRYLAVVHPI  660
KSAKWRRPRT AKMITMAVWG VSLLVILPIM IYAGLRSNQW GRSSCTINWP GESGAWYTGF  720
IIYTFILGFL VPLTIICLCY LFIIIKVKSS GIRVGSSKRK KSEKKVTRMV SIVVAVFIFC  780
WLPFYIFNVS SVSMAISPTP ALKGMFDFVV VLTYANSCAN PILYAFLSDN FKKSFQNVLC  840
LVKVSGTDDG ERSDSKQDKS RLNETTETQR TLLNGDLQTS I                    881

SEQ ID NO: 109         moltype = AA  length = 881
FEATURE                Location/Qualifiers
source                 1..881
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 109
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WMGWIKPNNG LANYAQKFQG RVTMTRDTSI STAYMELSRL RSDDTAVYYC ARSEFTTEFD  120
YWGQGTLVTV SSGGGGSGGG GSGGGGSDIV LTQSPDSLAV SLGERATINC KSSESVDSYA  180
NSFMHWYQQK PGQPPKLLIY RASTRESGVP DRFSGSGSRT DFTLTISSLQ AEDVAVYYCQ  240
QSKEDPLTFG GGTKVEIKAS TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL  300
DFACDFWVLV VVGGVLACYS LLVTVAFIIF WVRSKRSRLL HSDYMNMTPR RPGPTRKHYQ  360
PYAPPRDFAA YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVLD KRRGRDPEMG  420
GKPRRKNPQE GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM  480
QALPPRTSGS GATNFSLLKQ AGDVEENPGP SRMDMADEPL NGSHTWLSIP FDLNGSVVST  540
NTSNQTEPYY DLTSNAVLTF IYFVVCIIGL CGNTLVIYVI LRYAKMKTIT NIYILNLAIA  600
DELFMLGLPF LAMQVALVHW PFGKAICRVV MTVDGINQFT SIFCLTVMSI DRYLAVVHPI  660
KSAKWRRPRT AKMITMAVWG VSLLVILPIM IYAGLRSNQW GRSSCTINWP GESGAWYTGF  720
IIYTFILGFL VPLTIICLCY LFIIIKVKSS GIRVGSSKRK KSEKKVTRMV SIVVAVFIFC  780
WLPFYIFNVS SVSMAISPTP ALKGMFDFVV VLTYANSCAN PILYAFLSDN FKKSFQNVLC  840
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SEQ ID NO: 110          moltype = AA  length = 881
FEATURE                 Location/Qualifiers
source                  1..881
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 110
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WMGWIKPNNG LANYAQKFQG RVTMTRDTSI STAYMELSRL RSDDTAVYYC ARSEGTTEFD  120
YWGQGTLVTV SSGGGGSGGG GSGGGGSDIV LTQSPDSLAV SLGERATINC KSSESVDSYA  180
NSFMHWYQQK PGQPPKLLIY RASTRESGVP DRFSGSGSRT DFTLTISSLQ AEDVAVYYCQ  240
QSKEDPLTFG GGTKVEIKAS TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL  300
DFACDFWVLV VVGGVLACYS LLVTVAFIIF WVRSKRSRLL HSDYMNMTPR RPGPTRKHYQ  360
PYAPPRDFAA YRSLRVKFSR SADAPAYQQG QNQLYNELNL GRREEYDVLD KRRGRDPEMG  420
GKPRRKNPQE GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM  480
QALPPRTSGS GATNFSLLKQ AGDVEENPGP SRMDMADEPL NGSHTWLSIP FDLNGSVVST  540
NTSNQTEPYY DLTSNAVLTF IYFVVCIIGL CGNTLVIYVI LRYAKMKTIT NIYILNLAIA  600
DELFMLGLPF LAMQVALVHW PFGKAICRVV MTVDGINQFT SIFCLTVMSI DRYLAVVHPI  660
KSAKWRRPRT AKMITMAVWG VSLLVILPIM IYAGLRSNQW GRSSCTINWP GESGAWYTGF  720
IIYTFILGFL VPLTIICLCY LFIIIKVKSS GIRVGSSKRK KSEKKVTRMV SIVVAVFIFC  780
WLPFYIFNVS SVSMAISPTP ALKGMFDFVV VLTYANSCAN PILYAFLSDN FKKSFQNVLC  840
LVKVSGTDDG ERSDSKQDKS RLNETTETQR TLLNGDLQTS I                881

SEQ ID NO: 111          moltype = DNA  length = 1416
FEATURE                 Location/Qualifiers
source                  1..1416
                        mol_type = other DNA
                        organism = synthetic construct
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gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag  300
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aagaggagca ggctcctgca cagtgactac atgaacatga ctccccgccg ccccgggccc  1020
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agagtgaagt tcagcaggag cgcagacgcc cccgcgtacc agcagggcca gaaccagctc  1140
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cgggaccctg agatgggggg aaagccgaga aggaagaacc ctcaggaagg cctgtacaat  1260
gaactgcaga aagataagat ggcggaggcc tacagtgaga ttgggatgaa aggcgagcgc  1320
cggaggggca aggggcacga tggcctttac cagggtctca gtacagccac caaggacacc  1380
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SEQ ID NO: 112          moltype = DNA  length = 1416
FEATURE                 Location/Qualifiers
source                  1..1416
                        mol_type = other DNA
                        organism = synthetic construct
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ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
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atcgcgagcc agcccctgtc gctgaggccg gaagcatgcc gccctgccgc cggaggtgct  840
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acccgcaagc attaccagcc ctatgcccca ccacgcgact tcgcagccta tcgctccctg  1080
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cgggaccctg agatgggggg aaagccgaga aggaagaacc ctcaggaagg cctgtacaat  1260
gaactgcaga aagataagat ggcggaggcc tacagtgaga ttgggatgaa aggcgagcgc  1320
cggaggggca aggggcacga tggcctttac cagggtctca gtacagccac caaggacacc  1380
tacgacgccc ttcacatgca ggccctgccc cctcgc                            1416

SEQ ID NO: 113          moltype = DNA  length = 1416
FEATURE                 Location/Qualifiers
source                  1..1416
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 113
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ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
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agagtgaagt tcagcaggag cgcagacgcc cccgcgtacc agcagggcca gaaccagctc  1140
tataacgagc tcaatctagg acgaagagag gagtacgatg ttttggacaa gagacgtggc  1200
cgggaccctg agatgggggg aaagccgaga aggaagaacc ctcaggaagg cctgtacaat  1260
gaactgcaga aagataagat ggcggaggcc tacagtgaga ttgggatgaa aggcgagcgc  1320
cggaggggca aggggcacga tggcctttac cagggtctca gtacagccac caaggacacc  1380
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SEQ ID NO: 114          moltype = DNA  length = 1416
FEATURE                 Location/Qualifiers
source                  1..1416
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 114
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ccgggacagg ggctggagtg gatgggctgg atcaagccta acaacggtct ggccaactac  180
gcgcagaagt tccagggccg cgtgaccatg actcgggaca ccagcatctc taccgcgtac  240
atggagctgt cccgcctgcg ctccgatgac acggccgtgt actactgcgc ccgctcggag  300
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gcggtgtact actgtcaaca gtccaaggag gacccgctta ccttcggcgg cggcaccaag  720
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gtgcataccc ggggattgga cttcgcatgc gacttttggg tgctggtggt ggttggtgga  900
gtcctggctt gctatagctt gctagtaaca gtggccttta ttattttctg ggtgaggagt  960
aagaggagca ggctcctgca cagtgactac atgaacatga ctccccgccg ccccgggccc  1020
acccgcaagc attaccagcc ctatgcccca ccacgcgact tcgcagccta tcgctccctg  1080
agagtgaagt tcagcaggag cgcagacgcc cccgcgtacc agcagggcca gaaccagctc  1140
tataacgagc tcaatctagg acgaagagag gagtacgatg ttttggacaa gagacgtggc  1200
cgggaccctg agatgggggg aaagccgaga aggaagaacc ctcaggaagg cctgtacaat  1260
gaactgcaga aagataagat ggcggaggcc tacagtgaga ttgggatgaa aggcgagcgc  1320
cggaggggca aggggcacga tggcctttac cagggtctca gtacagccac caaggacacc  1380
tacgacgccc ttcacatgca ggccctgccc cctcgc                            1416

SEQ ID NO: 115          moltype = DNA  length = 1416
FEATURE                 Location/Qualifiers
source                  1..1416
                        mol_type = other DNA
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                         organism = synthetic construct
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ctccaaacca gtatctaa                                                 3918

SEQ ID NO: 119          moltype = DNA  length = 3918
FEATURE                 Location/Qualifiers
source                  1..3918
                        mol_type = other DNA
                        organism = synthetic construct
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SEQ ID NO: 120          moltype = DNA  length = 3918
FEATURE                 Location/Qualifiers
source                  1..3918
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 120
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SEQ ID NO: 121          moltype = AA  length = 15
FEATURE                 Location/Qualifiers
source                  1..15
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 121
GGGGSGGGGS GGGGS                                                   15

SEQ ID NO: 122          moltype = AA  length = 42
FEATURE                 Location/Qualifiers
source                  1..42
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 122
KRGRKKLLYI FKQPFMRPVQ TTQEEDGCSC RFPEEEEGGC EL                     42

SEQ ID NO: 123          moltype = AA  length = 41
FEATURE                 Location/Qualifiers
source                  1..41
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 123
RSKRSRLLHS DYMNMTPRRP GPTRKHYQPY APPRDFAAYR S                      41

SEQ ID NO: 124          moltype = AA  length = 42
FEATURE                 Location/Qualifiers
source                  1..42
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 124
ALYLLRRDQR LPPDAHKPPG GGSFRTPIQE EQADAHSTLA KI                     42

SEQ ID NO: 125          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
source                  1..112
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 125
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ELQKDKMAEA YSEIGMKGER RRGKGHDGLY QGLSTATKDT YDALHMQALP PR          112

SEQ ID NO: 126          moltype = AA  length = 45
FEATURE                 Location/Qualifiers
source                  1..45
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 126
TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACD                  45

SEQ ID NO: 127          moltype = AA  length = 47
FEATURE                 Location/Qualifiers
source                  1..47
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 127
RAAAIEVMYP PPYLDNEKSN GTIIHVKGKH LCPSPLFPGP SKPKDPK                47

SEQ ID NO: 128          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 128
SKYGPPCPSC P                                                       11
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SEQ ID NO: 129         moltype = AA  length = 24
FEATURE                Location/Qualifiers
source                 1..24
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 129
IYIWAPLAGT CGVLLLSLVI TLYC                                          24

SEQ ID NO: 130         moltype = AA  length = 24
FEATURE                Location/Qualifiers
source                 1..24
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 130
FWVLVVVGGV LACYSLLVTV AFII                                          24

SEQ ID NO: 131         moltype = AA  length = 24
FEATURE                Location/Qualifiers
source                 1..24
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 131
FWLPIGCAAF VVVCILGCIL ICWL                                          24

SEQ ID NO: 132         moltype = AA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 132
LGWLTVVLLA VAACVLLLTS AQLGL                                         25

SEQ ID NO: 133         moltype = AA  length = 1145
FEATURE                Location/Qualifiers
source                 1..1145
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 133
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RGSNYTSKYL GMTLATDPTD GSILACDPGL SRTCDQNTYL SGLCYLFRQN LQGPMLQGRP  120
GFQECIKGNV DLVFLFDGSM SLQPDEFQKI LDFMKDVMKK LSNTSYQFAA VQFSTSYKTE  180
FDFSDYVKWK DPDALLKHVK HMLLLTNTFG AINYVATEVF REELGARPDA TKVLIIITDG  240
EATDSGNIDA AKDIIRYIIG IGKHFQTKES QETLHKFASK PASEFVKILD TFEKLKDLFT  300
ELQKKIYVIE GTSKQDLTSF NMELSSSGIS ADLSRGHAVV GAVGAKDWAG GFLDLKADLQ  360
DDTFIGNEPL TPEVRAGYLG YTVTWLPSRQ KTSLLASGAP RYQHMGRVLL FQEPQGGGHW  420
SQVQTIHGTQ IGSYFGGELC GVDVDQDGET ELLLIGAPLF YGEQRGGRVF IYQRRQLGFE  480
EVSELQGDPG YPLGRFGEAI TALTDINGDG LVDVAVGAPL EEQGAVYIFN GRHGGLSPQP  540
SQRIEGTQVL SGIQWFGRSI HGVKDLEGDG LADVAVGAES QMIVLSSRPV VDMVTLMSFS  600
PAEIPVHEVE CSYSTSNKMK EGVNITICFQ IKSLYPQFQG RLVANLTYTL QLDGHRTRRR  660
GLFPGGRHEL RRNIAVTTSM SCTDFSFHFP VCVQDLISPI NVSLNFSLWE EEGTPRDQRA  720
QGKDIPPILR PSLHSETWEI PFEKNCGEDK KCEANLRVSF SPARSRALRL TAFASLSVEL  780
SLSNLEEDAY WVQLDLHFPP GLSFRKVEML KPHSQIPVSC EELPEESRLL SRALSCNVSS  840
PIFKAGHSVA LQMMFNTLVN SSWGDSVELH ANVTCNNEDS DLLEDNSATT IIPILYPINI  900
LIQDQEDSTL YVSFTPKGPK IHQVKHMYQV RIQPSIHDHN IPTLEAVVGV PQPPSEGPIT  960
HQWSVQMEPP VPCHYEDLER LPDAAEPCLP GALFRCPVVF RQEILVQVIG TLELVGEIEA  1020
SSMFSLCSSL SISFNSSKHF HLYGSNASLA QVVMKVDVVY EKQMLYLYVL SGIGGLLLLL  1080
LIFIVLYKVG FFKRNLKEKM EAGRGVPNGI PAEDSEQLAS GQEAGDPGCL KPLHEKDSES  1140
GGGKD                                                               1145

SEQ ID NO: 134         moltype = AA  length = 181
FEATURE                Location/Qualifiers
source                 1..181
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 134
VDLVFLFDGS MSLQPDEFQK ILDFMKDVMK KLSNTSYQFA AVQFSTSYKT EFDFSDYVKW  60
KDPDALLKHV KHMLLLTNTF GAINYVATEV FREELGARPD ATKVLIIITD GEATDSGNID  120
AAKDIIRYII GIGKHFQTKE SQETLHKFAS KPASEFVKIL DTFEKLKDLF TELQKKIYVI  180
E                                                                   181

SEQ ID NO: 135         moltype = AA  length = 369
FEATURE                Location/Qualifiers
source                 1..369
                       mol_type = protein
                       organism = Homo sapiens
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SEQUENCE: 135
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VDGINQFTSI FCLTVMSIDR YLAVVHPIKS AKWRRPRTAK MITMAVWGVS LLVILPIMIY  180
AGLRSNQWGR SSCTINWPGE SGAWYTGFII YTFILGFLVP LTIICLCYLF IIIKVKSSGI  240
RVGSSKRKKS EKKVTRMVSI VVAVFIFCWL PFYIFNVSSV SMAISPTPAL KGMFDFVVVL  300
TYANSCANPI LYAFLSDNFK KSFQNVLCLV KVSGTDDGER SDSKQDKSRL NETTETQRTL  360
LNGDLQTSI                                                         369

SEQ ID NO: 136         moltype = AA  length = 314
FEATURE                Location/Qualifiers
source                 1..314
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 136
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NTLVIYVILR YAKMKTITNI YILNLAIADE LFMLGLPFLA MQVALVHWPF GKAICRVVMT  120
VDGINQFTSI FCLTVMSIDR YLAVVHPIKS AKWRRPRTAK MITMAVWGVS LLVILPIMIY  180
AGLRSNQWGR SSCTINWPGE SGAWYTGFII YTFILGFLVP LTIICLCYLF IIIKVKSSGI  240
RVGSSKRKKS EKKVTRMVSI VVAVFIFCWL PFYIFNVSSV SMAISPTPAL KGMFDFVVVL  300
TYANSCANPI LYAF                                                   314

SEQ ID NO: 137         moltype = DNA  length = 1110
FEATURE                Location/Qualifiers
source                 1..1110
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 137
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tacctggctg tggtccaccc catcaagtcg gccaagtgga ggagaccccg gacggccaag  480
atgatcacca tggctgtgtg ggagtctct ctgctggtca tcttgcccat catgatatat  540
gctgggctcc ggagcaacca gtgggggaga agcagctgca ccatcaactg gccaggtgaa  600
tctggggctt ggtacacagg gttcatcatc tacactttca ttctggggtt cctggtaccc  660
ctcaccatca tctgtctttg ctacctgttc attatcatca aggtgaagtc ctctggaatc  720
cgagtgggct cctctaagag gaagaagtct gagaagaagg tcacccgaat ggtgtccatc  780
gtggtggctg tcttcatctt ctgctggctt cccttctaca tattcaacgt ttcttccgtc  840
tccatggcca tcagccccac cccagccctt aaaggcatgt ttgactttgt ggtggtcctc  900
acctatgcta acagctgtgc caaccctatc ctatatgcct tcttgtctga caacttcaag  960
aagagcttcc agaatgtcct ctgcttggtc aaggtgagcg gcacagatga tggggagcgg  1020
agtgacagta agcaggacaa atcccggctg aatgagacca cggagaccca gaggaccctc  1080
ctcaatggag acctccaaac cagtatctaa                                  1110

SEQ ID NO: 138         moltype = DNA  length = 936
FEATURE                Location/Qualifiers
source                 1..936
                       mol_type = genomic DNA
                       organism = Homo sapiens
SEQUENCE: 138
atggacatgg cggatgagcc actcaatgga agccacacat ggctatccat tccatttgac  60
ctcaatggct ctgtggtgtc aaccaacacc tcaaaccaga cagagccgta ctatgacctg  120
acaagcaatg cagtcctcac attcatctat tttgtggtct gcatcattgg gttgtgtggc  180
aacacacttg tcatttatgt catcctccgc tatgccaaga tgaagaccat caccaacatt  240
tacatcctca acctggccat cgcagatgag ctcttcatgc tgggtctgcc tttcttggct  300
atgcaggtgg ctctggtcca ctggcccttt ggcaaggcca tttgccgggt ggtcatgact  360
gtggatggca tcaatcagtt caccagcatc ttctgcctga cagtcatgag catcgaccga  420
tacctggctg tggtccaccc catcaagtcg gccaagtgga ggagaccccg gacggccaag  480
atgatcacca tggctgtgtg ggagtctct ctgctggtca tcttgcccat catgatatat  540
gctgggctcc ggagcaacca gtgggggaga agcagctgca ccatcaactg gccaggtgaa  600
tctggggctt ggtacacagg gttcatcatc tacactttca ttctggggtt cctggtaccc  660
ctcaccatca tctgtctttg ctacctgttc attatcatca aggtgaagtc ctctggaatc  720
cgagtgggct cctctaagag gaagaagtct gagaagaagg tcacccgaat ggtgtccatc  780
gtggtggctg tcttcatctt ctgctggctt cccttctaca tattcaacgt ttcttccgtc  840
tccatggcca tcagccccac cccagccctt aaaggcatgt ttgactttgt ggtggtcctc  900
acctatgcta acagctgtgc caaccctatc ctatat                            936

SEQ ID NO: 139         moltype = AA  length = 22
FEATURE                Location/Qualifiers
source                 1..22
                       mol_type = protein
                       organism = synthetic construct
```

-continued

```
SEQUENCE: 139
GSGATNFSLL KQAGDVEENP GP                                       22

SEQ ID NO: 140         moltype = AA  length = 23
FEATURE                Location/Qualifiers
source                 1..23
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 140
GSGQCTNYAL LKLAGDVESN PGP                                      23

SEQ ID NO: 141         moltype = AA  length = 21
FEATURE                Location/Qualifiers
source                 1..21
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 141
GSGEGRGSLL TCGDVEENPG P                                        21

SEQ ID NO: 142         moltype = AA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 142
GSGVKQTLNF DLLKLAGDVE SNPGP                                    25

SEQ ID NO: 143         moltype = AA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = protein
                       organism = synthetic construct
VARIANT                1..40
                       note = This sequence may encompass 1-10 GGGS repeating units
SEQUENCE: 143
GGGSGGGSGG GSGGGSGGGS GGGSGGGSGG GSGGGSGGGS                    40

SEQ ID NO: 144         moltype = DNA  length = 28
FEATURE                Location/Qualifiers
source                 1..28
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 144
ctgtgccacc gcgggctcct acgtgagc                                 28

SEQ ID NO: 145         moltype = DNA  length = 18
FEATURE                Location/Qualifiers
source                 1..18
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 145
tagtacacgg cgctgtcc                                            18

SEQ ID NO: 146         moltype = DNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 146
cgtgagccct gcggattatt ggggc                                    25

SEQ ID NO: 147         moltype = DNA  length = 18
FEATURE                Location/Qualifiers
source                 1..18
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 147
taggagccgt aggtggca                                            18

SEQ ID NO: 148         moltype = DNA  length = 23
FEATURE                Location/Qualifiers
source                 1..23
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 148
ctgtgccacc gtcggctcct acg                                      23
```

-continued

```
SEQ ID NO: 149          moltype = DNA  length = 18
FEATURE                 Location/Qualifiers
source                  1..18
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 149
tagtacacgg cgctgtcc                                              18

SEQ ID NO: 150          moltype = DNA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 150
ctgtgccacc ctcggctcct ac                                         22

SEQ ID NO: 151          moltype = DNA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 151
ctgtgccacc atcggctcct ac                                         22

SEQ ID NO: 152          moltype = DNA  length = 30
FEATURE                 Location/Qualifiers
source                  1..30
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 152
ctgtgccacc atgggctcct acgtgagccc                                 30

SEQ ID NO: 153          moltype = DNA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 153
ccgctcggag gccaccaccg ag                                         22

SEQ ID NO: 154          moltype = DNA  length = 18
FEATURE                 Location/Qualifiers
source                  1..18
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 154
gcgcagtagt acacggcc                                              18

SEQ ID NO: 155          moltype = DNA  length = 22
FEATURE                 Location/Qualifiers
source                  1..22
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 155
ctcggagatc gccaccgagt tc                                         22

SEQ ID NO: 156          moltype = DNA  length = 18
FEATURE                 Location/Qualifiers
source                  1..18
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 156
cgggcgcagt agtacacg                                              18

SEQ ID NO: 157          moltype = DNA  length = 18
FEATURE                 Location/Qualifiers
source                  1..18
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 157
gcgcagtagt acacggcc                                              18

SEQ ID NO: 158          moltype = DNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
```

-continued

```
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 158
ccgctcggag ggcaccaccg agttc                                      25

SEQ ID NO: 159          moltype = DNA  length = 23
FEATURE                 Location/Qualifiers
source                  1..23
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 159
ccgctcggag ttcaccaccg agt                                        23

SEQ ID NO: 160          moltype = DNA  length = 25
FEATURE                 Location/Qualifiers
source                  1..25
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 160
ccgctcggag tggaccaccg agttc                                      25

SEQ ID NO: 161          moltype = DNA  length = 1179
FEATURE                 Location/Qualifiers
source                  1..1179
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 161
ggctccggtg cccgtcagtg ggcagagcgc acatcgccca cagtccccga gaagttgggg  60
ggaggggtcg gcaattgaac cggtgcctag agaaggtggc gcggggtaaa ctgggaaagt  120
gatgtcgtgt actggctccg cctttttccc gagggtgggg gagaaccgta tataagtgca  180
gtagtcgccg tgaacgttct ttttcgcaac gggtttgccg ccagaacaca ggtaagtgcc  240
gtgtgtggtt cccgcgggcc tggcctcttt acgggttatg gcccttgcgt gccttgaatt  300
acttccacct ggctgcagta cgtgattctt gatcccgagc ttcgggttgg aagtgggtgg  360
gagagttcga ggccttgcgc ttaaggagcc ccttcgcctc gtgcttgagt tgaggcctgg  420
cctgggcgct ggggccgccg cgtgcgaatc tggtggcacc ttcgcgcctg tctcgctgct  480
ttcgataagt ctctagccat ttaaaatttt tgatgacctg ctgcgacgct ttttttctgg  540
caagatagtc ttgtaaatgc gggccaagat ctgcacactg gtatttcggt ttttggggcc  600
gcgggcggcg acggggcccg tgcgtcccag cgcacatgtt cggcgaggcg gggcctgcga  660
gcgcggccac cgagaatcgg acgggggtag tctcaagctg gccggcctgc tctggtgcct  720
ggtctcgcgc cgccgtgtat cgccccgccc tgggcggcaa ggctggcccg gtcggcacca  780
gttgcgtgag cggaaagatg gccgcttccc ggccctgctg cagggagctc aaaatggagg  840
acgcggcgct cgggagagcg ggcgggtgag tcacccacac aaaggaaaag ggcctttccg  900
tcctcagccg tcgcttcatg tgactccacg gagtaccggg cgccgtccag gcacctcgat  960
tagttctcga gcttttggag tacgtcgtct ttaggttggg gggaggggtt ttatgcgatg  1020
gagtttcccc acactgagtg ggtggagact gaagttaggc cagcttggca cttgatgtaa  1080
ttctccttgg aatttgccct ttttgagttt ggatcttggt tcattctcaa gcctcagaca  1140
gtggttcaaa gtttttttct tccatttcag gtgtcgtga                       1179

SEQ ID NO: 162          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 162
MALPVTALLL PLALLLHAAR P                                          21

SEQ ID NO: 163          moltype = DNA  length = 63
FEATURE                 Location/Qualifiers
source                  1..63
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 163
atggccttac cagtgaccgc cttgctcctg ccgctggcct tgctgctcca cgccgccagg  60
ccg                                                              63
```

1. A chimeric antigen receptor (CAR), comprising:

(a) an extracellular binding domain;

(b) a costimulatory domain; and

(c) an activation domain,

wherein the extracellular binding domain comprises a heavy chain variable (V_H) region comprising complementary determining 1 (HC CDR1), complementary determining 2 (HC CDR2), and complementary determining 3 (HC CDR3) regions; and a light chain variable (V_L) region comprising complementary determining 1 (LC CDR1), complementary determining 2 (LC CDR2), and complementary determining 3 (LC CDR3) regions, wherein:

(i) the V_H region comprises HC CDR1 and HC CDR2 regions comprising amino acid sequences set forth SEQ ID NOs: 1 and 2, respectively, and an HC CDR3 region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 3-8; and the V_L region comprises LC CDR1, LC CDR2, and LC CDR3 regions comprising amino acid sequences set forth in SEQ ID NOs: 9-11, respectively, or

(ii) the V_H region comprises HC CDR1 and HC CDR2 regions comprising amino acid sequences set forth in SEQ ID NOs: 70 and 71, respectively, and an HC CDR3 region comprising an amino acid sequence set forth in any one of SEQ ID NOs: 72-75; and the V_L region comprises LC CDR1, LC CDR2, and LC CDR3 regions comprising amino acid sequences set forth in SEQ ID NOs: 76-78, respectively.

2. The CAR of claim 1, wherein the V_H region comprises the amino acid sequences set forth in (i).

3. The CAR of claim 1, wherein the HC CDR3 comprises an amino acid sequence set forth in SEQ ID NO: 5 or SEQ ID NO: 8.

4. The CAR of claim 1, which comprises V_H and V_L region amino acid sequences selected from the group consisting of:

(a) SEQ ID NOs: 13 and 19, respectively;

(b) SEQ ID NOs: 14 and 19, respectively;

(c) SEQ ID NOs: 15 and 19, respectively;

(d) SEQ ID NOs: 16 and 19, respectively;

(e) SEQ ID NOs: 17 and 19, respectively; and

(f) SEQ ID NOs: 18 and 19, respectively;

5. The CAR of claim 1, wherein the extracellular domain is a single chain antibody fragment (scFv).

6. The CAR of claim 5, wherein scFv comprises a (GxS)n linker between the V_H and V_L regions.

7. The CAR of claim 5, wherein the scFv comprises an amino acid sequence set forth in any one of SEQ ID NOs: 29-34.

8. The CAR of claim 1, wherein the V_H region comprises the amino acid sequences set forth in (ii).

9. The CAR of claim 8, wherein the HC CDR3 comprises an amino acid sequence set forth in SEQ ID NO: 74 or 75.

10. The CAR of claim 8, which comprises VH and VL region amino acid sequences selected from the group consisting of:

(a) SEQ ID NOs: 80 and 84, respectively;

(b) SEQ ID NOs: 81 and 84, respectively;

(c) SEQ ID NOs: 82 and 84, respectively; and

(d) SEQ ID NOs: 83 and 84, respectively.

11. The CAR of claim 8, wherein the extracellular domain is an scFv.

12. The CAR of claim 11, wherein the scFv comprises a $(GxS)_n$ linker between the V_H and V_L regions.

13. The CAR of claim 11, wherein the scFv comprises an amino acid sequence set forth in any one of SEQ ID NOs: 92-95.

14. The CAR of claim 1, wherein the costimulatory domain is from 4-1BB (CD 137), CD28, OX40, ICOS, GITR, CD27, CD30, CD40, DAP 10, DAP12, BAFFR, HVEM, ICAM-1, lymphocyte function-associated antigen-1 (LFA-1), CD2, CDS, CD7, CD287, LIGHT, NKG2C, NKG2D, SLAMF7, NKp80, NKp30, NKp44, NKp46, CD 160, B7-H3, a ligand that specifically binds with CD83 or a combination thereof; optionally wherein the costimulatory domain comprises an amino acid sequence set forth in any one of SEQ ID NOs: 122-124.

15. The CAR of claim 14, wherein the costimulatory domain comprises 4-1BB or CD28.

16. The CAR of claim 1, wherein the extracellular binding domain comprises costimulatory domains from 4-1BB and CD28, or CD-28 and OX-40.

17. The CAR of claim 1, wherein the activation domain is from CD3 zeta, common FcR gamma (FCER1G), Fc gamma RIIa, FcR beta (Fc epsilon lb), CD3 gamma, CD3 delta, CD3epsilon, CD5, CD22, CD79a, CD79b, CD278 (ICOS), FcεRI, or CD66d.

18. The CAR of claim 17, wherein the activation domain is from CD3 zeta, optionally comprising the amino acid sequence of SEQ ID NO: 125.

19. The CAR of claim 1, further comprising a hinge domain, a transmembrane domain, or a combination thereof, which optionally is located between the extracellular binding domain and the costimulatory domain.

20. The CAR of claim 19 comprising a hinge domain from CD8 alpha, CD28, or IgG4; optionally wherein the hinge domain comprises an amino sequence set forth in any one of SEQ ID NOs: 126-128.

21. The CAR of claim 19, comprising a transmembrane domain from a cell surface receptor selected from the group consisting of an alpha, beta or zeta chain of a T cell receptor, CD8 alpha, CD28, ICOS, GITR, CD3 epsilon, CD45, CD4, CD5, CD8, CD9, CD16, CD22, CD33, CD37, CD64, CD80, CD86, CD134, CD137, CD154, CD271, TNFRSF19, Killer Cell Immunoglobulin-Like Receptor (KIR), and a combination thereof.

22. The CAR of claim 21, wherein the transmembrane domain is from CD8 alpha, CD28, ICOS, and GITR; optionally wherein the transmembrane domain comprises an amino acid sequence set forth in any one SEQ ID NOs: 129-131.

23. The CAR of claim 1, further comprising a signal peptide located at the N-terminus of a CAR precursor protein; optionally wherein the signal peptide is a CD8 alpha chain signal peptide.

24. The CAR of claim 1, further comprising a second extracellular antigen binding domain.

25. The CAR of claim 23, wherein the second extracellular antigen binding domain comprises a second scFv.

26. The CAR of claim 24, wherein the second extracellular antigen binding domain comprises an ICAM-1 binding domain.

27. The CAR of claim 26, wherein the ICAM-1 binding domain comprises an αL subunit I domain of human lymphocyte function-associated antigen-1 (LFA-1); optionally

wherein the ICAM-1 binding domain comprises an αL subunit I domain of human lymphocyte function-associated antigen-1 (LFA-1).

28. The CAR of claim **1**, wherein the CAR comprises an amino acid sequence set forth in any one of SEQ ID NOs: 43-48 or 102-105.

29-36. (canceled)

37. A population of immune cells expressing the CAR of claim **1**.

38. (canceled)

39. The population of immune cells of claim **37**, wherein the immune cells are T cells, natural killer (NK) cells, tumor infiltrating lymphocytes, dendritic cells, macrophages, B cells, neutrophils, eosinophils, basophils, mast cells, myeloid-derived suppressor cells, stem cells, precursors thereof, subtypes thereof, or a combination thereof; optionally wherein the immune cell is a human immune cell.

40. The population of immune cells of claim **39**, wherein the immune cells are T cells.

41. The population of immune cells of claim **40**, wherein the T cells express SSTR2.

42. The population of immune cells of claim **37**, further comprising a second population of immune cells, wherein the second population of immune cells expresses a second CAR or another polypeptide of interest.

43. (canceled)

44. The population of immune cells of claim **42**, wherein the second CAR comprises an ICAM-1 binding domain; optionally wherein the ICAM-1 binding domain comprises an αL subunit I domain of human LFA-1.

45. A cell therapy-based method of treating cancer, comprising administering to a subject in need thereof the population of immune cells of claim **37**.

46-66. (canceled)

67. A method for treating cancer and monitoring CAR-T cell distribution in a patient, comprising:

incubating the population of CAR-T cells of claim **41** with a radioactive label that binds to SSTR2,

intravenously infusing the labeled CAR-T cells into a patient in an amount of 10^4-10^8 cells/kg patient, and

detecting the labeled CAR-T cell distribution by PET/CT imaging,

wherein the labeled CAR-T cells are infiltrated into cancer cells to kill the cancer cells.

68-70. (canceled)

71. A method for treating cancer and monitoring CAR-T cell distribution in a patient, comprising:

intravenously infusing the population of CAR-T cells of claim **41** into a patient, wherein the CAR-T cells express or have been transduced to express at least 100,000 molecules of SSTR2 per T cell

injecting into the patient a radioactive label that binds to SSTR2 at least one hour prior to PET/CT imaging, and

detecting the labeled CAR-T cell distribution by PET/CT imaging,

wherein the labeled CAR-T cells are infiltrated into cancer cells to kill the cancer cells.

72. (canceled)

73. A method of producing a population of genetically engineered immune cells, the method comprising:

(a) providing a population of immune cells; and

(b) introducing into the immune cells a nucleic acid coding for the CAR claim **1** to produce a population of genetically engineered immune cells.

74-78. (canceled)

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